SOFARSOLAR Co., Ltd.

Modbus-G3 Protocol

V1.38

In January, 2025

Revision History

No.	Modify date	Version	Modify Information
1	2024/07/12	V1.31	Initial version
2	2024/07/24	V1.32	1. Added definition of "Centralized PID Control Word" register (0x10AF). 2. Modify the maximum value of the "Active output percentage" register (0x0901). 3. Modify the maximum value of the "Export maximum active power percentage" register (0x1106). 4. Correct the units of reactive power and apparent power in the "Grid-connected output parameter address" data block. 5. Correct the units of reactive power and apparent power in the "Grid-disconnected output parameter address" data block. 6. Revise the description of the "PV branch software over-current" fault.
3	2024/08/14	V1.33	1. Added definition of “Battery usage configuration” register (0x1073) 2. Added definition of “Description of port temperature detection fault” 3. Added definition of “Port Temperature Detection Data Area” 4. Added ID309 definition of “Abnormal DC port temperature (including temperature and NTC abnormalities)” for “Description of system fault information” 5. Added ID310 definition of “Abnormal AC port temperature (including temperature and NTC abnormalities)” for “Description of system fault information”
4	2024/09/12	V1.34	1. Correct “Description of port temperature detection fault” fault item address. 2. Added valid value description of “Communication protocol” register(0x1046) 3. Modify the address range of the "Port Temperature Detection Data Area" to 256 registers 4. Modify the value range of the “Set the sampling period for port temperature”(0x4AE1) register 5. Added definition of “Port NTC fault detection time” register (0x4AE6) 6. Added definition of “Port temperature fault detection time” register (0x4AE7) 7. Added definition of “Port temperature secondary fault detection time” register (0x4AE8) 8. Added definition of “Port temperature slope fault detection time” register (0x4AE9)
5	2024/10/16	V1.35	1. Added ID381 definition of “DC witch 1 trip” for “Description of system fault information”. 2. Added ID382 definition of “DC witch 2 trip” for “Description of system fault information”. 3. Added ID383 definition of “DC witch 3 trip” for “Description of system fault information”. 4. Added ID384 definition of “DC witch 4 trip” for “Description of system fault information”. 5. Added ID462 definition of “Live wire and ground wire short circuit fault” for “Description of system fault information”. 6. Added definition of “System files export by USB drive” register(0x1074) 7. Correct the the unit of “Intelligent load activation power” register(0x106F)

6	2024/11/15	V1.36	1. Added definition of “Power generation efficiency” register(0x04BF) 2. Added definition of “Total battery capacity” register(0x066B) 3. Added definition of “Power enable control” register(0x10CC) 4. Added definition of “Active power output control” register(0x10CD,0x10CE) 5. Added definition of “Reactive power output control” register(0x10CF,0x10D0) 6. Add a definition for the data area of "Battery cluster 1 address" 7. Add a definition for the data area of "Battery cluster 1 address" 8. Added ID334 definition of “Boost short circuit fault” for “Description of system fault information” 9. Added ID335 definition of “PV+ ground short circuit fault” for “Description of system fault information” 10. Added ID463 definition of “Permanent fault of live wire and ground wire short circuit” for “Description of system fault information” 11. Added ID040 definition of “Heating film heating in progress” for “Description of system status information” 12. Added ID041 definition of “Heating film enable” for “Description of system status information” 13. Added ID042 definition of “BMS requests heating” for “Description of system status information” 14. Added ID043 definition of “DSP allows heating” for “Description of system status information” 15. Added ID044 definition of “PV meter communication disconnected” for “Description of system status information” 16. Modify the value range of the “Dry contact control”(0x1064) register 17. Modify the value range of the “Battery usage configuration”(0x1073) register
7	2024/12/19	V1.37	1. Added definition of “Fan self-test control” register(0x10E0) 2. Added definition of “Fan noise mode” register(0x10E1) 3. Added definition of “Fan alarm detection mode” register(0x10E2) 4. Added definition of “Fan detection sensitivity” register(0x10E3) 5. Added definition of “Fan Speed Control” register(0x10E4)

8	2025/1/17	V1.38	1. Added definition of "Communication interruption control" register(0x10C0) 2. Added definition of "Communication interruption timeout parameter" register(0x10C1) 3. Added definition of "Preset value for communication interruption" register(0x10C2) 4. Added definition of "Communication module state" register(0x10AA) 5. Added definition of "External device state" register(0x10AB) 6. Added definition of "Fixed Unbalanced Power Control" register(0x10EB) 7. Added definition of "Unbalanced power percentage of R-phase" register(0x10EC) 8. Added definition of "Unbalanced power percentage of S-phase" register(0x10ED) 9. Added definition of "Unbalanced power percentage of T-phase" register(0x10EE) 10. Added definition of "Battery channel state" register(0x0646~0x0662) 11. Added definition of "BMS Current state" register(0x9437~0x9717) 12. Added definition of "Parameter Configuration Area 2" 13. Modify some registers in the "time-sharing control" area(0x1120~0x1126) 14. Added definition of "Target SOC" register(0x112A) 15. Added definition of "Target power" register(0x112B) 16. Added definition of "Power feeding priority parameter address" register(0x113A~0x113F) 17. Added definition of "Buying electricity to charge batteries" register(0x1134) 18. Added definition of "Battery backup SOC" register(0x1135) 19. Added definition of "Maximum charging SOC" register(0x105B) 20. Added definition of "Relay configuration" register(0x10F0) 21. Added definition of "Standby monitoring control" register(0x10F1) 22. Added definition of "Configuration of port types on the grid side" register(0x1075) 23. Added definition of "Configuration of port types on the generator side" register(0x1076) 24. Added ID096 definition of "EPS short circuit protection" 25. Modify the value range of the "Input channel type selection"(0x1011~0x1020) register 26. Modify the value range of the "Active load shedding rate"(0x0902) register 27. Modify the value range of the "Active power limit change rate"(0x110A) register 28. Modify the value type of the "Anti-back-flow power"(0x1024) register 29. Modify the value range of the "System running state"(0x0404) register

Table of Contents

1. Introduction

This Agreement applies to the communication between our equipment and the external monitoring and ac quisition terminal. Based on the MODBUS-RTU communication protocol, the equipment address data and fault s tatus defined by the protocol can be read in real time, and the working mode and working parameters of the equipment can be set by setting parameter points.

2. Physical Interface

Data protocol is based on RS485, default baud rate: 9600/115200bps, parity: none, data bit: 8, stop bit: 1.

3. Terms Defined

The descriptions of the terms involved in this Agreement are shown in the table below：

Term	Description
U16	Unsigned 16 digits
S16	Signed 16 digits
U32	Unsigned 32 digits
S32	Signed 32 digits
BCD16	The 16-bit BCD number code
ASCII	A string ending with a '\0'
OVP	Over voltage protection
UVP	Under voltage protection
OFP	Over frequency protection
UFP	Under frequency protection
GFCI	Ground Fault Circuit Interrupter
OVRT	Over Voltage Ride-Through Error
LVRT	Low Voltage Ride-Through Error
AD	Analog-to-Digital
AC	Alternating Current
DC	Direct Current
MPPT	Maximum Power Point Tracking
SW	Software
HW	Hardware
RMS	Root Mean Square
OCP	Over current protection
LLC	Resonant Converters
DCI	Direct Current Injection
SN	Serial Number
SPD	Surge protection Device
ID	Identification
PV	Photovoltaic

Term	Description
EPS	Emergency Power Supply
ISO	Insulation
PE	protecting earthing
SC	Short Circuit
BMS	Battery Management System
IGBT	Insulated Gate Bipolar Transistor
ZVP	Zero Voltage Protection
CAN	Controller Area Network
BCU	Battery Controller Unit
BMU	Battery Management Unit
PCC	Point of Common Coupling
Gen	Generator
N/A	Uninvolved

4. Protocol Format Description

4.1Master multi-register write request（0x10）

Domain	Length	Content	Note
Sub device address	1 byte	Device address：0x00 ~ 0xF7	Depending on the slave address(0x00 for the broadcast address)
Function code	1 byte	0x10	
Starting address	2 bytes	high byte	Range：0x0000 ~ 0xFFFF
		low byte	
Quantity of registers	2 bytes	high byte	Range：0x0001 ~ 0x007D
		low byte	
Byte count n	1 byte	n = 2 * Quantity of registers	Range：0x02~0xFA
Registers value	n bytes	high byte 1	For 32-bit and 64-bit registers,send in big-endian mode. For example, the 32-bit data transmission order is：byte3,byte2,byte1,byte0
		low byte 1	
		...	
		high byte n	
		low byte n	
CRC Check	2 bytes	low byte	Use Modbus-CRC16 calculate method
		high byte	

4.2The slave multi-register write response

Domain	Length	Content	Note
Sub device address	1 byte	Device address：0x01 ~ 0xF7	Depending on the slave address
Function code	1 byte	0x10	
Starting address	2 bytes	high byte	Range：0x0000 ~ 0xFFFF
		low byte	
Quantity of registers	2 bytes	high byte	Range：0x0001 ~ 0x007D
		low byte	
CRC Check	2 bytes	low byte	Use Modbus-CRC16 calculate method
		high byte	

4.3Master single-register write request（0x06）

Domain	Length	Content	Note
Sub device address	1 byte	Device address：0x01 ~ 0xF7	Depending on the slave address
Function code	1 byte	0x06	
Starting address	2 bytes	high byte	Range：0x0000 ~ 0xFFFF
		low byte	
Register value	2 bytes	high byte	Range：0x0000 ~ 0xFFFF
		low byte	

CRC Check	2 bytes	low byte	Use Modbus-CRC16 calculate method
		high byte	

4.4Write the response from the stand register

Domain	Length	Content	Note
Sub device address	1 byte	Device address：0x01 ~ 0xF7	Depending on the slave address
Function code	1 byte	0x06	
Starting address	2 bytes	high byte	Range：0x0000 ~ 0xFFFF
		low byte	
Register value	2 bytes	high byte	Range：0x0000 ~ 0xFFFF
		low byte	
CRC Check	2 bytes	low byte	Use Modbus-CRC16 calculate method
		high byte	

4.5Master multiple registers read the request（0x03）

Domain	Length	Content	Note
Sub device address	1 byte	Device address：0x01 ~ 0xF7	Depending on the slave address
Function code	1 byte	0x03	
Starting address	2 bytes	high byte	Range：0x0000 ~ 0xFFFF
		low byte	
Quantity of registers	2 bytes	high byte	Range：0x0001 ~ 0x007D
		low byte	
CRC Check	2 bytes	low byte	Use Modbus-CRC16 calculate method
		high byte	

4.6Read the multiple register responses from the machine

Domain	Length	Content	Note
Sub device address	1 byte	Device address：0x01 ~ 0xF7	Depending on the slave address
Function code	1 byte	0x03	
Byte count n	1 byte	n = 2 * Quantity of registers	Range：0x02~0xFA
Register value	n bytes	high byte 1	For 32-bit and 64-bit registers,send in big-endian mode. For example, the 32-bit data transmission order is： byte3,byte2,byte1,byte0
		low byte 1	
		...	
		high byte n	
		low byte n	
CRC Check	2 bytes	low byte	Use Modbus-CRC16 calculate method
		high byte	

4.7Machine answer exception format description

In addition to the standard MODBUS-RTU protocol response exception code, the command request operation of some models' slave devices uniformly returns the following exception format messages:

Domain	Length	Content	Note
Sub device address	1 byte	Device address：0x01 ~ 0xF7	Slave address
Function code	1 byte	0x90	
Error code	1 byte	0x01：Illegal function code 0x02：Illegal data address 0x03：Illegal data content 0x04：Slave device fault 0x07：Slave device busy	
CRC Check	2 bytes	low byte	Use Modbus-CRC16 calculate method
		high byte	

4.8Description of the register transfer encoding format

Data Encode		
Register type	Value	Send the order(Break it up into bytes and start transferring from the left)
U16/S16	0x1234	0x12,0x34
U32/S32	0x12345678	0x56,0x78,0x12,0x34
ASCII string	ABCD	A,B,C,D

5. Agreement address definition

Register address property description：

- RW Property：Support for 0x03,0x06,0x10 commands.
- WO Property：The 0x03 command is not supported, but supports the 0x06,0x10 commands.
- RO Property：Support for 0x03 command，but the 0x06,0x10 commands is not supported.

5.0 General Data Area

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Version number of this protocol	0044	BCD16	The protocol version number corresponds to the version number in the jacket of this protocol.	N/A	RO
2	Reserve	0045 - 0065	N/A		N/A	N/A

5.1Real-time Data Area

The register data defined in this data area is read-only data items, which are divided into 12 data blocks. Each data block does not support reading across blocks, and supports reading data from 60 register addresses at most once.

5.1.1System information parameter address 1

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	System running state	0404	U16	0：Waiting status 1：Detection status 2：Grid-connected status 3：Emergency power supply status 4：Recoverable fault status 5：Permanent fault status 6：Upgrading status 7：Self-charging status 8：SVG status 9：PID status 10：Limit the load status 11：Standby monitoring status	N/A	RO
2	Fault 1	0405	U16	More information refer “Description of system fault information” Example: Reading a value of 0x0402 from 0x0405 indicates that the inverter has a fault of under voltage in the power grid	N/A	RO
3	Fault 2	0406	U16		N/A	RO
4	Fault 3	0407	U16		N/A	RO
5	Fault 4	0408	U16		N/A	RO
6	Fault 5	0409	U16		N/A	RO
7	Fault 6	040A	U16		N/A	RO
8	Fault 7	040B	U16		N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
9	Fault 8	040C	U16		N/A	RO
10	Fault 9	040D	U16		N/A	RO
11	Fault 10	040E	U16		N/A	RO
12	Fault 11	040F	U16		N/A	RO
13	Fault 12	0410	U16		N/A	RO
14	Fault 13	0411	U16		N/A	RO
15	Fault 14	0412	U16		N/A	RO
16	Fault 15	0413	U16		N/A	RO
17	Fault 16	0414	U16		N/A	RO
18	Fault 17	0415	U16		N/A	RO
19	Fault 18	0416	U16		N/A	RO
20	Grid-connected waiting time	0417	U16	System startup countdown	1S	RO
21	Ambient temperature 1	0418	S16		1℃	RO
22	Ambient temperature 2	0419	S16		1℃	RO
23	Radiator temperature 1	041A	S16		1℃	RO
24	Radiator temperature 2	041B	S16		1℃	RO
25	Radiator temperature 3	041C	S16		1℃	RO
26	Radiator temperature 4	041D	S16		1℃	RO
27	Radiator temperature 5	041E	S16		1℃	RO
28	Radiator temperature 6	041F	S16		1℃	RO
29	Module temperature 1	0420	S16		1℃	RO
30	Module temperature 2	0421	S16		1℃	RO
31	Module temperature 3	0422	S16		1℃	RO
32	Module temperature 4	0423	S16		1℃	RO
33	Module temperature 5	0424	S16		1℃	RO
34	Module temperature 6	0425	S16		1℃	RO
35	Power generation time of the day	0426	U16	The power generation time of the inverter today starts the timing after the grid connection	1Min	RO
36	Total power generation time	0427 - 0428	U32	After the inverter is connected to the grid, start counting	1Min	RO
37	Total run time	0429 - 042A	U32	After the inverter is powered on,start counting	1Min	RO
38	Insulation impedance	042B	U16	Work in the inverter detection state	1KΩ	RO
39	System time (Year)	042C	U16	Range： [0, 99], actual year need to be added 2000 Example：0x0018 means 2024 year	N/A	RO
40	System time (Month)	042D	U16	Range：[1, 12]	N/A	RO
41	System Time (Day)	042E	U16	Range：[1, 31]	N/A	RO
42	System time (Hour)	042F	U16	Range：[0, 23]	N/A	RO
43	System time (Minute)	0430	U16	Range：[0, 59]	N/A	RO
44	System time (Seconds)	0431	U16	Range：[0, 59]	N/A	RO
45	Fault 19	0432	U16	More information refer "Description of system status information" Example: A reading from 0x0433 of 0x0001 indicates that the inverter has a fault with a control board over temperature alarm	N/A	RO
46	Fault 20	0433	U16		N/A	RO
47	Fault 21	0434	U16		N/A	RO
48	Fault 22	0435	U16		N/A	RO
49	Fault 23	0436	U16		N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
50	Fault 24	0437	U16		N/A	RO
51	Fault 25	0438	U16		N/A	RO
52	Fault 26	0439	U16		N/A	RO
53	Fault 27	043A	U16		N/A	RO
54	Fault 28	043B	U16		N/A	RO
55	Fault 29	043C	U16		N/A	RO
56	Fault 30	043D	U16		N/A	RO
57	Fan speed	043E	U16		1r/min	RO
58	Reserve	043F	N/A		N/A	N/A

5.1.2System Information Parameter Address 2

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Device serial number 1	0445	ASCII	Device serial number characters 1 to 16,17 to 20 characters in register addresses 0x0470 and 0x0471. notes： Register high 8 bits for serial number 1 Register low 8 bits for serial number 2 Example：0x5353 means "SS"	N/A	RO
		0446	ASCII		N/A	RO
		0447	ASCII		N/A	RO
		0448	ASCII		N/A	RO
		0449	ASCII		N/A	RO
		044A	ASCII		N/A	RO
		044B	ASCII		N/A	RO
		044C	ASCII		N/A	RO
2	Hardware version number	044D	ASCII	Example： 0x5630 means "V0",0x3033 means "03"	N/A	RO
		044E	ASCII		N/A	RO
3	Communication chip software version number	044F	ASCII	e.g: The value 0x3047 read from register 0x044F means "0G". The value 0x3130 read from register 0x0450 means "10". The value 0x3030 read from 0x0451 means "00".	N/A	RO
		0450	ASCII		N/A	RO
		0451	ASCII		N/A	RO
		0452	ASCII		N/A	RO
4	Master controller chip software version number	0453	ASCII	e.g: The value 0x3056 read from register 0x0453 means "0V". The value 0x3130 read from register 0x0454 means "10". The value 0x3030 read from 0x0455 means "00".	N/A	RO
		0454	ASCII		N/A	RO
		0455	ASCII		N/A	RO
		0456	ASCII		N/A	RO
5	Slave controller chip software version number	0457	ASCII	e.g: The value 0x0447 read from register 0x0453 means "0V". The value 0x3030 read from register 0x0458 means "00". The value 0x3030 read from 0x0459 means "00". The value 0x3031 read from 0x45A means "01".	N/A	RO
		0458	ASCII		N/A	RO
		0459	ASCII		N/A	RO
		045A	ASCII		N/A	RO
6	Safety library version number	045B	ASCII	Example：0x3036 means master version 06	N/A	RO
		045C	ASCII	Example：0x3038 means slave version 08	N/A	RO
7	The BOOT version of the ARM	045D	U16		N/A	RO
8	The BOOT version number of the master DSP	045E	U16		N/A	RO
9	The BOOT version number of the slave DSP	045F	U16		N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
10	Safety certification software version	0460	ASCII	Refer to explain of “Communication chip software version number”	N/A	RO
		0461	ASCII		N/A	RO
		0462	ASCII		N/A	RO
		0463	ASCII		N/A	RO
11	Safety certification hardware version	0464	ASCII	Refer to explain of “Hardware version number”	N/A	RO
		0465	ASCII		N/A	RO
12	AFCI version	0466	ASCII	High 8-bit ignore, low 8-bit is version number	N/A	RO
		0467	ASCII	The software version of AFCI	N/A	RO
		0468	ASCII	The hardware version of AFCI	N/A	RO
		0469	ASCII	The hardware serial number of AFCI	N/A	RO
		046A	ASCII	The communication version of AFCI	N/A	RO
13	Safety package version	046B	ASCII	Refer to explain of “Communication chip software version number”	N/A	RO
		046C	ASCII		N/A	RO
14	Reserve	046D - 046F	N/A		N/A	N/A
15	Device serial number2	0470 - 0471	ASCII	Device serial number from 17th to 20th characters	N/A	RO
16	Reserve	0472 - 0476	N/A		N/A	N/A
17	Status 1	0477	U16	More information refer “Description of system status information”. Example： A reading from 0x0477 of 0x0005 indicates that the status of the inverter "Grid low voltage drop load" and "Bus over voltage drop load" are triggered	N/A	RO
18	Status 2	0478	U16		N/A	RO
19	Status 3	0479	U16		N/A	RO
20	Status 4	047A	U16		N/A	RO
21	Status 5	047B	U16		N/A	RO
22	Status 6	047C	U16		N/A	RO
23	Status 7	047D	U16		N/A	RO
24	Status 8	047E	U16		N/A	RO
25	Status 9	047F	U16		N/A	RO

5.1.3Grid-connected output parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Power grid frequency	0484	U16	Example：0x138B means 50.03Hz	0.01Hz	RO
2	Total active power	0485	S16	The output power of the inverter is positive in discharge, charge is negative Example：0x006A means 1.06KW	0.01kW	RO
3	Total reactive power	0486	S16	The value is positive when current leads voltage and negative when voltage lags current	0.01kVar	RO
4	Total apparent power	0487	S16	The current maximum active power, the value is positive when discharging and negative when charging	0.01kVA	RO
5	Total PCC active power	0488	S16	Some models need to be connected to the electricity meter effectively, selling electricity is positive, and buying electricity is negative	0.01kW	RO
6	Total PCC reactive power	0489	S16	Inverter equipment advance is positive and lag is	0.01kVar	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				negative		
7	Total PCC apparent power	048A	S16	Selling electricity is positive, while buying electricity is negative	0.01kVA	RO
8	Total PCC active power 2	048B	S16	If 0x0488 address data overflows, this address is used	0.1kW	RO
9	Reserve	048C	N/A		N/A	N/A
10	Grid voltage of phase R	048D	U16	Example：0x093E means 236.6V	0.1V	RO
11	Output current of phase R	048E	U16	Example：0x0049 means 0.73A	0.01A	RO
12	Output active power of phase R	048F	S16	The value is positive when discharging and negative when charging	0.01kW	RO
13	Output reactive power of phase R	0490	S16	Inverter equipment advance is positive and lag is negative	0.01kVar	RO
14	Power factor of phase R	0491	S16	Inverter equipment advance is positive and lag is negative	0.001p.u.	RO
15	PCC current of phase R	0492	U16	Some models need to be connected to the electricity meter effectively	0.01A	RO
16	PCC active power of phase R	0493	S16		0.01kW	RO
17	PCC reactive power of phase R	0494	S16		0.01kVar	RO
18	PCC power factor of phase R	0495	S16		0.001p.u.	RO
19	Reserve	0496 - 0497	N/A		N/A	N/A
20	Grid voltage of phase S	0498	U16	Refer to the relevant explanation of the R phase mentioned above	0.1V	RO
21	Output current of phase S	0499	U16		0.01A	RO
22	Output active power of phase S	049A	S16		0.01kW	RO
23	Output reactive power of phase S	049B	S16		0.01kVar	RO
24	Power factor of phase S	049C	S16		0.001p.u.	RO
25	PCC current of phase S	049D	U16		0.01A	RO
26	PCC active power of phase S	049E	S16		0.01kW	RO
27	PCC reactive power of phase S	049F	S16		0.01kVar	RO
28	PCC power factor of phase S	04A0	S16		0.001p.u.	RO
29	Reserve	04A1 - 04A2	N/A		N/A	N/A
30	Grid voltage of phase T	04A3	U16	Refer to the relevant explanation of the R phase mentioned above	0.1V	RO
31	Output current of phase T	04A4	U16		0.01A	RO
32	Output active power of phase T	04A5	S16		0.01kW	RO
33	Output reactive power of phase T	04A6	S16		0.01kVar	RO
34	Power factor of phase T	04A7	S16		0.001p.u.	RO
35	PCC current of phase T	04A8	U16		0.01A	RO
36	PCC active power of phase T	04A9	S16		0.01kW	RO
37	PCC reactive power of phase T	04AA	S16		0.01kVar	RO
38	PCC power factor of phase T	04AB	S16		0.001p.u.	RO
39	Reserve	04AC - 04AD	N/A		N/A	N/A
40	External power generation	04AE	U16		0.01kW	RO
41	Total system load power	04AF	U16	Need to connect effective load, the load consumption power of Energy flow diagram	0.01kW	RO
42	Grid L1 to N voltage RMS	04B0	U16	Effective value of power grid L1 on N voltage	0.1V	RO
43	L1 output current RMS	04B1	U16	L1 output current effective value	0.01A	RO
44	Active power output on L1 line	04B2	S16	Active power is output on the L1 line	0.01kW	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
45	RMS CT current on line L1	04B3	U16	Effective value of the CT current on the L1 line	0.01A	RO
46	PCC active power on line L1	04B4	S16	The PCC active power on the L1 line	0.01kW	RO
47	Grid L2 to N voltage RMS	04B5	U16	Effective value of power grid L2 for N voltage	0.1V	RO
48	L2 output current RMS	04B6	U16	L2 output current effective value	0.01A	RO
49	Active power output on L2 line	04B7	S16	Active power is output on the L2 line	0.01kW	RO
50	RMS CT current on line L2	04B8	U16	Effective value of the CT current on the L2 line	0.01A	RO
51	PCC active power on line L2	04B9	S16		0.01kW	RO
52	Line voltage L1	04BA	U16	voltage between phases R/S	0.1V	RO
53	Line voltage L2	04BB	U16	voltage between phases S/T	0.1V	RO
54	Line voltage L3	04BC	U16	voltage between phases T/R	0.1V	RO
55	Total power factor	04BD	S16	The value is positive when current leads voltage and negative when voltage lags current	0.001p.u.	RO
56	High-voltage side LLC charge and discharge current	04BE	S16		0.01A	RO
57	Power generation efficiency	04BF	U16		0.01%	RO
58	Reserve	04C0 - 04FF	N/A		N/A	N/A

Note: The units of "Total active power", "Total apparent power", "Total PCC active power", and "Total PCC apparent power" in this data block depend on the specific model. The units marked in the table are applicable to equipment with a power below 320kW, and for equipment with a power of 320kW or more, these register units are defaulted to 0.1kW.

5.1.4Grid-disconnected output parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Load active power	0504	S16	The value of Load consumption is positive. It's negative if feedback	0.01kW	RO
2	Load reactive power	0505	S16		0.01kVar	RO
3	Load apparent power	0506	S16	The value of Load consumption is positive. It's negative if feedback	0.01kVA	RO
4	Output voltage frequency	0507	U16	Voltage frequency output by the inverter	0.01Hz	RO
5	Reserve	0508 - 0509	N/A		N/A	N/A
6	Inverter output voltage of phase R	050A	U16		0.1V	RO
7	Load current of phase R	050B	S16	Current between R phase and load	0.01A	RO
8	Load active power of phase R	050C	S16	The value of Load consumption is positive. It's negative if feedback	0.01kW	RO
9	Load reactive power of phase R	050D	S16	The value is positive when current leads voltage and negative when voltage lags current	0.01kVar	RO
10	Load apparent power of phase R	050E	S16	The value of Load consumption is positive. It's negative if feedback	0.01kVA	RO
11	The load peak ratio of phase R	050F	U16		0.01p.u.	RO
12	Emergency load voltage of phase R	0510	U16		0.1V	RO
13	Reserve	0511	N/A		N/A	N/A
14	Inverter output voltage of phase S	0512	U16	Refer to the relevant explanation of the R phase mentioned above	0.1V	RO
15	Load current of phase S	0513	S16		0.01A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
16	Load active power of phase S	0514	S16		0.01kW	RO
17	Load reactive power of phase S	0515	S16		0.01kVar	RO
18	Load apparent power of phase S	0516	S16		0.01kVA	RO
19	The load peak ratio of phase S	0517	U16		0.01p.u.	RO
20	Emergency load voltage of phase S	0518	U16		0.1V	RO
21	Reserve	0519	N/A		N/A	N/A
22	Inverter output voltage of phase T	051A	U16	Refer to the relevant explanation of the R phase mentioned above	0.1V	RO
23	Load current of phase T	051B	S16		0.01A	RO
24	Load active power of phase T	051C	S16		0.01kW	RO
25	Load reactive power of phase T	051D	S16		0.01kVar	RO
26	Load apparent power of phase T	051E	S16		0.01kVA	RO
27	The load peak ratio of phase T	051F	U16		0.01p.u.	RO
28	Emergency load voltage of phase T	0520	U16		0.1V	RO
29	Reserve	0521	N/A		N/A	N/A
30	Voltage effective value of inverter L1 to N	0522	U16		0.1V	RO
31	Effective current value of L1 load	0523	S16		0.01A	RO
32	Active power value of load L1 to N	0524	S16		0.01kW	RO
33	Voltage effective value of inverter L2 to N	0525	U16		0.1V	RO
34	Effective current value of L2 load	0526	S16		0.01A	RO
35	Active power value of load L2 to N	0527	S16		0.01kW	RO
36	Reserve	0528 - 057F	N/A		N/A	N/A

5.1.5PV Input channel parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	PV voltage of channel 1	0584	U16	Example：0x0906 means 231.0V	0.1V	RO
2	PV current of channel 1	0585	S16	Example：0x01BB means 4.43V	0.01A	RO
3	PV power of channel 1	0586	U16	Example：0x0066 means 1.02KW	0.01kW	RO
4	PV voltage of channel 2	0587	U16	Refer to the first PV item description	0.1V	RO
5	PV current of channel 2	0588	S16		0.01A	RO
6	PV power of channel 2	0589	U16		0.01kW	RO
7	PV voltage of channel 3	058A	U16	Refer to the first PV item description	0.1V	RO
8	PV current of channel 3	058B	S16		0.01A	RO
9	PV power of channel 3	058C	U16		0.01kW	RO
10	PV voltage of channel 4	058D	U16	Refer to the first PV item description	0.1V	RO
11	PV current of channel 4	058E	S16		0.01A	RO
12	PV power of channel 4	058F	U16		0.01kW	RO
13	PV voltage of channel 5	0590	U16	Refer to the first PV item description	0.1V	RO
14	PV current of channel 5	0591	S16		0.01A	RO
15	PV power of channel 5	0592	U16		0.01kW	RO
16	...	...	...	...	...	...

Index	Name	Address（HEX）	Type	Description	Unit	Properties
17	PV voltage of channel 12	05A5	U16	Refer to the first PV item description	0.1V	RO
18	PV current of channel 12	05A6	S16		0.01A	RO
19	PV power of channel 12	05A7	U16		0.01kW	RO
20	PV voltage of channel 13	05A8	U16	Refer to the first PV item description	0.1V	RO
21	PV current of channel 13	05A9	S16		0.01A	RO
22	PV power of channel 13	05AA	U16		0.01kW	RO
23	PV voltage of channel 14	05AB	U16	Refer to the first PV item description	0.1V	RO
24	PV current of channel 14	05AC	S16		0.01A	RO
25	PV power of channel 14	05AD	U16		0.01kW	RO
26	PV voltage of channel 15	05AE	U16	Refer to the first PV item description	0.1V	RO
27	PV current of channel 15	05AF	S16		0.01A	RO
28	PV power of channel 15	05B0	U16		0.01kW	RO
29	PV voltage of channel 16	05B1	U16	Refer to the first PV item description	0.1V	RO
30	PV current of channel 16	05B2	S16		0.01A	RO
31	PV power of channel 16	05B3	U16		0.01kW	RO
32	Reserve	05B4 - 05BF	N/A		N/A	N/A

Note： The data of this block contains 16 PV channels, and the address order is arranged. The number of PV routes depends on the specific equipment model.

5.1.6PV Total power parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Total PV power	05C4	U16	The sum of all PV powers, Example： 0x0007 means 0.7kw	0.1KW	RO
2	Reserve	05C5 - 05FF	N/A		N/A	N/A

5.1.7Battery pack basic information parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Battery pack voltage of channel 1	0604	U16	Example：0x016C means 36.4V	0.1V	RO
2	Battery pack charging and discharging current of channel 1	0605	S16	The value is positive when charging and negative when discharging	0.01A	RO
3	Battery pack charging and discharging power of channel 1	0606	S16	Example：0x0232 means 5.62kW	0.01kW	RO
4	Battery pack environmental temperature of channel 1	0607	S16	Peripheral temperature of battery, negative number indicates sub zero temperature	1℃	RO
5	Battery pack SOC of channel 1	0608	U16	Remaining battery power	1%	RO
6	Battery pack SOH of channel 1	0609	U16	The current battery's ability to store electrical energy relative to new batteries	1%	RO
7	Battery pack cycle times of channel 1	060A	U16	The number of battery charging and discharging cycles	1cycle	RO
8	Battery pack voltage of channel 2	060B	U16	Refer to the first channel description	0.1V	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
9	Battery pack charging and discharging current of channel 2	060C	S16		0.01A	RO
10	Battery pack charging and discharging power of channel 2	060D	S16		0.01kW	RO
11	Battery pack environmental temperature of channel 2	060E	S16		1℃	RO
12	Battery pack SOC of channel 2	060F	U16		1%	RO
13	Battery pack SOH of channel 2	0610	U16		1%	RO
14	Battery pack cycle times of channel 2	0611	U16		1cycle	RO
15	...	...	...	...	...	...
16	Battery pack voltage of channel 7	062E	U16	Refer to the first channel description	0.1V	RO
17	Battery pack charging and discharging current of channel 7	062F	S16		0.01A	RO
18	Battery pack charging and discharging power of channel 7	0630	S16		0.01kW	RO
19	Battery pack environmental temperature of channel 7	0631	S16		1℃	RO
20	Battery pack SOC of channel 7	0632	U16		1%	RO
21	Battery pack SOH of channel 7	0633	U16		1%	RO
22	Battery pack cycle times of channel 7	0634	U16		1cycle	RO
23	Battery pack voltage of channel 8	0635	U16	Refer to the first channel description	0.1V	RO
24	Battery pack charging and discharging current of channel 8	0636	S16		0.01A	RO
25	Battery pack charging and discharging power of channel 8	0637	S16		0.01kW	RO
26	Battery pack environmental temperature of channel 8	0638	S16		1℃	RO
27	Battery pack SOC of channel 8	0639	U16		1%	RO
28	Battery pack SOH of channel 8	063A	U16		1%	RO
29	Battery pack cycle times of channel 8	063B	U16		1cycle	RO
30	Reserve	063C - 063F	N/A		N/A	N/A

Note: This block of data contains a total of 8 battery pack data channels, arranged in address order. The number of battery pack channels depends on the specific device model.

5.1.8Battery pack current upper limit parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Battery pack charging current upper limit of channel 1	0644	U16		0.01A	RO
2	Battery pack discharging current upper limit of channel 1	0645	U16		0.01A	RO
3	Battery state of channel 1	0646	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
4	Battery pack Reserve of channel 1	0647	N/A		N/A	N/A
5	Battery pack charging current upper limit of channel 2	0648	U16		0.01A	RO
6	Battery pack discharging current upper limit of channel 2	0649	U16		0.01A	RO
7	Battery state of channel 2	064A	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
8	Battery pack Reserve of channel 2	064B	N/A		N/A	N/A
9	Battery pack charging current upper limit of channel 3	064C	U16		0.01A	RO
10	Battery pack discharging current upper limit of channel 3	064D	U16		0.01A	RO
11	Battery state of channel 3	064E	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
12	Battery pack Reserve of channel 3	064F	N/A		N/A	N/A
13	Battery pack charging current upper limit of channel 4	0650	U16		0.01A	RO
14	Battery pack discharging current upper limit of channel 4	0651	U16		0.01A	RO
15	Battery state of channel 4	0652	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
16	Battery pack Reserve of channel 4	0653	N/A		N/A	N/A
17	Battery pack charging current upper limit of channel 5	0654	U16		0.01A	RO
18	Battery pack discharging current upper limit of channel 5	0655	U16		0.01A	RO
19	Battery state of channel 5	0656	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
20	Battery pack Reserve of channel 5	0657	N/A		N/A	N/A
21	Battery pack charging current upper limit of channel 6	0658	U16		0.01A	RO
22	Battery pack discharging current upper limit of channel 6	0659	U16		0.01A	RO
23	Battery state of channel 6	065A	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
24	Battery pack Reserve of channel 6	065B	N/A		N/A	N/A
25	Battery pack charging current upper limit of channel 7	065C	U16		0.01A	RO
26	Battery pack discharging current upper limit of channel 7	065D	U16		0.01A	RO
27	Battery state of channel 7	065E	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
28	Battery pack Reserve of channel 7	065F	N/A		N/A	N/A
29	Battery pack charging current upper limit of channel 8	0660	U16		0.01A	RO
30	Battery pack discharging current upper limit of channel 8	0661	U16		0.01A	RO
31	Battery state of channel 8	0662	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
32	Battery pack Reserve of channel 8	0663	N/A		N/A	N/A
33	Reserve	0664 - 0666	N/A		N/A	N/A

Index	Name	Address（HEX）	Type	Description	Unit	Properties
34	Total battery charging and discharging power	0667	S16	The value is positive when charging and negative when discharging	0.1kW	RO
35	The average SOC of the battery pack	0668	U16	Average electricity level	1%	RO
36	Battery pack SOH	0669	U16	The current battery's ability to store electrical energy relative to new batteries	1%	RO
37	Real time battery pack count	066A	U16		N/A	RO
38	Total battery capacity	066B	U16		1Ah	RO
39	Reserve	066C - 067F	N/A		N/A	N/A

5.1.9Electricity statistics parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Power generation for the day	0684 - 0685	U32	Example：0x0000 0x00EF means 2.39KWh	0.01kWh	RO
2	Total power generation	0686 - 0687	U32	Example：0x0000 0x0017 means 2.3KWh	0.1kWh	RO
3	Load power consumption for the day	0688 - 0689	U32		0.01kWh	RO
4	Total load power consumption	068A - 068B	U32		0.1kWh	RO
5	Power bought for the day	068C - 068D	U32		0.01kWh	RO
6	Total power bought	068E - 068F	U32		0.1kWh	RO
7	Power sold on the day	0690 - 0691	U32		0.01kWh	RO
8	Total power sold	0692 - 0693	U32		0.1kWh	RO
9	Daily battery charging capacity	0694 - 0695	U32		0.01kWh	RO
10	Total battery charging capacity	0696 - 0697	U32		0.1kWh	RO
11	Daily battery discharging capacity	0698 - 0699	U32		0.01kWh	RO
12	Total battery discharging capacity	069A - 069B	U32		0.1kWh	RO
13	Electricity generated this month	069C - 069D	U32		0.1kWh	RO
14	Electricity generation this year	069E - 069F	U32		0.1kWh	RO
15	Load consumption for the month	06A0 - 06A1	U32		0.1kWh	RO
16	Load consumption for the year	06A2 - 06A3	U32		0.1kWh	RO
17	Electricity bought this month	06A4 - 06A5	U32		0.1kWh	RO
18	Electricity bought this year	06A6 - 06A7	U32		0.1kWh	RO
19	Electricity sold this month	06A8 - 06A9	U32		0.1kWh	RO
20	Electricity sold this year	06AA - 06AB	U32		0.1kWh	RO
21	Battery charging amount this month	06AC - 06AD	U32		0.1kWh	RO
22	Battery charging amount this year	06AE - 06AF	U32		0.1kWh	RO
23	Battery discharging amount this month	06B0 - 06B1	U32		0.1kWh	RO
24	Battery discharging amount this year	06B2 - 06B3	U32		0.1kWh	RO
25	Reserve	06B4 - 06BF	N/A		N/A	N/A

Note: Low addresses are high bits, high addresses are low bits. The final data is obtained by multiplying the read number by the unit precision.

5.1.10Internal information parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Leakage current	06C4	U16	Example: 0x0005 means 5mA	1mA	RO
2	Balance current	06C5	S16	To achieve a balanced three-phase	0.01A	RO
3	DC component of R-phase current	06C6	S16		1mA	RO
4	DC component of S-phase current	06C7	S16		1mA	RO
5	DC component of T-phase current	06C8	S16		1mA	RO
6	DC component of R-phase voltage	06C9	S16		1mV	RO
7	DC component of S-phase voltage	06CA	S16		1mV	RO
8	DC component of T-phase voltage	06CB	S16		1mV	RO
9	Total BUS voltage	06CC	U16		0.1V	RO
10	BUS positive voltage	06CD	U16	The voltage to ground is positive	0.1V	RO
11	BUS negative voltage	06CE	U16	The voltage to ground is negative	0.1V	RO
12	BUS negative voltage	06CF	U16		0.1V	RO
13	Buck Boost current	06D0	S16		0.01A	RO
14	BUS positive half voltage	06D1	U16		0.1V	RO
15	BUS negative half voltage	06D2	U16		0.1V	RO
16	PV 1 Fly Across Capacitor Voltage	06D3	U16		0.1V	RO
17	PV 2 Fly Across Capacitor Voltage	06D4	U16		0.1V	RO
18	PV 3 Fly Across Capacitor Voltage	06D5	U16		0.1V	RO
19	PV 4 Fly Across Capacitor Voltage	06D6	U16		0.1V	RO
20	PV 5 Fly Across Capacitor Voltage	06D7	U16		0.1V	RO
21	PV 6 Fly Across Capacitor Voltage	06D8	U16		0.1V	RO
22	PV 7 Fly Across Capacitor Voltage	06D9	U16		0.1V	RO
23	PV 8 Fly Across Capacitor Voltage	06DA	U16		0.1V	RO
24	PV 9 Fly Across Capacitor Voltage	06DB	U16		0.1V	RO
25	PV 10 Fly Across Capacitor Voltage	06DC	U16		0.1V	RO
26	PV 11 Fly Across Capacitor Voltage	06DD	U16		0.1V	RO
27	PV 12 Fly Across Capacitor Voltage	06DE	U16		0.1V	RO
28	PV 13 Fly Across Capacitor Voltage	06DF	U16		0.1V	RO
29	PV 14 Fly Across Capacitor Voltage	06E0	U16		0.1V	RO
30	PV 15 Fly Across Capacitor Voltage	06E1	U16		0.1V	RO
31	PV 16 Fly Across Capacitor Voltage	06E2	U16		0.1V	RO
32	Reserve	06E3 - 06EC	N/A		N/A	N/A
33	Rated power of inverter	06ED	U16		0.1kW	RO
34	Number of valid points for IV curve scan	06EE	U16	Range：[1, 150]	N/A	RO
35	BUS negative to ground voltage	06EF	S16	Belonging to some model parameters	0.1V	RO
36	Reserve	06F0 - 06FF	N/A		N/A	N/A

5.1.11Confluence information parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Group 1 confluence voltage	0704	U16	Equal to PV1 voltage, 0x0866 means 215.0V	0.1V	RO
2	Group 1, 1st string current	0705	S16	Equal to PV1 current, 0x015F means 3.51A	0.01A	RO
3	Group 1, 2nd string current	0706	S16	Availability of this data is model dependent	0.01A	RO
4	Group 2 confluence voltage	0707	U16	Refer to the group 1 of parameter explanations	0.1V	RO
5	Group 2, 1st string current	0708	S16		0.01A	RO
6	Group 2, 2nd string current	0709	S16		0.01A	RO
7	Group 3 confluence voltage	070A	U16	Refer to the group 1 of parameter explanations	0.1V	RO
8	Group 3, 1st string current	070B	S16		0.01A	RO
9	Group 3, 2nd string current	070C	S16		0.01A	RO
10	Group 4 confluence voltage	070D	U16	Refer to the group 1 of parameter explanations	0.1V	RO
11	Group 4, 1st string current	070E	S16		0.01A	RO
12	Group 4, 2nd string current	070F	S16		0.01A	RO
13	Group 5 confluence voltage	0710	U16	Refer to the group 1 of parameter explanations	0.1V	RO
14	Group 5, 1st string current	0711	S16		0.01A	RO
15	Group 5, 2nd string current	0712	S16		0.01A	RO
16	...	...	...	...	...	...
17	Group 12 confluence voltage	0725	U16	Refer to the group 1 of parameter explanations	0.1V	RO
18	Group 12, 1st string current	0726	S16		0.01A	RO
19	Group 12, 2nd string current	0727	S16		0.01A	RO
20	Group 13 confluence voltage	0728	U16	Refer to the group 1 of parameter explanations	0.1V	RO
21	Group 13, 1st string current	0729	S16		0.01A	RO
22	Group 13, 2nd string current	072A	S16		0.01A	RO
23	Group 14 confluence voltage	072B	U16	Refer to the group 1 of parameter explanations	0.1V	RO
24	Group 14, 1st string current	072C	S16		0.01A	RO
25	Group 14, 2nd string current	072D	S16		0.01A	RO
26	Group 15 confluence voltage	072E	U16	Refer to the group 1 of parameter explanations	0.1V	RO
27	Group 15, 1st string current	072F	S16		0.01A	RO
28	Group 15, 2nd string current	0730	S16		0.01A	RO
29	Group 16 confluence voltage	0731	U16	Refer to the group 1 of parameter explanations	0.1V	RO
30	Group 16, 1st string current	0732	S16		0.01A	RO
31	Group 16, 2nd string current	0733	S16		0.01A	RO
32	Reserve	0734 - 073F	N/A		N/A	N/A

Note: This block of data contains a total of 16 sets of channel data, arranged in address order, and the number of channels depends on the specific device model.

5.1.12Group string current parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Group 1, 3rd string current	0744	S16	Availability of these data are model dependent, example：0x0186 means 3.90A	0.01A	RO
2	Group 1, 4th string current	0745	S16		0.01A	RO
3	Group 2, 3rd string current	0746	S16	Refer to the group 1 of parameter explanations	0.01A	RO
4	Group 2, 4th string current	0747	S16		0.01A	RO
5	Group 3, 3rd string current	0748	S16	Refer to the group 1 of parameter explanations	0.01A	RO
6	Group 3, 4th string current	0749	S16		0.01A	RO
7	Group 4, 3rd string current	074A	S16	Refer to the group 1 of parameter explanations	0.01A	RO
8	Group 4, 4th string current	074B	S16		0.01A	RO
9	Group 5, 3rd string current	074C	S16	Refer to the group 1 of parameter explanations	0.01A	RO
10	Group 5, 4th string current	074D	S16		0.01A	RO
11	Group 6, 3rd string current	074E	S16	Refer to the group 1 of parameter explanations	0.01A	RO
12	Group 6, 4th string current	074F	S16		0.01A	RO
13	Group 7, 3rd string current	0750	S16	Refer to the group 1 of parameter explanations	0.01A	RO
14	Group 7, 4th string current	0751	S16		0.01A	RO
15	Group 8, 3rd string current	0752	S16	Refer to the group 1 of parameter explanations	0.01A	RO
16	Group 8, 4th string current	0753	S16		0.01A	RO
17	Group 9, 3rd string current	0754	S16	Refer to the group 1 of parameter explanations	0.01A	RO
18	Group 9, 4th string current	0755	S16		0.01A	RO
19	Group 10, 3rd string current	0756	S16	Refer to the group 1 of parameter explanations	0.01A	RO
20	Group 10, 4th string current	0757	S16		0.01A	RO
21	Group 11, 3rd string current	0758	S16	Refer to the group 1 of parameter explanations	0.01A	RO
22	Group 11, 4th string current	0759	S16		0.01A	RO
23	Group 12, 3rd string current	075A	S16	Refer to the group 1 of parameter explanations	0.01A	RO
24	Group 12, 4th string current	075B	S16		0.01A	RO
25	Group 13, 3rd string current	075C	S16	Refer to the group 1 of parameter explanations	0.01A	RO
26	Group 13, 4th string current	075D	S16		0.01A	RO
27	Group 14, 3rd string current	075E	S16	Refer to the group 1 of parameter explanations	0.01A	RO
28	Group 14, 4th string current	075F	S16		0.01A	RO
29	Group 15, 3rd string current	0760	S16	Refer to the group 1 of parameter explanations	0.01A	RO
30	Group 15, 4th string current	0761	S16		0.01A	RO
31	Group 16, 3rd string current	0762	S16	Refer to the group 1 of parameter explanations	0.01A	RO
32	Group 16, 4th string current	0763	S16		0.01A	RO
33	Reserve	0764 - 077F	N/A		N/A	N/A

Note: This block of data contains a total of 16 sets of channel data, arranged in address order, and the number of channels depends on the specific device model.

5.2Safety Parameter Area

5.2.1Safety-Power on parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Waiting time before grid connection	0800	U16	Range：[1, 65535]	1S	RW
2	Grid connection rise rate	0801	U16	100 indicates the rate corresponding to a 1-minute rise to full load, other values are multiplied and divided on this basis. Range：[1, 3000]	1%Pn/min	RW
3	Waiting time before reconnecting to the grid after power grid fault recovery	0802	U16	Range：[1, 65535]	1S	RW
4	Rising speed after power grid fault recovery	0803	U16	100 indicates the rate at which 1 minute rises to full load, other values are multiplied and divided by this. Range：[1, 3000]	1%Pn/min	RW
5	Upper limit of grid start-up voltage	0804	U16	The grid start-up voltage must not exceed this setting. Range：[0, 65535]	0.1V	RW
6	Lower limit of grid start-up voltage	0805	U16	The voltage at start-up cannot be lower than this setting. Range：[0, 65535]	0.1V	RW
7	Start grid frequency upper limit	0806	U16	Range：[4000, 7000]	0.01Hz	RW
8	Start grid frequency lower limit	0807	U16	Range：[4000, 7000]	0.01Hz	RW
9	Reconnect grid voltage upper limit	0808	U16	Range：[0, 65535]	0.1V	RW
10	Reconnect grid voltage lower limit	0809	U16	Range：[0, 65535]	0.1V	RW
11	Upper limit of reconnect grid frequency	080A	U16	The voltage and frequency at reconnect start cannot exceed this setting. Rage：[4000, 7000]	0.01Hz	RW
12	Lower limit of reconnect grid frequency	080B	U16	The voltage and frequency at reconnect start cannot be lower than this setting. Rage：[4000, 7000]	0.01Hz	RW
13	Reserve	080C - 083F	N/A		N/A	N/A

5.2.2Safety-Voltage protection parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Protection enable control	0840	U16	Bit0:Over-voltage level 1 protection enable bit Bit1:Over-voltage level 2 protection enable bit Bit2:Over-voltage level 3 protection enable bit Bit3:Under-voltage level 1 protection enable bit Bit4:Under-voltage level 2 protection enable bit Bit5:Under-voltage level 3 protection enable bit Bit6:10 minutes over-voltage protection enable bit Note: Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
2	Rated grid voltage	0841	U16	Range：[0, 65535]	0.1V	RW
3	Level 1over-voltage protection value	0842	U16	Range：[0, 65535]	0.1V	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
4	Level 1 over-voltage protection time	0843	U16	Range：[1, 65535]	10ms	RW
5	Level 2 over-voltage protection value	0844	U16	Range：[0, 65535]	0.1V	RW
6	Level 2 over-voltage protection time	0845	U16	Range：[1, 65535]	10ms	RW
7	Level 3 over-voltage protection value	0846	U16	Range：[0, 65535]	0.1V	RW
8	Level 3 over-voltage protection time	0847	U16	Range：[1, 65535]	10ms	RW
9	Level 1 under-voltage protection value	0848	U16	Range：[0, 65535]	0.1V	RW
10	Level 1 under-voltage protection time	0849	U16	Range：[1, 65535]	10ms	RW
11	Level 2 under-voltage protection value	084A	U16	Range：[0, 65535]	0.1V	RW
12	Level 2 under-voltage protection time	084B	U16	Range：[1, 65535]	10ms	RW
13	Level 3 under-voltage protection value	084C	U16	Range：[0, 65535]	0.1V	RW
14	Level 3 under-voltage protection time	084D	U16	Range：[1, 65535]	10ms	RW
15	Grid 10-minute over-voltage protection value	084E	U16	Range：[0, 65535]	0.1V	RW
16	Level 1 over-voltage protection time high	084F	U16	This register is used in combination with level 1 over-voltage protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
17	Level 2 over-voltage protection time high	0850	U16	This register is used in combination with level 2 over-voltage protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
18	Level 3 over-voltage protection time high	0851	U16	This register is used in combination with level 3 over-voltage protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
19	Reserve	0852 - 087F	N/A		N/A	N/A

5.2.3Safety-Frequency protection parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Protection enable control	0880	U16	Bit0:Over-frequency level 1 protection enable bit Bit1:Over-frequency level 2 protection enable bit Bit2:Over-frequency level 3 protection enable bit Bit3:Under-frequency level 1 protection enable bit Bit4:Under-frequency level 2 protection enable bit Bit5:Under-frequency level 3 protection enable bit Bit6:Frequency rate of change protection enable bit Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
2	Rated grid frequency	0881	U16	Range: [4000, 7000]	0.01Hz	RW
3	Level 1 over-frequency protection value	0882	U16	Over-Frequency protection threshold, exceeds trigger protection mechanism. Range: [4000, 7000]	0.01Hz	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
4	Level 1 over-frequency protection time	0883	U16	Over-Frequency time protection threshold, exceeds trigger protection mechanism. Range: [1, 65535]	10ms	RW
5	Level 2 over-frequency protection value	0884	U16	Refer to level 1 over-frequency protection explanations	0.01Hz	RW
6	Level 2 over-frequency protection time	0885	U16		10ms	RW
7	Level 3 over-frequency protection value	0886	U16		0.01Hz	RW
8	Level 3 over-frequency protection time	0887	U16		10ms	RW
9	Level 1 under-frequency protection value	0888	U16	Under-Frequency protection threshold, lower than trigger protection mechanism. Range: [4000, 7000]	0.01Hz	RW
10	Level 1 under-frequency protection time	0889	U16	Under-Frequency time protection threshold, lower than trigger protection mechanism. Range: [1, 65535]	10ms	RW
11	Level 2 under-frequency protection value	088A	U16	Refer to level 1 under-frequency protection explanations	0.01Hz	RW
12	Level 2 under-frequency protection time	088B	U16		10ms	RW
13	Level 3 under-frequency protection value	088C	U16		0.01Hz	RW
14	Level 3 under-frequency protection time	088D	U16		10ms	RW
15	Level 1 over-frequency protection time high	088E	U16	This register is used in combination with level 1 over-frequency protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
16	Level 2 over-frequency protection time high	088F	U16	This register is used in combination with level 2 over-frequency protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
17	Level 3 over-frequency protection time high	0890	U16	This register is used in combination with level 3 over-frequency protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
18	Level 1 under-frequency protection time high	0891	U16	This register is used in combination with level 1 under-frequency protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
19	Level 2 under-frequency protection time high	0892	U16	This register is used in combination with level 2 under-frequency protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
20	Level 3 under-frequency protection time high	0893	U16	This register is used in combination with level 3 under-frequency protection time to form a 32-bit register, Range: [0, 255]	N/A	RW
21	Rate of frequency change protection value	0894	U16	Range: [100, 1000]	0.01Hz/s	RW
22	Rate of frequency change protection time	0895	U16	Range: [1, 65535]	10ms	RW
23	Reserve	0896 - 08BF	N/A		N/A	N/A

5.2.4Safety-DCI protection parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Protection enable control	08C0	U16	Bit0:DCI Level 1 protection enable bit Bit1:DCI Level 2 protection enable bit Bit2:DCI Level 3 protection enable bit Bit3:DCI Test enable bit Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
2	DCI Level 1 protection value	08C1	U16	DC component level 1 protection threshold, exceeds trigger protection mechanism,Range:[1, 65535]	1mA	RW
3	DCI Level 1 protection time	08C2	U16	DC component level 1 time protection threshold, exceeding the trigger protection mechanism, Range:[1, 65535]	10ms	RW
4	DCI Level 2 protection value	08C3	U16	Refer to DCI level 1 protection explanations	1mA	RW
5	DCI Level 2 protection time	08C4	U16		10ms	RW
6	DCI Level 3 protection value	08C5	U16		1mA	RW
7	DCI Level 3 protection time	08C6	U16		10ms	RW
8	DCI R-phase test value	08C7	S16	Range:[-32768, 32767]	1mA	RW
9	DCI S-phase test value	08C8	S16	Range:[-32768, 32767]	1mA	RW
10	DCI T-phase test value	08C9	S16	Range:[-32768, 32767]	1mA	RW
11	DCI Level 1 protection ratio	08CA	U16	Range:[1, 65535]	0.01%	RW
12	DCI Level 2 protection ratio	08CB	U16	Range:[1, 65535]	0.01%	RW
13	DCI Level 3 protection ratio	08CC	U16	Range:[1, 65535]	0.01%	RW
14	Reserve	08CD - 08FF	N/A		N/A	N/A

5.2.5Safety-Action over-under voltage load-down parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Enable control	0900	U16	Bit0:Active power load shedding bit Bit1:Remote power on/off enable bit Bit2:Over-voltage load-down enable bit Bit3:Over-voltage charge load-down enable bit Bit4:Under-voltage charge load-down enable bit Bit5:DRM0 enable bit Bit6:Logic interface(DRM1-8) enable bit Bit7:Anti-back-flow over-load enable bit Bit8:Voltage drop-down PT curve enable bit Bit9:Remote shutdown load-down enable bit(0:default direct shutdown,1:ramp load-down shutdown) Bit10:CLS Anti-back-flow enable bit Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
2	Active output percentage	0901	U16	Range:[0, 1100]	0.1%	RW
3	Active load shedding rate	0902	U16	Range:[[0, 65535]	1%Pn/min	RW
4	Grid over-voltage load shedding start point	0903	U16	Range:[0, 65535]	0.1V	RW
5	Grid over-voltage derating cut-off point	0904	U16	Range:[0, 65535]	0.1V	RW
6	Grid over-voltage derating cut-off power	0905	S16	Range:[-100, 100]	1%	RW
7	Over-voltage derating rate	0906	U16	Range:[1, 300]	1%Pn/min	RW
8	Grid under-voltage load shedding start point	0907	U16	Range:[0, 65535]	0.1V	RW
9	Grid under-voltage load shedding cut-off point	0908	U16	Range:[0, 65535]	0.1V	RW
10	Grid under-voltage cut-off power	0909	S16	Range:[-100, 100]	1%	RW
11	Power corresponding to logic interfaces 1	090A	S16	Range:[-100, 100]	1%	RW
12	Power corresponding to logic interfaces 2	090B	S16	Range:[-100, 100]	1%	RW
13	Power corresponding to logic interfaces 3	090C	S16	Range:[-100, 100]	1%	RW
14	Power corresponding to logic interfaces 4	090D	S16	Range:[-100, 100]	1%	RW
15	Power corresponding to logic interfaces 5	090E	S16	Range:[-100, 100]	1%	RW
16	Power corresponding to logic interfaces 6	090F	S16	Range:[-100, 100]	1%	RW
17	Power corresponding to logic interfaces 7	0910	S16	Range:[-100, 100]	1%	RW
18	Power corresponding to logic	0911	S16	Range:[-100, 100]	1%	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
	interfaces 8					
19	Anti-back-flow power	0912	U16	VDE4105 safety regulation grid-connected power limit, greater than 0	0.01%	RW
20	Anti-back-flow overload time	0913	U16	VDE4105 safety regulation back-flow overload off-grid time, greater than 0	0.01%	RW
21	Logical interface load-down rate	0914	U16	Range:[1, 65535]	1%Pn/min	RW
22	Logical interface return load rate	0915	U16	Range:[1, 65535]	1%Pn/min	RW
23	Voltage drop time	0916	U16	Range:[0, 65535]	1s	RW
24	Remote shutdown down rate	0917	U16	Range:[1, 3000]	1%Pn/min	RW
25	CLS Remote control	0918	U16	Bit0:CLS home device selection bit(0:home device, 1:non-home device) Bit1:CLS low-power state-locked enable bit(low-power: current limit is 0.1% of rated current) Bit2:CLS can automatically restore the enable bit in the low-power state Bit3~Bit15：Reserved Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
26	CLS Anti-back-flow output current limit percentage	0919	U16	Range:[0, 100]	1%	RW
27	CLS Anti-back-flow input current limit percentage	091A	U16	Range:[0, 100]	1%	RW
28	CLS Anti-back-flow PCC current limit time	091B	U16	Range:[1, 65535]	10ms	RW
29	CLS Anti-back-flow PCC current protection time	091C	U16	Range:[1, 65535]	10ms	RW
30	Reserve	091D - 093F	N/A		N/A	N/A

5.2.6Safety-Overfrequency load shedding parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Enable control	0940	U16	Bit0:Over-frequency output load-down enable bit Bit1:Under-frequency input load-down enable bit Bit2:Over-frequency back enable bit Bit3:Under-frequency back enable bit Bit4:Over-frequency load-down from rated power bit Bit5:Current power calculation ramp down enable bit Bit6:Energy storage segment curve enable bit Bit7:Maximum power calculation enable bit Bit8:Under-frequency load-down from rated power enable bit	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				Bit9:Over-under frequency and frequency dependent load back rate enable bit Bit12~Bit13:Load down mode(0:slope mode,1:frequency range mode) Bit14:Over-frequency load shedding power cut-off enable bit Note:Enable bit 1 is enabled, set 0 is prohibited		
2	Over-frequency load shedding starting frequency	0941	U16	Range:[4000, 7000]	0.01Hz	RW
3	Over-frequency load shedding cut-off frequency	0942	U16	Valid in frequency range mode, Range:[4000, 7000]	0.01Hz	RW
4	Over-frequency ramp down percentage	0943	U16	Valid in slope mode, Range:[1, 300]	1%Pn/Hz	RW
5	Over-frequency load-down start wait time	0944	U16	Range:[0, 65535]	10ms	RW
6	Waiting time before over-frequency load back	0945	U16	Range:[0, 65535]	10ms	RW
7	Over-frequency load-down return frequency	0946	U16	Effective when over-frequency reload is prohibited,Range:[4000, 7000]	0.01Hz	RW
8	Percentage of over-frequency reload rate	0947	U16	Range:[1, 300]	1%Pn/min	RW
9	Under-frequency load-down start frequency	0948	U16	Range:[4000, 7000]	0.01Hz	RW
10	Under-frequency cut-off frequency	0949	U16	Valid in frequency range mode, Range:[4000, 7000]	0.01Hz	RW
11	Under-frequency derate slope percentage	094A	U16	Valid in slope mode, Range:[1, 300]	1%Pn/Hz	RW
12	Under-voltage load-down start wait time	094B	U16	Range:[0, 65535]	10ms	RW
13	Waiting time before under-frequency reload	094C	U16	Range:[0, 65535]	10ms	RW
14	Under-frequency load-down return frequency	094D	U16	Effective when under-frequency reload is prohibited,Range:[4000, 7000]	0.01Hz	RW
15	Under-frequency reload rate percentage	094E	U16	Range:[1, 300]	1%Pn/min	RW
16	Exit the highest frequency of load shedding	094F	U16	It's the same for over frequency and under frequency protection. It's effective when reloading after over frequency Range:[4000, 7000]	0.01Hz	RW
17	Exit the lowest frequency of load shedding	0950	U16	It's the same for over frequency and under frequency protection. It's effective when reloading after over frequency Range:[4000, 7000]	0.01Hz	RW
18	(Energy storage)Starting frequency of over-frequency and load shedding	0951	U16	Range:[4000, 7000]	0.01Hz	RW
19	(Energy storage)Over frequency and	0952	U16	Valid in frequency range mode, Range:[4000,	0.01Hz	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
	load shedding cut-off frequency			7000]		
20	(Energy storage)Percentage of over-frequency load-down slope	0953	U16	Valid in slope mode, Range:[1, 300]	1%Pn/Hz	RW
21	(Energy storage)Over-frequency load-down start-up waiting time	0954	U16	Range:[0, 65535]	10ms	RW
22	(Energy storage)Waiting time before over-frequency return to load	0955	U16	Range:[0, 65535]	10ms	RW
23	(Energy storage)Over-frequency load-down return frequency	0956	U16	Effective when over-frequency back load is prohibited, Range:[4000, 7000]	0.01Hz	RW
24	(Energy storage)Percentage of over-frequency back load rate	0957	U16	Range:[1, 300]	1%Pn/min	RW
25	(Energy storage)Starting frequency of under-frequency unloading discharge	0958	U16	Range:[4000, 7000]	0.01Hz	RW
26	(Energy storage)Under-frequency shedding cut-off frequency	0959	U16	Valid in frequency range mode, Range:[4000, 7000]	0.01Hz	RW
27	(Energy storage)Percentage of under-frequency load shedding slope	095A	U16	Valid in slope mode, Range:[1, 300]	1%Pn/Hz	RW
28	(Energy storage)Under-frequency load reduction startup waiting time	095B	U16	Range:[0, 65535]	10ms	RW
29	(Energy storage)Waiting time before under-frequency reload	095C	U16	Range:[0, 65535]	10ms	RW
30	(Energy storage)Under-frequency shedding and reloading frequency	095D	U16	Effective when under-frequency reload is prohibited, Range:[4000, 7000]	0.01Hz	RW
31	(Energy storage)Percentage of under-frequency reload rate	095E	U16	Range:[1, 300]	1%Pn/min	RW
32	(Energy storage)Maximum frequency to exit the load shedding	095F	U16	It's the same for over frequency and under frequency protection. Range:[4000, 7000]	0.01Hz	RW
33	(Energy storage)Exit the lowest frequency of the load shedding	0960	U16	It's the same for over frequency and under frequency protection. Range:[4000, 7000]	0.01Hz	RW
34	(Energy storage)Overdrive power crossing frequency point	0961	U16	Valid when 0x0940 Bit6 is enabled，Range:[4000, 7000]	0.01Hz	RW
35	(Energy storage)Under-frequency power down crossing frequency point	0962	U16	Valid when 0940 Bit6 is enabled, Range:[4000, 7000]	0.01Hz	RW
36	(Energy storage)Under-frequency load shedding charging starting frequency	0963	U16	Range:[4000, 7000]	0.01Hz	RW
37	Reserve	0964 - 097F	N/A		N/A	N/A

5.2.7Safety-Reactive power parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Enable control	0980	U16	Bit0:Reactive power enable bit Bit1~Bit3:Reactive mode Bit8: Voltage enter enable bit(Reactive mode 3) Bit9:Charging reactive enable bit Bit10:Austria enable bit(Reactive mode 4) Bit14:Over-frequency load shedding power cut-off enable bit Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
2	Power factor	0981	S16	Used for reactive power mode 1, Range:[-10000, 10000]	0.0001p.u.	RW
3	Fixed reactive power percentage	0982	S16	Used for reactive power mode 2, Range:[-6000, 6000]	0.01%	RW
4	The first point power factor value	0983	S16	Used for reactive power mode 3, Range:[-10000, 10000]	0.0001p.u.	RW
5	First point power percentage	0984	S16	Used for reactive power mode 3, Range:[-100, 100]	1%	RW
6	Second point power factor value	0985	S16	Used for reactive power mode 3, Range:[-10000, 10000]	0.0001p.u.	RW
7	Second point power percentage	0986	S16	Used for reactive power mode 3, Range:[-100, 100]	1%	RW
8	Third point power factor value	0987	S16	Used for reactive power mode 3, Range:[-10000, 10000]	0.0001p.u.	RW
9	Third point power percentage	0988	S16	Used for reactive power mode 3, Range:[-100, 100]	1%	RW
10	Fourth power factor point	0989	S16	Used for reactive power mode 3, Range:[-10000, 10000]	0.0001p.u.	RW
11	Fourth power point percentage	098A	S16	Used for reactive power mode 3, Range:[-100, 100]	1%	RW
12	Percentage of LockinV voltage value	098B	U16	Used for reactive power mode 3, Range:[0, 200]	1%	RW
13	Percentage of voltage value LockoutV	098C	U16	Used for reactive power mode 3, Range:[0, 200]	1%	RW
14	Percentage of high voltage starting	098D	U16	Used for reactive power mode 4, Range:[0, 200]	1%	RW
15	High voltage termination voltage percentage	098E	U16	Used for reactive power mode 4, Range:[0, 200]	1%	RW
16	Low voltage starting voltage percentage	098F	U16	Used for reactive power mode 4, Range:[0, 200]	1%	RW
17	Low voltage termination voltage percentage	0990	U16	Used for reactive power mode 4, Range:[0, 200]	1%	RW
18	Lockin power value percentage	0991	U16	Used for reactive power mode 4, Range:[0, 200]	1%	RW
19	Lockout power value percentage	0992	U16	Used for reactive power mode 4, Range:[0, 200]	1%	RW
20	Maximum leading reactive power percentage	0993	U16	Used for reactive power mode 4, Range:[0, 10000]	0.01%Pn	RW
21	Reactive power response waiting	0994	U16	Used for reactive power mode 4, Range:[0,	10ms	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
	time			65535]		
22	Percentage of reactive power offset	0995	S16	Used for reactive power mode 4, Range:[-10000, 10000]	0.01%Qmax	RW
23	50% of the percentage change in reactive power from low voltage starting voltage to high voltage starting voltage point	0996	S16	Used for reactive power mode 4, Range:[-10000, 10000]	0.01%Pn	RW
24	Percentage of high voltage starting	0997	U16	Used for reactive power mode 5, Range:[0, 200]	1%	RW
25	High voltage termination voltage percentage	0998	U16	Used for reactive power mode 5, Range:[0, 200]	1%	RW
26	Low voltage starting voltage percentage	0999	U16	Used for reactive power mode 5, Range:[0, 200]	1%	RW
27	Low voltage termination voltage percentage	099A	U16	Used for reactive power mode 5, Range:[0, 200]	1%	RW
28	Lockin power value percentage	099B	U16	Used for reactive power mode 5, Range:[0, 200]	1%	RW
29	Lockout power value percentage	099C	U16	Used for reactive power mode 5, Range:[0, 200]	1%	RW
30	Maximum reactive power percentage	099D	U16	Used for reactive power mode 5, Range:[0, 10000]	0.01%	RW
31	Reactive power response waiting time	099E	U16	Used for reactive power mode 5, Range:[0, 65535]	10ms	RW
32	Phase type	099F	U16	Used for reactive power mode 6, Range:[0, 2] 0:Zero reactive power 1:Hysteresis reactive power 2:Overrun reactive power	N/A	RW
33	Reactive power regulation duration setting	09A0	U16	Used for reactive power mode 1、2、3、4、5、6, Range:[0, 65535]	1s	RW
34	Maximum lag reactive power percentage	09A1	U16	Used for reactive power mode 4, Range:[0, 10000]	0.01%Pn	RW
35	Reserve	09A2 - 09BF	N/A		N/A	N/A

5.2.8Safety-Voltage ride through parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Enable control	09C0	U16	Bit0:LVRT enable bit Bit1:OVRT enable bit Bit2:Zero current mode enable bit Bit3:Inherit reactive current before crossing Bit4:Current calculation using pre-drop voltage Bit5:Negative sequence current enable bit Bit6:High-through active enable bit Bit7:Low wear active enable bit Bit8~Bit11:Voltage crossing current calculation mode(0:invalid;1:German 4110 calculation mode, Other reservations) Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
2	Percentage of voltage value entering LVRT	09C1	U16	Valid when register 0x09C0 bit0 enabled Range:[0, 65535]	1%	RW
3	Percentage of voltage at the 1st point of low voltage crossing	09C2	U16		1%	RW
4	The 1st point time of low voltage crossing	09C3	U16		1ms	RW
5	Percentage of voltage at the 2nd point of low voltage crossing	09C4	U16		1%	RW
6	The 2nd point time of low voltage crossing	09C5	U16		1ms	RW
7	Percentage of voltage at the 3rd point of low voltage crossing	09C6	U16		1%	RW
8	The 3rd point time of low voltage crossing	09C7	U16		1ms	RW
9	Percentage of voltage at the 4th point of low voltage crossing	09C8	U16		1%	RW
10	The 4th point time of low voltage crossing	09C9	U16		1ms	RW
11	Low voltage crossing reactive current factor K	09CA	U16		0.1p.u.	RW
12	Waiting time after low voltage crossing recovery	09CB	U16		1ms	RW
13	Low voltage crossing power return rate	09CC	U16	Range:[1, 65535]	1%Pn/min	RW
14	Percentage of voltage value into OVRT	09CD	U16	Valid when register 0x09C0 bit1 enabled Range:[0, 65535]	1%	RW
15	Percentage of voltage at the 1st point of high voltage crossing	09CE	U16		1%	RW
16	The 1st point time of high voltage crossing	09CF	U16		10ms	RW
17	Percentage of voltage at the 2nd	09D0	U16		1%	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
	point of high voltage crossing					
18	The 2nd point time of high voltage crossing	09D1	U16		10ms	RW
19	Percentage of voltage at the 3rd point of high voltage crossing	09D2	U16		1%	RW
20	The 3rd point time of high voltage crossing	09D3	U16		10ms	RW
21	Percentage of voltage at the 4th point of high voltage crossing	09D4	U16		1%	RW
22	The 4th point time of high voltage crossing	09D5	U16		10ms	RW
23	High voltage crossing reactive current factor K	09D6	U16		0.1p.u.	RW
24	Waiting time after High voltage crossing recovery	09D7	U16		1ms	RW
25	High voltage crossing power return rate	09D8	U16	Range:[1, 65535]	1%Pn/min	RW
26	Zero current mode into low voltage percentage	09D9	U16	Range:[0, 65535]	1%	RW
27	Zero-current mode entry high voltage percentage	09DA	U16	Range:[0, 65535]	1%	RW
28	Low voltage crossing exit voltage value percentage	09DB	U16	Range:[0, 65535]	1%	RW
29	Percentage of starting voltage value for low voltage ride-through reactive current calculation	09DC	U16	Range:[0, 65535]	1%	RW
30	Percentage of high voltage crossing exit voltage value	09DD	U16	Range:[0, 65535]	1%	RW
31	Percentage of calculated starting voltage value for high voltage crossing reactive current	09DE	U16	Range:[0, 65535]	1%	RW
32	Percentage of maximum reactive current for unbalanced crossing	09DF	U16	Range:[0, 65535]	1%	RW
33	Percentage of the starting voltage value of the high wear active power	09E0	U16	Range:[0, 65535]	1%	RW
34	Percentage of low wear active power starting voltage	09E1	U16	Range:[0, 65535]	1%	RW
35	Reserve	09E2 - 09FF	N/A		N/A	N/A

5.2.9Safety-Island、GFCI、ISO parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Island enable control	0A00	U16	Bit0:Island enable register Bit1:Single-phase island enable(for 3-phase inverters) Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
2	GFCI enable control	0A01	U16	Bit0:GFCI enable register Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
3	ISO enable control	0A02	U16	Bit0:Insulation impedance enable register Bit1:Grounding detection enable register Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
4	Insulation impedance protection value	0A03	U16	Range:[30, 65535]	1kΩ	RW
5	Component leakage current limit to ground	0A04	U16	Range:[1, 50]	1mA	RW
6	Leakage current limit	0A05	U16	Temporarily not open, setting invalid,Range:[1, 50]	1mA/kVA	RW
7	PE and N control registers	0A06	U16	Bit0:Off-grid PEN short enable bit Bit1:N line detection enable bit Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
8	Sensitivity to island detection	0A07	U16		N/A	RW
9	Ground fault judgement threshold	0A08	U16	Range:[1000, 65535]	1kΩ	RW
10	Power calculation control	0A09	U16	Bit0:Calculate active and reactive power based on battery power Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
11	Reserve	0A0A - 0A3F	N/A		N/A	N/A

5.3Parameter Configuration Area

Some of the parameter blocks involved in parameter configuration are equipped with setting effective enab le registers (marked with *), which have no special instructions. When writing, they are fixed to write 1, and when reading, they return the status of the last write operation, including:

Value	Description
0x0000	Operation successful
0x0001	In operation
0xFFFB	Operation failed, controller refused to respond
0xFFFC	Operation failed, controller does not respond
0xFFFD	Operation failed, current function disabled
0xFFFE	Operation failed, parameter storage failed
0xFFFF	Operation failed, input parameters are incorrect

5.3.1System parameter configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	System date-year	1004	U16	Range:[0, 99], the actual year needs to be added with 2000	N/A	RW
2	System date-month	1005	U16	Range:[1, 12]	N/A	RW
3	System date-day	1006	U16	Range:[1, 31]	N/A	RW
4	System date-hour	1007	U16	Range:[0, 23]	N/A	RW
5	System date-minute	1008	U16	Range:[0, 59]	N/A	RW
6	System date-second	1009	U16	Range:[0, 59]	N/A	RW
7	*Date configuration enable	100A	U16	When the write value is 1, the value in the system time shadow register is updated to the actual system time	N/A	RW
8	RS485 Address configuration	100B	U16	Range:[1, 247]	N/A	RW
9	RS485 Baud rate selection	100C	U16	0：4800bps 1：9600bps(default) 2：19200bps 3：38400bps 4：57600bps 5：115200bps 10：1200bps 11：2400bps	N/A	RW
10	RS485 Stop bit selection	100D	U16	0：1 stop bit(default) 1：1.5 stop bit 2：2 stop bits	N/A	RW
11	RS485 Parity selection	100E	U16	0：None(default) 1：Even 2：Odd 3：Mark 4：Space	N/A	RW
12	*RS485 Configuration enable	100F	U16	When the write value is 1, the value in the RS485	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				configuration shadow register is updated to the system RS485 configuration		
13	PV Input mode selection	1010	U16	0：Parallel mode 1：Independent mode(default)	N/A	RW
14	Input channel 0 type selection	1011	U16	Range:[0, 383] Note: A value of 0 indicates that the current channel is not in use; Values from 1 to 127 indicate that the current channel is input from a photovoltaic panel; Values between 128 and 255 indicate that the current channel is a battery input; The value between 256 and 383 indicates that the current channel is being used as a fan input; If the values of two or more channels are the same and greater than zero, it indicates that these channels are input in parallel; The modification of input channels must be done when the inverter is not connected to or disconnected from the grid.	N/A	RW
15	Input channel 1 type selection	1012	U16		N/A	RW
16	Input channel 2 type selection	1013	U16		N/A	RW
17	Input channel 3 type selection	1014	U16		N/A	RW
18	Input channel 4 type selection	1015	U16		N/A	RW
19	Input channel 5 type selection	1016	U16		N/A	RW
20	Input channel 6 type selection	1017	U16		N/A	RW
21	Input channel 7 type selection	1018	U16		N/A	RW
22	Input channel 8 type selection	1019	U16		N/A	RW
23	Input channel 0 type selection	101A	U16		N/A	RW
24	Input channel 10 type selection	101B	U16		N/A	RW
25	Input channel 11 type selection	101C	U16		N/A	RW
26	Input channel 12 type selection	101D	U16		N/A	RW
27	Input channel 13 type selection	101E	U16		N/A	RW
28	Input channel 14 type selection	101F	U16		N/A	RW
29	Input channel 15 type selection	1020	U16		N/A	RW
30	*PV Input configuration enable	1021	U16	When the write value is 1, the value in the shadow register of the input channel type is updated to the system input channel type configuration	N/A	RW
31	*Enable export of USB security parameters	1022	U16	When the write value is 1, it is used for the communication board to retrieve safety parameters from the USB drive	N/A	RW
32	Anti-back-flow enable control	1023	U16	0：Prohibit anti-back-flow function 1：Default anti-back-flow mode 2：Three phase average power anti-back-flow mode 3：Phase power mode 4：Master slave anti-back-flow mode	N/A	RW
33	Anti-back-flow power	1024	S16	Range:[-32768, 32767]	100W	RW
34	IV Curve scanning enable control	1025	U16	0：Prohibit, 1：enable	N/A	RW
35	IV Curve scanning period	1026	U16	Range:[5, 65535]	1Min	RW
36	*IV Curve scanning activate enable	1027	U16	When the write value is 1, activate one IV curve scan	N/A	RW
37	IV curve scanning feedback data channel	1028	U16	Used to specify the PV channel value corresponding to the IV scan result feedback, and the written value cannot exceed the maximum PV channel of the inverter. Range:[0, 31]	N/A	RW
38	Emergency power supply enable control	1029	U16	0：Turn off emergency power supply(default) 1：Activate emergency power supply and prohibit cold start	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				2：Activate emergency power supply to enable cold start		
39	Waiting time for emergency power supply startup	102A	U16	Reserved function, Range:[0, 65535]	1s	RW
40	Battery automatic activation enable control	102B	U16	0：Prohibit, 1：enable	N/A	RW
41	*Battery automatic activation configuration enable	102C	U16	When the write value is 1, activate the battery once	N/A	RW
42	*CT Automatic calibration enable	102D	U16	When the write value is 1, activate CT automatic correction once	N/A	RW
43	Italy automatic testing	102E	U16	write in： 0x0001：Perform standard testing(S) 0x0002：Perform rapid testing(F) When reading, return the status of the last write operation： 0x0000：Operation successful 0x0001：Operating standard testing 0x0002：Operating rapid testing 0xFFFB：Operation failed, controller refused to respond 0xFFFC：Operation failed, controller does not respond 0xFFFD：Operation failed, current function disabled 0xFFFE：Operation failed, parameter storage failed 0xFFFF：Operation failed, input parameters are incorrect	N/A	RW
44	Heating film enable	102F	U16	0：Prohibit, 1：enable	N/A	RW
45	Setting the return date for electricity statistics	1030	U16	This register specifies the year of the returned electrical energy data. The year refers to the Nth year closest to the time of the inverter system. N is the register value. 0：System time year. 1：1 year before the system time. … 19：19 years prior to system time.	N/A	RW
46	Electricity statistics return month setting register	1031	U16	Set the month for the specified electricity data to be returned, Range:[1, 12]	N/A	RW
47	Electric energy statistics return date setting register	1032	U16	Specify the date of the returned electrical energy data, Range:[1, 31]	N/A	RW
48	Electric energy statistics feedback setting register	1033	U16	High byte：Date and time setting for returning data. 0x01：Daily, the first 24 pieces of data in the feedback area are valid； 0x02：Every month, the first 31 pieces of data in	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				the feedback area are valid; 0x03：Every year, the first 12 pieces of data in the feedback area are valid; 0x04： Life-cycle, the first 20 data in the feedback area are valid; Other： Invalid。 Low byte：Physical quantity settings for returning data. 0x01： Photovoltaic generation electricity; 0x02： Load consumption electricity; 0x03： System purchased electricity; 0x04： System sold electricity; 0x05： Battery charging electricity; 0x06： Battery discharging electricity; Other： Invalid.		
49	Inverter menu language setting	1034	U16	0：Chinese 1：English 2：Italian 3：German 4：French 5：Portuguese 6：Ukrainian 7：Slovak 8：Spanish 9：Finnish 10：Polish 11：Korean 12：Japanese 13：Russian 14：Czech	N/A	RW
50	Local parallel control register	1035	U16	0：Disable parallel operation function 1：Enable AC parallel operation function 2：Enable AC+BAT parallel operation function	N/A	RW
51	Local parallel mode register	1036	U16	0：This machine is configured as a slave 1：This machine is configured as a host(default)	N/A	RW
52	Local parallel address register	1037	U16	Range:[0, 10]	N/A	RW
53	3-phase 4-wire mode power grid imbalance support control register	1038	U16	0：Disable unbalanced support function(default) 1：Enable unbalanced support function	N/A	RW
54	Power generation multiplier	1039	U16	Range:[800, 3600], default value: 1000	0.001p.u.	RW
55	Buy power multiplier	103A	U16	Range:[800, 1500], default value: 1000	0.001p.u.	RW
56	Sell power multiplier	103B	U16	Range:[800, 1500], default value: 1000	0.001p.u.	RW
57	Charge amount multiplier	103C	U16	Range:[800, 1500], default value: 1000	0.001p.u.	RW
58	Discharge volume multiplier	103D	U16	Range:[800, 1500], default value: 1000	0.001p.u.	RW
59	Logical Interface Control Register	103E	U16	0：Prohibit, 1：enable	N/A	RW
60	Logical interface value	103F	U16	Range:[0, 127] Note:	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				0: indicates that no logical interface is enabled or logical interface 0 is enabled 1~8 ： corresponding logical interface DRMS1-8127：indicates that logical interfaces 1-8 are enabled, but the value is invalid		

5.3.2Battery parameter configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Battery serial number	1044	U16	Range:[0, 7] Note：The serial number specified in this register corresponds to the actual battery input interface of the inverter. After a successful write to this register, the battery parameters will be updated to the parameters corresponding to the written serial number。	N/A	RW
2	Battery Address	1045	U16	Range:[0, 99] Note： If multiple batteries are allowed to be accessed in the system, this register is used to mark the battery address of the physical interface corresponding to the battery serial number	N/A	RW
3	Communication protocol	1046	U16	0：Sofar built-in BMS/DEFAULT 1：Pie Energy protocol/PYLON 2：Sofar protocol/GENERAL 3：AMASS 4：LG 5：Alpha.ESS 6：CATL 7：Weco 8：Fronus 9：EMS 10：Nilar 11：BTS 5K 12：Move for 13：STRONG C 14：Reserve 15：NEOOM	N/A	RW
4	Over-voltage protection value	1047	U16	Range:[0, 65535]	0.1V	RW
5	Charging voltage	1048	U16	Range:[0, 65535]	0.1V	RW
6	Under-voltage protection voltage	1049	U16	Lead-acid batteries visible, Range:[0, 65535]	0.1V	RW
7	Minimum discharge voltage	104A	U16	Range:[0, 65535]	0.1V	RW
8	Maximum charge current limit	104B	U16	Range:[0, 65535]	0.01A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
9	Maximum discharge current limit	104C	U16	Range:[0, 65535]	0.01A	RW
10	DOD indicates the maximum discharge when the grid is normal	104D	U16	Range:[1, 90]	1%	RW
11	EOD indicates the maximum off-grid discharge	104E	U16	Range:[1, 90]	1%	RW
12	Battery capacity	104F	U16	Range:[1, 65535]	1Ah	RW
13	Rated battery voltage	1050	U16	Range:[0, 65535]	0.1V	RW
14	Cell type	1051	U16	0: Lead acid (custom) 1: Lithium iron phosphate 2: Ternary 3: Lithium titanate 4: AGM 5: Gel 6: Flooded	N/A	RW
15	Off-grid recovery discharge hysteresis	1052	U16	Range:[5, 100]	1%	RW
16	*Battery parameter write control	1053	U16	A write value of 1 updates the value in the battery parameter shadow register to the system battery parameter configuration; a write value of 2 updates the value in the battery parameter shadow register to the default value of BatConfig Protocol;	N/A	RW
17	Battery address 2	1054	U16	Note： If multiple batteries are allowed to be accessed in the system, this register is used to mark the battery address of the physical interface corresponding to the battery serial number, Range:[0, 99]	N/A	RW
18	Battery address 3	1055	U16		N/A	RW
19	Battery address 4	1056	U16		N/A	RW
20	Lead acid battery temperature compensation coefficient	1057	S16	Range:[-60, 0]	0.1mV/Cell	RW
21	Lead acid battery recovery discharge voltage increment	1058	U16	Range:[0, 65535] Note: Stop discharging when the battery voltage is below the minimum discharge voltage;Resume discharge when battery voltage>minimum discharge voltage+incremental recovery discharge voltage	0.1V	RW
22	Function Control Bits	1059	U16	Bit0：SOC control disabled Note: Use SOC when set to 0, disable SOC when set to 1	N/A	RW
23	Lead-acid battery float voltage	105A	U16		0.1V	RW
24	Maximum charging SOC	105B	U16	Range:[60, 100]，default value: 100，When the battery SOC is greater than or equal to the maximum charging SOC, the device stops charging	1%	RW
25	Reserve	105B - 105F	N/A		N/A	N/A

5.3.3Function parameter configuration address 1

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	PCC power acquisition device configuration	1060	U16	1：Electricity meter 2：CT 3：ARPC 4：Logger	N/A	RW
2	Resonance detection sensitivity	1061	U16	Range:[1, 100]	N/A	RW
3	CT ratio setting	1062	U16	Range:[0, 65535]	N/A	RW
4	Hard anti-back-flow switch	1063	U16	0：Prohibit, 1：Enable	N/A	RW
5	Dry contact control	1064	U16	0：Disable dry contact 1：Generator mode 2：Relay mode 1(off-grid output high, rest low) 3：Relay mode 2(off-grid output low, rest high) 4：Intelligent load control (parameters located between 106C and 106F)	N/A	RW
6	Wet contact control (reserved)	1065	U16	0：Prohibit, 1：Enable	N/A	RW
7	PV branch over-current protection	1066	U16	Range:[0, 65535]	0.1A	RW
8	Grid detection and control	1067	U16	Bit0：Phase loss detection enabled Other reservations. Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
9	The value of the missing phase detection domain	1068	U16	Range:[0, 1000]	0.1%	RW
10	PCC bias power	1069	S16	Range:[-300, 300]	1%	RW
11	Purchase total power limit enable control	106A	U16	0:The power limit function of buying electricity is prohibited 1：Enable the power limit function for buying electricity	N/A	RW
12	Total power limit for purchasing electricity	106B	U16	Range:[0, 65535]	0.1kW	RW
13	Intelligent load mode selection	106C	U16	0：Turn off intelligent load 1：Turn on intelligent load 2：Timer mode 3：Intelligent mode	N/A	RW
14	Intelligent load startup time	106D	U16	High byte represent hours,Range:[0, 23] Low byte represent minutes,Range:[0, 59]	N/A	RW
15	End time of intelligent load	106E	U16	High byte represent hours,Range:[0, 23] Low byte represent minutes,Range:[0, 59]	N/A	RW
16	Intelligent load activation power	106F	U16	Range:[0, 65535]	0.1kW	RW
17	Parallel Module Number	1070	U16	Range:[1, 65535]	N/A	RW
18	DRMs Shutdown Control	1071	U16	Bit0：DRM0 Control(1:shutdown, 0:startup) Bit1~Bit15：reserved	N/A	RW
19	File transfer function configuration	1072	U16	Bit0：Read file instruction bit(0x14 command) Bit1：Write file instruction bit(0x15 command) Note: 1 is support, 0 is non support	N/A	RO
20	Battery usage configuration	1073	U16	0：Inverter does not use batteries	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				1： Inverter uses batteries,single channel use (default channel 1) 2： Battery channels 1 and 2 can be used independently 3：Battery channels 1 and 2 are used in parallel Other: invalid Note: When the inverter is not using the battery, it will not report battery communication errors.		
21	*System files export by USB drive	1074	U16	write in： 0x0001：Fault waveform file (refers to 10 sets of historical fault waveform data stored in ARM) 0x0002：Local fault waveform file (refers to the 2 latest sets of fault waveform data stored in DSP) 0x0003： Historical event files Other：Reserve When reading, return the status of the last write operation： 0x0000~0x0064：Progress of file export	N/A	RW
22	Configuration of port types on the grid side	1075	U16	Default value:0 0:Prohibit grid side port 1:Configure as a power grid 2:Configure as a generator Other:Invalid	N/A	RW
23	Configuration of port types on the generator side	1076	U16	Default value:1 0:Prohibit generator side port 1:Configure as a generator 2:Configure as a smart load Other:Invalid	N/A	RW
24	Reserve	1077 - 107F	N/A		N/A	N/A

5.3.4Function parameter configuration address 2

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Enable arc detection function	1084	U16	0x01：Enable	N/A	RW
2	Peak to peak value setting	1085	U16	Range:[0x0064, 0xFFFF]	N/A	RW
3	Variance setting	1086	U16	Range:[0x0095, 0xFFFF]	N/A	RW
4	Harmonic energy setting	1087	U16	Range:[0x0001, 0xFFFF]	N/A	RW
5	Amplitude variance setting	1088	U16	Range:[0x0001, 0xFFFF]	N/A	RW
6	Time domain weight setting	1089	U16	Range:[0x0001, 0xFFFF]	N/A	RW
7	Frequency domain weight setting	108A	U16	Range:[0x0005, 0xFFFF]	N/A	RW
8	Arc detection sensitivity	108B	U16	Range:[0x0014, 0xFFFF]	N/A	RW
9	Arc alarm counting shutdown threshold	108C	U16	Range:[0x0005, 0xFFFF]	N/A	RW
10	Self check enabled	108D	U16	Write 1 to start self-test (self-test all channels),	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				auto-zero at the end of the self-test.		
11	Remove arc alarm	108E	U16	Write 0x1234 Clear Arc Alarm (all channels)	N/A	RW
12	Arc alarm count	108F	U16		N/A	RW
13	Arc handshake result	1090	U16		N/A	RO
14	Channel self-check results	1091	U16	Bit0: Channel 1 self-check result Bit1: Channel 2 self-check result Bit2: Channel 3 self-check result Bit3: Channel 4 self-check result Bit4: Channel 5 self-check result Bit5: Channel 6 self-check result 0: Self-check success, 1: Self-check failure or no self-check	N/A	RO
15	Channel self-check status	1092	U16	0x00: No self-check 0x01: Self-check in progress 0x02: Self-check complete	N/A	RO
16	Reserve	1093 - 109F	N/A		N/A	N/A
17	PLC setting operation control	10A0	U16	0x00：Not allow，0x01：Allow	N/A	RW
18	PLC communication attribute control	10A1	U16	Bit0~Bit1：【0: 1 stop bit, 1: 1.5 stop bit】 Bit2~Bit4：【0: No parity, 1: odd parity, 2: even parity】 Bit5~Bit7：【0: 8 bits data, 1: 9 bits data】 Bit8~Bit15：【0: 4800bsp, 1：9600bps, 2: 19200bps, 3: 38400bps, 4: 115200bps】	N/A	RW
19	Reserve	10A2 - 10A4	N/A		N/A	N/A
20	BLE Bluetooth settings running	10A5	U16	0x00：Not allow，0x01：Allow	N/A	RW
21	BLE Bluetooth communication attribute control	10A6	U16	Bit0~Bit1：【0: 1 stop bit, 1: 1.5 stop bit】 Bit2~Bit4：【0: No parity, 1: odd parity, 2: even parity】 Bit5~Bit7：【0: 8 bits data, 1: 9 bits data】 Bit8~Bit15：【0: 4800bsp, 1：9600bps, 2: 19200bps, 3: 38400bps, 4: 115200bps】	N/A	RW
22	Reserve	10A7 - 10A9	N/A		N/A	N/A
23	Communication module state	10AA	U16	Bit0:System network register state Bit1:System connection to cloud server state Bit2:Bluetooth communication state Bit3~Bit15:Reserve Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
24	External device state	10AB	U16	0:No external devices found 1:USB device detected 2:Collector device detected	N/A	R
25	Reserve	10AC - 10AE	N/A		N/A	N/A
26	Centralized PID Control Word	10AF	U16	0: Disable, 1: Enable	N/A	RW
27	PID Start auto-run	10B0	U16	0：Prohibit, 1：Enable	N/A	RW
28	PID Operation start time	10B1	BCD16	Range:[0x0000, 0x2359] Minimum value 0x0000 Indicate 00:00 Maximum value 0x2359 Indicate 23:59	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
29	PID Operating duration	10B2	U16	Range:[0, 65535]	1min	RW
30	PID Start PV value threshold	10B3	U16	Range:[0, 65535] PID Start is only allowed when the PV voltage is below the threshold value.	0.1V	RW
31	Restart wait time after PID fault shutdown	10B4	U16	Range:[0, 65535]	1min	RW
32	CLS Anti-back-flow state	10B5	U16	0: Normal state; 1: Criticality state; 2: Low power state; 3: Fault state	N/A	RO
33	CLS State remote control	10B6	U16	Bit0: CLS locked state clean Bit1: CLS state reset Note：Enable is set to 1 for on, set to 0 for off	N/A	RW
34	Password reset control	10B7	U16	0x01: Reset password Other: Invalid	N/A	WO
35	Reserve	10B8	N/A		N/A	N/A
36	Generator control	10B9	U16	Bit0: Mode control 【0: Manual mode, 1: Automatic mode】 Bit1: Manual control【0: Manual off, 1: Manual on】 Bit2: Generator locked【0: disable, 1: enable】	N/A	RW
37	Peak generator power	10BA	U16	Range:[1, 65535]	0.1kW	RW
38	Generator startup voltage	10BB	U16	Range:[0, 65535]	0.1V	RW
39	Generator stop voltage	10BC	U16	Range:[0, 65535]	0.1V	RW
40	Generator SOC	10BD	U16	High byte indicate start SOC，Range:[0, 100] Low byte indicate end SOC，Range:[0, 100]	1%	RW
41	Reserve	10BE- 10BF	N/A		N/A	N/A
42	Communication interruption control	10C0	U16	Control actions after communication interruption 0：Prohibit this function 1：Execute shutdown 2：Operate according to the state before communication interruption 3：Operate according to the preset fixed active percentage 4：Operate according to the preset fixed reactive percentage 5：Operate according to the preset power factor parameters Note：The preset value is defined in the 0x10C2 register	N/A	RW
43	Communication interruption timeout parameter	10C1	U16	Range:[0, 999]	1S	RW
44	Preset value for communication interruption	10C2	S16	When operating with a fixed active percentage, the effective value range is 0~1100, representing 0%~110%	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				When operating with a fixed reactive percentage, the effective value range is 0~1100, representing 0%~110% When operating the power factor parameter, the effective range of values is ±1000~±800 or, which represents+1~±0.8		
45	Reserve	10C3	N/A		N/A	N/A
46	Bluetooth channel protocol configuration	10C4	U16	0：MODBUS-G3, 1：MODBUS-G4	N/A	RW
47	PLC channel protocol configuration	10C5	U16	0：MODBUS-G3, 1：MODBUS-G4	N/A	RW
48	USB channel protocol configuration	10C6	U16	0：MODBUS-G3, 1：MODBUS-G4	N/A	RW
49	RS485-1 channel protocol configuration	10C7	U16	0：MODBUS-G3, 1：MODBUS-G4	N/A	RW
50	RS485-2 channel protocol configuration	10C8	U16	0：MODBUS-G3, 1：MODBUS-G4	N/A	RW
51	Reserve	10C9 - 10CB	N/A		N/A	RW
52	Power enable control	10CC	U16	Bit0: Active power output enable Bit1: Reactive power output enable Bit2~Bit15: Reserve Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
53	Active power output control	10CD- 10CE	S32	Control the power output of the current active power	0.01kW	RW
54	Reactive power output control	10CF - 10D0	S32	Control the power output of the current reactive power	0.01kVar	RW
55	Reserve	10D1 - 10DF	N/A		N/A	RW
56	Fan self-test control	10E0	U16	0：Prohibit, 1：Enable	N/A	RW
57	Fan noise mode	10E1	U16	0：Standard mode, 1：Low noise mode(The default value is 0)	N/A	RW
58	Fan alarm detection mode	10E2	U16	0：Fault condition detection mode(The system actively controls the fan rotation state to detect fan faults, and eliminates faults when the fan rotation state is not controlled) 1：Fault maintenance detection mode(After detecting a fault, the fan maintains the fault until it is resolved and eliminated)	N/A	RW
59	Fan detection sensitivity	10E3	U16		N/A	RW
60	Fan Speed Control	10E4	U16	Maximum percentage of control speed	1%	RW
61	Reserve	10E5 - 10EA	N/A		N/A	N/A
62	Fixed Unbalanced Power Control	10EB	U16	0：Prohibit, 1：Enable	N/A	RW
63	Unbalanced power percentage of R-phase	10EC	S16	Range:[-500, 500]，Representing -50%~50%	0.1%	RW
64	Unbalanced power percentage of S-phase	10ED	S16		0.1%	RW
65	Unbalanced power percentage of T-phase	10EE	S16		0.1%	RW
66	Reserve	10EF	N/A		N/A	N/A

Index	Name	Address（HEX）	Type	Description	Unit	Properties
67	Relay configuration	10F0	U16	When the power is 0, the relay control (default value 0) 0: Relay does not disconnect 1: Relay disconnected	N/A	RW
68	Standby monitoring control	10F1	U16	In the event of a power outage at night, the relay will engage to draw power from the grid and maintain the device from losing power 0: Prohibited 1: Enable	N/A	RW

5.3.5Remote control parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Remote power on/off control	1104	U16	0x0000: Remote shutdown， 0x0001: Remote boot up	N/A	RW
2	Power control	1105	U16	Bit0: Active power enable Bit1: Reactive power enable Bit2: Reactive power mode selection position(0: Reactive power, 1: Power factor) Bit3: SVG Enable Bit4: SVG Reactive mode selection(0: Fixed reactive power, 1: Reactive power) Bit5: SVG Power factor compensation enable Bit6~Bit7: Reserve Bit8: Night operation enable Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
3	Export maximum active power percentage	1106	U16	Range:[0 ~ 1100]	0.1%	RW
4	Import maximum active power percentage	1107	U16	Range:[0 ~ 1000]	0.1%	RW
5	Reactive power percentage	1108	S16	Range:[-1000 ~ 1000] The maximum reactive power is limited by the specific model.	0.1%	RW
6	Power factor	1109	S16	Range:[-100 ~ 100] The minimum power factor is limited by the specific model.	0.01p.u.	RW
7	Active power limit change rate	110A	U16	Range:[0, 65535], 0 indicates that the device is unloading at its maximum rate	1%Pn/min	RW
8	Reactive power setting response time	110B	U16	Range:[0, 65535]	0.1S	RW
9	Fixed reactive power magnitude in SVG mode	110C	S16	Range:[-32768, 32767]	0.1kVar	RW
10	SVG power factor compensation reactive power regulation quantity	110D	S16	Range:[-32768, 32767]	0.1kVar	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
11	Reserve	110E - 110F	N/A		N/A	N/A
12	Energy storage operating mode setting	1110	U16	0: Self-generation mode 1: Time-sharing tariff mode 2: Timed charging and discharging mode 3: Passive mode 4: Peak-shaving mode 5: Off-grid mode 6: Generator mode	N/A	RW

5.3.6Timed charging and discharging parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Rule serial number	1111	U16	Range:[0 ~ 3] The lower the serial number, the higher the priority. When this register is successfully written, the timed charging parameters will be updated to the parameters corresponding to the serial number written.	N/A	RW
2	Enable control	1112	U16	Bit0: Charge enable Bit1: Discharge enable Note:Enable bit 1 is enabled, set 0 is prohibited	N/A	RW
3	Charging start time	1113	U16	High byte indicate hour, Range:[0, 23] Low byte indicate minute, Range:[0, 59]	N/A	RW
4	Charging end time	1114	U16	High byte indicate hour, Range:[0, 23] Low byte indicate minute, Range:[0, 59]	N/A	RW
5	Discharging start time	1115	U16	High byte indicate hour, Range:[0, 23] Low byte indicate minute, Range:[0, 59]	N/A	RW
6	Discharging end time	1116	U16	High byte indicate hour, Range:[0, 23] Low byte indicate minute, Range:[0, 59]	N/A	RW
7	Charging power	1117 - 1118	U32	Range:[1, 4294967296]	1W	RW
8	Discharge power	1119 - 111A	U32	Range:[1, 4294967296]	1W	RW
9	Reserve	111B - 111E	N/A		N/A	N/A
10	*Timed charge and discharge write control	111F	U16	On a write value of 1, the value in the timed-charge shadow register is updated to the system timed-charge configuration	N/A	RW

5.3.7Time sharing control parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Rule serial number	1120	U16	Range:[0,255] The smaller the serial number, the higher the	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				priority. After a successful write to this register, the time-of-use tariff parameter will be updated to the parameter corresponding to the serial number written.		
2	Enable control	1121	U16	0: Prohibit, 1: Enable	N/A	RW
3	Start time	1122	U16	High byte indicate hour, Range:[0, 23] Low byte indicate minute, Range:[0, 59]	N/A	RW
4	End time	1123	U16	High byte indicate hour, Range:[0, 23] Low byte indicate minute, Range:[0, 59]	N/A	RW
5	Target SOC	1124	U16	Range:[10, 100] (The following modes are determined by the 0x112A register) When the mode is forced charging - charging cut-off SOC When the mode is forced discharge - discharge cut-off SOC When the mode is peak shaving mode - stop peak shaving SOC When the mode is spontaneous self use - charging cut-off SOC When the mode is charging maintenance - charging cut-off SOC When the mode is discharge maintenance - discharge cut-off SOC	1%	RW
6	Target power	1125 - 1126	U32	(The following modes are determined by the 0x112A register) When the mode is forced charging - battery forced charging power When the mode is forced discharge - battery forced discharge power When the mode is peak shaving mode - PCC peak shaving power When the mode is power feeding priority - PCC power feeding priority When the mode is spontaneous self use - additional functions correspond to power When the mode is charging maintenance - battery maintenance charging power When the mode is discharge maintenance - battery maintenance discharge power	1W	RW
7	Effective date of the rule	1127	U16	High byte indicates effective month, Range:[1, 12] Low byte indicates effective day, Range:[1, 31]	N/A	RW
8	Expiration date of the rule	1128	U16	High byte indicates expiration month, Range:[1, 12]	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				Low byte indicates expiration day, Range:[1, 31]		
9	Rule effective week	1129	U16	Bit0: Monday Bit1：Tuesday Bit2: Wednesday Bit3: Thursday Bit4: Friday Bit5: Saturday Bit6: Sunday Note：This register is represented as a bit field, with a bit 0 indicating an invalid week and a bit 1 indicating a valid week	N/A	RW
10	Control mode configuration	112A	U16	0：Forced charging 1：Forced discharging 2：Peak shaving 3：Power feeding priority 4：Spontaneous self use 5：Charging maintenance 6：Discharging maintenance	N/A	RW
11	Additional feature configuration	112B	U16	(The following modes are determined by the 0x112A register) When the mode is peak shaving mode 0: Disable buying electricity from the grid to charge the battery 1: Enable purchasing electricity from the power grid to charge batteries When the mode is for spontaneous self use 0: Disable additional features 1: Enable mandatory purchase of electricity from the grid to charge batteries 2: Reserved When the mode is charging maintenance mode 0: Target SOC control 1: End time control When in discharge maintenance mode 0: Target SOC control 1: End time control	N/A	RW
12	Reserve	112C - 112E	N/A		N/A	N/A
13	*Write enable	112F	U16	When the value written is 1, the value in the shadow register of the time of use electricity price is updated to the system's time of use electricity price configuration	N/A	RW

5.3.8Peak shaving mode parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Upper limit of purchasing power	1130 - 1131	U32	Range:[100, 65535] Upper limit of the power allowed to the system to buy power from the grid	1W	RW
2	Upper limit of selling power	1132 - 1133	U32	Range:[100, 65535] Upper limit of the power allowed to sell electricity to the grid	1W	RW
3	Buying electricity to charge batteries	1134	U16	0: Prohibited,1: Enable	N/A	RW
4	Battery backup SOC	1135	U16	Range：[20, 100], after SOC exceeds the set value, it will be converted to spontaneous self use	1%	RW
5	Reserve	1136 - 1139	N/A		N/A	N/A

5.3.9Power feeding priority parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	The power of feeding into the grid	113A - 113B	U32	Range:[100, 4294967295]	1W	RW
2	Reserve	113C - 113F	N/A		N/A	N/A

5.3.10Off-grid mode parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Charging source selection	1144	U16	0: Grid 1: Generator 2: Reserved	N/A	RW
2	Grid draw power	1145	U16	Range:[0, 65535]	0.01kW	RW
3	Maximum generator power draw	1146	U16	Range:[0, 65535]	0.01kW	RW
4	Reserve	1147 - 117F	N/A		N/A	N/A

5.3.11Energy storage mode parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Passive mode - Timeout control	1184	U16	Range:[0, 65535] Sets the passive mode communication timeout time. When the inverter does not receive any communication for longer than the time set in this register, the inverter forces a timeout action.	1s	RW
2	Passive mode - Timeout action	1185	U16	0: forces standby 1: forced return to the energy storage mode before entering the passive mode	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				Other: Invalid		
3	Reserve	1186	N/A		N/A	N/A
4	Manual mode desired grid power	1187 - 1188	S32	Positive values indicate the power direction from grid to system". Negative values indicate the power direction "from the system to the grid".	1W	RW
5	Manual mode battery minimum charge and discharge power	1189 - 118A	S32	Positive values indicate charging. Negative values indicate discharging.	1W	RW
6	Manual mode maximum charge/discharge power	118B - 118C	S32	Positive values indicate charging. Negative values indicate discharging.	1W	RW
7	Manual mode permissible power selling	118D - 118E	S32	Positive values indicate the power direction from grid to system". Negative values indicate the power direction "from the system to the grid".	1W	RW
8	Manual mode allows the purchase of power	118F - 1190	S32		1W	RW
9	Scheduler mode anticipation grid power	1191 - 1192	S32	Scheduler task parameters, valid within the scheduler time period. Positive value indicates the power direction "from the grid to the system". Negative value indicates the direction of power "from the system to the grid".	1W	RW
10	Scheduler mode battery minimum charge or discharge power	1193 - 1194	S32		1W	RW
11	Scheduler mode battery maximum charge or discharge power	1195 - 1196	S32		1W	RW
12	Scheduler mode allowable selling power	1197 - 1198	S32		1W	RW
13	Scheduler mode allowable purchasing power	1199 - 119A	S32		1W	RW
14	Scheduler mode anticipation grid power	119B - 119C	S32		1W	RW
15	Scheduler mode battery minimum charge or discharge power	119D - 119E	S32		1W	RW
16	Scheduler mode battery maximum charge or discharge power	119F - 11A0	S32		1W	RW
17	Scheduler allowable selling power	11A1 - 11A2	S32		1W	RW
18	Scheduler mode allowable purchasing power	11A3 - 11A4	S32		1W	RW
19	Scheduler mode start time of setting expected parameter values for application	11A5 - 11A6	U32	Range:[0, 4294967296]	1s	RW
20	Scheduler mode duration of expected parameter settings for application	11A7 - 11A8	U32	Range:[0, 4294967296]	1s	RW
21	Scheduler management mode select	11A9	U16	0: Spontaneous self use mode 1: Manual Mode 2: Scheduler Mode Other: Invalid	N/A	RW
22	Reserve	11AA - 11BF	N/A		N/A	N/A

5.3.12Power station communication parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Communication interruption protection function enable register	11C4	U16	0：Protection function prohibited(default) 1：Protection function enabled	N/A	RW
2	Communication interruption protection recovery method	11C5	U16	0：Automatic recovery（default） 1：Manual recovery	N/A	RW
3	Communication interruption threshold time setting	11C6	U16	Range：[10, 120]	1min	RW
4	Communication interruption data monitoring source setting	11C7	U16	0：485（default） 1：WIFI 3：PLC	N/A	RW
5	Communication interruption alarm clearing register	11C8	U16	Writing a value of "1" indicates clearing the communication interruption alarm status.	N/A	WO
6	Reserve	11C9 - 11FF	N/A		N/A	N/A

5.3.13EMS Time period enable configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Time period enable control bit	1204	U16	Bit0：Time 1 enable position Bit1：Time 2 enable position Bit2：Time 3 enable position Bit3：Time 4 enable position Bit4：Time 5 enable position Bit5：Time 6 enable position Note: Enable position 1 is enabled, set 0 is prohibited	N/A	RW

5.3.14EMS Time period 1 configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Start time	1205	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
2	End time	1206	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
3	Work mode	1207	U16	0：Photovoltaic mode 1：Constant power mode 2：Spontaneous self use mode 3：Peak shaving and valley filling mode 4：Demand control mode 5：Battery forced charge mode 6：Battery forced discharge mode	N/A	RW
4	Grid connection point power upper limit	1208	S16		0.01kW	RW
5	Grid connection point power lower limit	1209	S16		0.01kW	RW
6	Battery allow charge upper limit	120A	U16	Range：[0, 100]	1%	RW
7	Battery allow discharge lower limit	120B	U16	Range：[0, 100]	1%	RW
8	Allow grid to charge for system	120C	U16	0：Not allow, 1：Allow	N/A	RW
9	Inverter's power	120D	S16		0.01kW	RW
10	Force charge and discharge power	120E	S16	Total battery port power	0.01kW	RW
11	Spontaneous self use of excess power for grid connection	120F	U16	0：Not allow, 1：Allow	N/A	RW
12	Reserve	1210 - 1211	N/A		N/A	N/A

5.3.15EMS Time period 2 configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Start time	1212	U16	High byte indicates hour，range：[0, 23]	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				Low byte indicates minute，range：[0, 59]		
2	End time	1213	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
3	Work mode	1214	U16	0：Photovoltaic mode 1：Constant power mode 2：Spontaneous self use mode 3：Peak shaving and valley filling mode 4：Demand control mode 5：Battery forced charge mode 6：Battery forced discharge mode	N/A	RW
4	Grid connection point power upper limit	1215	S16		0.01kW	RW
5	Grid connection point power lower limit	1216	S16		0.01kW	RW
6	Battery allow charge upper limit	1217	U16	Range：[0, 100]	1%	RW
7	Battery allow discharge lower limit	1218	U16	Range：[0, 100]	1%	RW
8	Allow grid to charge for system	1219	U16	0：Not allow, 1：Allow	N/A	RW
9	Inverter's power	121A	S16		0.01kW	RW
10	Force charge and discharge power	121B	S16	Total battery port power	0.01kW	RW
11	Spontaneous self use of excess power for grid connection	121C	U16	0：Not allow, 1：Allow	N/A	RW
12	Reserve	121D - 121E	N/A		N/A	N/A

5.3.16EMS Time period 3 configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Start time	121F	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
2	End time	1220	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
3	Work mode	1221	U16	0：Photovoltaic mode 1：Constant power mode 2：Spontaneous self use mode 3：Peak shaving and valley filling mode 4：Demand control mode 5：Battery forced charge mode 6：Battery forced discharge mode	N/A	RW
4	Grid connection point power upper limit	1222	S16		0.01kW	RW
5	Grid connection point power lower limit	1223	S16		0.01kW	RW
6	Battery allow charge upper limit	1224	U16	Range：[0, 100]	1%	RW
7	Battery allow discharge lower limit	1225	U16	Range：[0, 100]	1%	RW
8	Allow grid to charge for system	1226	U16	0：Not allow, 1：Allow	N/A	RW
9	Inverter's power	1227	S16		0.01kW	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
10	Force charge and discharge power	1228	S16	Total battery port power	0.01kW	RW
11	Spontaneous self use of excess power for grid connection	1229	U16	0：Not allow, 1：Allow	N/A	RW
12	Reserve	122A - 122B	N/A		N/A	N/A

5.3.17EMS Time period 4 configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Start time	122C	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
2	End time	122D	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
3	Work mode	122E	U16	0：Photovoltaic mode 1：Constant power mode 2：Spontaneous self use mode 3：Peak shaving and valley filling mode 4：Demand control mode 5：Battery forced charge mode 6：Battery forced discharge mode	N/A	RW
4	Grid connection point power upper limit	122F	S16		0.01kW	RW
5	Grid connection point power lower limit	1230	S16		0.01kW	RW
6	Battery allow charge upper limit	1231	U16	Range：[0, 100]	1%	RW
7	Battery allow discharge lower limit	1232	U16	Range：[0, 100]	1%	RW
8	Allow grid to charge for system	1233	U16	0：Not allow, 1：Allow	N/A	RW
9	Inverter’s power	1234	S16		0.01kW	RW
10	Force charge and discharge power	1235	S16	Total battery port power	0.01kW	RW
11	Spontaneous self use of excess power for grid connection	1236	U16	0：Not allow, 1：Allow	N/A	RW
12	Reserve	1237 - 1238	N/A		N/A	N/A

5.3.18EMS Time period 5 configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Start time	1244	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
2	End time	1245	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
3	Work mode	1246	U16	0：Photovoltaic mode 1：Constant power mode 2：Spontaneous self use mode 3：Peak shaving and valley filling mode 4：Demand control mode	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				5：Battery forced charge mode 6：Battery forced discharge mode		
4	Grid connection point power upper limit	1247	S16		0.01kW	RW
5	Grid connection point power lower limit	1248	S16		0.01kW	RW
6	Battery allow charge upper limit	1249	U16	Range：[0, 100]	1%	RW
7	Battery allow discharge lower limit	124A	U16	Range：[0, 100]	1%	RW
8	Allow grid to charge for system	124B	U16	0：Not allow, 1：Allow	N/A	RW
9	Inverter's power	124C	S16		0.01kW	RW
10	Force charge and discharge power	124D	S16	Total battery port power	0.01kW	RW
11	Spontaneous self use of excess power for grid connection	124E	U16	0：Not allow, 1：Allow	N/A	RW
12	Reserve	124F - 1250	N/A		N/A	N/A

5.3.19EMS Time period 6 configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Start time	1251	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
2	End time	1252	U16	High byte indicates hour，range：[0, 23] Low byte indicates minute，range：[0, 59]	N/A	RW
3	Work mode	1253	U16	0：Photovoltaic mode 1：Constant power mode 2：Spontaneous self use mode 3：Peak shaving and valley filling mode 4：Demand control mode 5：Battery forced charge mode 6：Battery forced discharge mode	N/A	RW
4	Grid connection point power upper limit	1254	S16		0.01kW	RW
5	Grid connection point power lower limit	1255	S16		0.01kW	RW
6	Battery allow charge upper limit	1256	U16	Range：[0, 100]	1%	RW
7	Battery allow discharge lower limit	1257	U16	Range：[0, 100]	1%	RW
8	Allow grid to charge for system	1258	U16	0：Not allow, 1：Allow	N/A	RW
9	Inverter's power	1259	S16		0.01kW	RW
10	Force charge and discharge power	125A	S16	Total battery port power	0.01kW	RW
11	Spontaneous self use of excess power for grid connection	125B	U16	0：Not allow, 1：Allow	N/A	RW
12	Reserve	125C - 125D	N/A		N/A	N/A

5.4Data Return Area

5.4.1Italian Automatic Testing Data Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	One level over-voltage protection value	1304	U16		0.1V	RO
2	One level over-voltage protection time	1305	U16		1ms	RO
3	One level over-voltage protection value test result	1306	U16		0.1V	RO
4	One level over-voltage protection time test result	1307	U16		1ms	RO
5	Two level over-voltage protection value	1308	U16		0.1V	RO
6	Two level over-voltage protection time	1309	U16		1ms	RO
7	Two level over-voltage protection value test result	130A	U16		0.1V	RO
8	Two level over-voltage protection time test result	130B	U16		1ms	RO
9	One level under-voltage protection value	130C	U16		0.1V	RO
10	One level under-voltage protection time	130D	U16		1ms	RO
11	One level under-voltage protection value test result	130E	U16		0.1V	RO
12	One level under-voltage protection time test result	130F	U16		1ms	RO
13	Two level under-voltage protection value	1310	U16		0.1V	RO
14	Two level under-voltage protection time	1311	U16		1ms	RO
15	Two level under-voltage protection value test result	1312	U16		0.1V	RO
16	Two level under-voltage protection time test result	1313	U16		1ms	RO
17	One level over-frequency protection value	1314	U16		0.01Hz	RO
18	One level over-frequency protection time	1315	U16		1ms	RO
19	One level over-frequency protection value test result	1316	U16		0.01Hz	RO
20	One level over-frequency protection time test result	1317	U16		1ms	RO
21	Two level over-frequency protection value	1318	U16		0.01Hz	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
22	Two level over-frequency protection time	1319	U16		1ms	RO
23	Two level over-frequency protection value test result	131A	U16		0.01Hz	RO
24	Two level over-frequency protection time test result	131B	U16		1ms	RO
25	One level under-frequency protection value	131C	U16		0.01Hz	RO
26	One level under-frequency protection time	131D	U16		1ms	RO
27	One level under-frequency protection value test result	131E	U16		0.01Hz	RO
28	One level under-frequency protection time test result	131F	U16		1ms	RO
29	Two level under-frequency protection value	1320	U16		0.01Hz	RO
30	Two level under-frequency protection time	1321	U16		1ms	RO
31	Two level under-frequency protection value test result	1322	U16		0.01Hz	RO
32	Two level under-frequency protection time test result	1323	U16		1ms	RO
33	Italian automatic test result 33	1324	U16		N/A	RO
34	Italian automatic test result 34	1325	U16		N/A	RO
35	Italian automatic test result 35	1326	U16		N/A	RO
36	Italian automatic test result 36	1327	U16		N/A	RO
37	Italian automatic test result 37	1328	U16		N/A	RO
38	Italian automatic test result 38	1329	U16		N/A	RO
39	Italian automatic test result 39	132A	U16		N/A	RO
40	Italian automatic test result 40	132B	U16		N/A	RO
41	Italian automatic test result 41	132C	U16		N/A	RO
42	Italian automatic test result 42	132D	U16		N/A	RO
43	Italian automatic test result 43	132E	U16		N/A	RO
44	Italian automatic test result 44	132F	U16		N/A	RO
45	Italian automatic test result 45	1330	U16		N/A	RO
46	Italian automatic test result 46	1331	U16		N/A	RO
47	Italian automatic test result 47	1332	U16		N/A	RO
48	Italian automatic test result 48	1333	U16		N/A	RO
49	Reserve	1334 - 133F	N/A		N/A	N/A

5.4.2IV Scan Data 1 Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	IV curve scan voltage 1	1344	U16		0.1V	RO
2	IV curve scan current 1	1345	U16		0.01A	RO
3	IV curve scan voltage 2	1346	U16		0.1V	RO
4	IV curve scan current 2	1347	U16		0.01A	RO
5	IV curve scan voltage 3	1348	U16		0.1V	RO
6	IV curve scan current 3	1349	U16		0.01A	RO
7	...	...	...	...	...	...
8	IV curve scan voltage 28	137A	U16		0.1V	RO
9	IV curve scan current 28	137B	U16		0.01A	RO
10	IV curve scan voltage 29	137C	U16		0.1V	RO
11	IV curve scan current 29	137D	U16		0.01A	RO
12	IV curve scan voltage 30	137E	U16		0.1V	RO
13	IV curve scan current 30	137F	U16		0.01A	RO

Note: This block of data contains a total of 30 IV scan data items, arranged in address order.

5.4.3IV Scan Data 2 Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	IV curve scan voltage 31	1384	U16		0.1V	RO
2	IV curve scan current 31	1385	U16		0.01A	RO
3	IV curve scan voltage 32	1386	U16		0.1V	RO
4	IV curve scan current 32	1387	U16		0.01A	RO
5	IV curve scan voltage 33	1388	U16		0.1V	RO
6	IV curve scan current 33	1389	U16		0.01A	RO
7	...	...	...	...	...	...
8	IV curve scan voltage 58	13BA	U16		0.1V	RO
9	IV curve scan current 58	13BB	U16		0.01A	RO
10	IV curve scan voltage 59	13BC	U16		0.1V	RO
11	IV curve scan current 59	13BD	U16		0.01A	RO
12	IV curve scan voltage 60	13BE	U16		0.1V	RO
13	IV curve scan current 60	13BF	U16		0.01A	RO

Note: This block of data contains a total of 30 IV scan data items, arranged in address order.

5.4.4IV Scan Data 3 Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	IV curve scan voltage 61	13C4	U16		0.1V	RO
2	IV curve scan current 61	13C5	U16		0.01A	RO
3	IV curve scan voltage 62	13C6	U16		0.1V	RO
4	IV curve scan current 62	13C7	U16		0.01A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
5	IV curve scan voltage 63	13C8	U16		0.1V	RO
6	IV curve scan current 63	13C9	U16		0.01A	RO
7	...	...	...	...	...	...
8	IV curve scan voltage 88	13FA	U16		0.1V	RO
9	IV curve scan current 88	13FB	U16		0.01A	RO
10	IV curve scan voltage 89	13FC	U16		0.1V	RO
11	IV curve scan current 89	13FD	U16		0.01A	RO
12	IV curve scan voltage 90	13FE	U16		0.1V	RO
13	IV curve scan current 90	13FF	U16		0.01A	RO

Note: This block of data contains a total of 30 IV scan data items, arranged in address order.

3.4.5IV Scan Data 4 Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	IV curve scan voltage 91	1404	U16		0.1V	RO
2	IV curve scan current 91	1405	U16		0.01A	RO
3	IV curve scan voltage 92	1406	U16		0.1V	RO
4	IV curve scan current 92	1407	U16		0.01A	RO
5	IV curve scan voltage 92	1408	U16		0.1V	RO
6	IV curve scan current 92	1409	U16		0.01A	RO
7	...	...	...	...	...	...
8	IV curve scan voltage 118	143A	U16		0.1V	RO
9	IV curve scan current 118	143B	U16		0.01A	RO
10	IV curve scan voltage 119	143C	U16		0.1V	RO
11	IV curve scan current 119	143D	U16		0.01A	RO
12	IV curve scan voltage 120	143E	U16		0.1V	RO
13	IV curve scan current 120	143F	U16		0.01A	RO

Note: This block of data contains a total of 30 IV scan data items, arranged in address order.

5.4.6IV Scan Data 5 Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	IV curve scan voltage 121	1444	U16		0.1V	RO
2	IV curve scan current 121	1445	U16		0.01A	RO
3	IV curve scan voltage 122	1446	U16		0.1V	RO
4	IV curve scan current 122	1447	U16		0.01A	RO
5	IV curve scan voltage 123	1448	U16		0.1V	RO
6	IV curve scan current 123	1449	U16		0.01A	RO
7	...	...	...	...	...	...
8	IV curve scan voltage 148	147A	U16		0.1V	RO
9	IV curve scan current 148	147B	U16		0.01A	RO
10	IV curve scan voltage 149	147C	U16		0.1V	RO
11	IV curve scan current 149	147D	U16		0.01A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
12	IV curve scan voltage 150	147E	U16		0.1V	RO
13	IV curve scan current 150	147F	U16		0.01A	RO

Note: This block of data contains a total of 30 IV scan data items, arranged in address order.

5.4.7Historical Events Data Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	History event ID 1st most recent	1480	U16		N/A	RO
2	The year and month of the most recent 1st historical event	1481	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
3	The day and hour of the most recent 1st historical event	1482	U16	High byte indicates day, range：[1~31] Low byte indicate hour,range：[0~23]	N/A	RO
4	The minute and second of the most recent 1st historical event	1483	U16	High byte indicates minute, range：[0~59] Low byte indicate second,range：[0~59]	N/A	RO
5	History event ID 2nd most recent	1484	U16		N/A	RO
6	The year and month of the most recent 2nd historical event	1485	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
7	The day and hour of the most recent 2nd historical event	1486	U16	High byte indicates day, range：[1~31] Low byte indicate hour,range：[0~23]	N/A	RO
8	The minute and second of the most recent 2nd historical event	1487	U16	High byte indicates minute, range：[0~59] Low byte indicate second,range：[0~59]	N/A	RO
9	...	...	...	...	...	...
10	History event ID 99th most recent	1608	U16		N/A	RO
11	The year and month of the most recent 99th historical event	1609	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
12	The day and hour of the most recent 99th historical event	160A	U16	High byte indicates day, range：[1~31] Low byte indicate hour,range：[0~23]	N/A	RO
13	The minute and second of the most recent 99th historical event	160B	U16	High byte indicates minute, range：[0~59] Low byte indicate second,range：[0~59]	N/A	RO
14	History event ID 100th most recent	160C	U16		N/A	RO
15	The year and month of the most recent 100th historical event	160D	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
16	The day and hour of the most recent 100th historical event	160E	U16	High byte indicates day, range：[1~31] Low byte indicate hour,range：[0~23]	N/A	RO
17	The minute and second of the most recent 100th historical event	160F	U16	High byte indicates minute, range：[0~59] Low byte indicate second,range：[0~59]	N/A	RO

Note: This block of data contains a total of 100 historical event data, arranged in address order. For detailed explanations of historical event IDs, please refer to the fault description corresponding to the fault ID in the "Description of system fault information"。

5.4.8Historical Energy Data Address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Historical energy statistics 1st	1610 - 1611	U32		0.01kW	R
2	Historical energy statistics 2nd	1612 - 1613	U32		0.01kW	R
3	Historical energy statistics 3rd	1614 - 1615	U32		0.01kW	R
4	Historical energy statistics 4th	1616 - 1617	U32		0.01kW	R
5	Historical energy statistics 5th	1618 - 1619	U32		0.01kW	R
6	...	...	...	...	...	...
7	Historical energy statistics 36th	1656 - 1657	U32		0.01kW	R
8	Historical energy statistics 37th	1658 - 1659	U32		0.01kW	R
9	Historical energy statistics 38th	165A - 165B	U32		0.01kW	R
10	Historical energy statistics 39th	165C - 165D	U32		0.01kW	R
11	Historical energy statistics 40th	165E - 165F	U32		0.01kW	R

Note: This block of data contains a total of 40 historical energy record data, arranged in address order, and the precision of the read back register value multiplied by the actual recorded value. Before reading, it is necessary to set the values from 0x1030 to 0x1033. If it is a year, the first 12 pieces of data represent the electricity consumption data for the past 12 months; If it is a month, then the first 31 pieces of data represent the electricity consumption data for the 31 days.

5.4.9Core functional configuration parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Safety regulation country	2004	U16	High byte indicate country code Low byte indicate region code	N/A	RW
2	Safety regulation version	2005	U16		N/A	RW
3	*User data reset	2006	U16	Bit0: Clears the power level Bit1: Clears the event log Bit2: All function configuration parameters return to their default values Writing 1 to the corresponding bit indicates executing the operation.	N/A	RW
4	Safety country name	2007 - 200B	ASCII		N/A	RW
5	Reserve	200C			N/A	N/A
6	Pair up communication protocol settings	200D	U16	0x0000:modbus_RTU protocol; 0x0001:modbus_Korea protocol; 0x0002:sofar_sgp_normal standard version protocol; 0x0003:sofar_sgp_advance advanced protocol; 0x0004:sunspec protocol; 0x0005:solarmax protocol;	N/A	RW
7	Safety country(Safety database)	200E	U16	High byte indicate country code Low byte indicate region code	N/A	RW
8	Safety version(Safety database)	200F	U16		N/A	RW
9	Reserve	2010 - 2033	U16		N/A	N/A

Index	Name	Address（HEX）	Type	Description	Unit	Properties
10	Remote download control register(extension)	2034	U16	High byte writing to 0xA5 is valid The low byte contains the combination of chips to be upgraded and each bit is explained as follows: Bit0: DCDC-ARM; Bit1: DCDC-DSP;	N/A	WO
11	Remote upgrade control register(extension)	2035	U16	High byte writing to 0xA5 is valid. The low byte contains the combination of chips to be upgraded and each bit is explained as follows: Bit0: DCDC-ARM; Bit1: DCDC-DSP;	N/A	WO
12	Local upgrade control(extension)1	2036	U16	High byte writing to 0xA5 is valid. The low byte contains the combination of chips to be upgraded and each bit is explained as follows: Bit0: DCDC-ARM; Bit1: DCDC-DSP;	N/A	WO
13	Local Upgrade Control(extension)2	2037	U16	High bytes: A single BIN file is upgraded to a write 0xA5 The sofar package file is written to 0x80 Low bytes: Low bytes contain combinations of chips that need to be upgraded, see Chip Corner Color and file type encoding" document, if it is a sofar file is fixed Write to 0xFF	N/A	WO
14	Current upgrade progress(extension)	2038	U16	The high byte indicates the target file type for the current upgrade: 0x00:APP: 0x01: CORE; 0x02: Safety documents; 0xff: Sofar bag Low bytes indicate the current upgrade progress, Range[0, 100]	N/A	RO
15	Total number of upgrade targets	2039	U16	The value of this register indicates the total number of upgrade targets	N/A	RO
16	Upgrade result bit table 1	203A	U16	A bit value of 0 indicates an incomplete or failed upgrade; A bit value of 1 indicates a successful upgrade. The lowest bit indicates an upgrade target numbered 1. The highest bit indicates an upgrade target numbered 16.	N/A	RO
17	Upgrade result bit table 2	203B	U16	A bit value of 0 indicates an incomplete or failed upgrade; A bit value of 1 indicates a successful upgrade.	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				The lowest bit indicates an upgrade target numbered 17. The highest bit indicates the upgrade target numbered 32.		
18	Remote Download Control Register	203C	U16	High byte writing to 0xA5 is valid. The low byte contains the chip combination that needs to be upgraded, and the explanation for each bit is as follows： Bit0：Communication board； Bit1：Master DSP； Bit2：Slave DSP； Bit3：Fuse； Bit4：BMS； Bit5：PCU； Bit6：BDU； Bit7：AFCI；	N/A	WO
19	Remote Upgrade Control Register	203D	U16	High Byte: Single BIN file upgrade writes 0xA5 sofar packaged files are written to 0x80 Low byte: The low byte contains the combination of chips to be upgraded, refer to the document Chip Roles and File Type Codes", if it is a sofar file fixed write 0xFF"	N/A	WO
20	Local Upgrade Control Register	203E	U16	High byte writing to 0xA5 is valid. The low byte contains the chip combination that needs to be upgraded, and the explanation for each bit is as follows： Bit0：Communication board； Bit1：Master DSP； Bit2：Slave DSP； Bit3：Fuse； Bit4：BMS； Bit5：PCU； Bit6：BDU； Bit7：AFCI；	N/A	WO
21	Current upgrade progress	203F	U16	The high byte indicates the current upgrade target chip： 0x00: not in upgrade status; 0x01: master DSP upgrade; 0x02: slave DSP upgrade; 0x03: ARM upgrade; 0x04: FUSE upgrade; 0x05: BMS upgrade; 0x06: PCU upgrade; 0x07: BDU upgrade; 0x28: DC/DC ARM upgrade;	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				0x29: DC/DC DSP upgrade; 0x42: AFCI upgrade Low byte indicates the upgrade progress, Range[0, 100]		
22	Timed upgrade time(year)	2040	U16	The value of the time register is all 0XFF to cancel the timed upgrade	N/A	RW
23	Timed upgrade time(month)	2041	U16		N/A	RW
24	Timed upgrade time(day)	2042	U16		N/A	RW
25	Timed upgrade time(hour)	2043	U16		N/A	RW
26	Timed upgrade time(minute)	2044	U16		N/A	RW
27	Timed upgrade time(second)	2045	U16		N/A	RW
28	Upgrade result	2046	U16	Bit0~Bit7: The number of devices upgraded this time Bit8~Bit15: Whether the result of this upgrade is completed 0xA5：completed Other values: results are being counted The value of this field is 0xA5 when the upgrade result of register 0x2047 is read. This register will reset after receiving the upgrade command.	N/A	RO
29	Upgrade result 2	2047	U16	Number of successful device upgrades This register will reset after receiving the upgrade command.	N/A	RO
30	Upgrade result query control register	2048	U16	The APP and monitoring platform write the code corresponding to the upgrade object to be queried to this register. Write to this register and then retrieve the upgrade result from registers 0x2046 and 0x2047	N/A	WO
31	Current upgrade progress of the safety regulations file.	2049	U16	High byte indicates upgrade status： 0x00: Not in upgrade 0x01: Upgrading 0x02: Upgrade failure 0x03: Upgrade success Other: Invalid Low byte indicates the upgrade progress, Range[0, 100]	N/A	RO
32	Current upgrade progress of the sofar pack file	204A	U16		N/A	RO
33	Current upgrade progress of core	204B	U16		N/A	RO
34	Current upgrade progress of APP	204C	U16		N/A	RO
35	Current upgrade progress of ARM	204D	U16		N/A	RO
36	Current upgrade progress of master DSP	204E	U16		N/A	RO
37	Current upgrade progress of slave DSP	204F	U16		N/A	RO
38	Current upgrade progress of BMS	2050	U16		N/A	RO
39	Current upgrade progress of FUSE	2051	U16		N/A	RO
40	Current upgrade progress of PCU	2052	U16		N/A	RO
41	Current upgrade progress of BDU	2053	U16		N/A	RO
42	Current upgrade progress of DC/DC ARM	2054	U16		N/A	RO
43	Current upgrade progress of DC/DC	2055	U16		N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
	DSP					
44	Current upgrade progress of AFCI	2056	U16		N/A	RO
45	Reserve	2057 - 207F	N/A		N/A	N/A

Note: The 0x2004 and 0x2005 registers only affect the display of security country and version, and the actual security parameters are not affected by these two registers. If you need to modify the security parameters, please modify the 0x200E and 0x200F registers.

5.5Parameter Configuration Area 2

5.5.1System network parameter configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Network card MAC address byte 1~2	2504	U16	For example, if the MAC address is DC-91-66-A2-AE-25, then the register value of 0x0066 is 0xDC91, the register value of 0x0067 is 0x66A2, and the register value of 0x0068 is 0xAE25	N/A	RO
2	Network card MAC address byte 3~4	2505	U16		N/A	RO
3	Network card MAC address byte 5~6	2506	U16		N/A	RO
4	IP allocation method	2507	U16	0：Static IP, 1：DHCP allocation	N/A	RW
5	Device IP address	2508 -- 2509	U32	If the IP address is 192.168.1.25, then The value of the 0x2089 register is: 0xC0A8 The value of the 0x208A register is 0x0119	N/A	RW
6	Device gateway address	250A -- 250B	U32	If the gateway address is 192.168.1.1, then The value of register 0x208B is: 0xC0A8 The value of the 0x208C register is 0x0101	N/A	RW
7	Device Mask Address	250C -- 250D	U32	If the mask is 255.255.255.0, then The value of the 0x208D register is 0xFFFF The value of the 0x208E register is 0xFF00	N/A	RW
8	Reserve	250E -- 2511	N/A		N/A	N/A
9	Modbus-TCP server port	2512	U16	Range：[1, 65535]	N/A	RW

5.6Port Temperature Detection Data Area

5.6.1System information address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Software version	4A00	ASCII	e.g: The value 0x3056 read from register 0x4A00 means "0V". The value 0x3130 read from register 0x4A01 means "10". The value 0x3030 read from 0x4A02 means "00".	N/A	RO
		4A01	ASCII		N/A	RO
		4A02	ASCII		N/A	RO
		4A03	ASCII		N/A	RO
		4A04	ASCII		N/A	RO
		4A05	ASCII		N/A	RO
		4A06	ASCII		N/A	RO
		4A07	ASCII		N/A	RO
		4A08	ASCII		N/A	RO
		4A09	ASCII		N/A	RO
		4A0A	ASCII		N/A	RO
2	Hardware version	4A0B	ASCII	As version number "V101", register value: 0x5631 3031	N/A	RO
		4A0C	ASCII		N/A	RO
3	Reserve	4A0D -- 4A3F	N/A		N/A	N/A

Note: The "Software version" and "Hardware version" need to be determined based on the value read back from the "inverter port temperature detection status" register address: 0x4A40. If the returned value is 2, it indicates that the version information is valid.

5.6.2Real-time data address1

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Inverter port temperature detection status	4A40	U16	0: Inverter does not support port temperature detection function 1: Control board DSP integrated port temperature detection circuit 2: External port temperature detection board 3: Control board DSP integrated port temperature detection circuit acquisition failed 4: External temperature detection board failed to collect data 5: Control board DSP integrated port temperature detection circuit loses power at night	N/A	RO
2	Fault 1	4A41	U16	More information refer “Description of port temperature detection fault”	N/A	RO
3	Fault 2	4A42	U16		N/A	RO
4	Fault 3	4A43	U16		N/A	RO
5	Fault 4	4A44	U16		N/A	RO
6	Fault 5	4A45	U16		N/A	RO
7	Fault 6	4A46	U16		N/A	RO
8	Fault 7	4A47	U16		N/A	RO
9	Fault 8	4A48	U16		N/A	RO
10	Fault 9	4A49	U16		N/A	RO
11	Fault 10	4A4A	U16		N/A	RO
12	Fault 11	4A4B	U16		N/A	RO
13	Fault 12	4A4C	U16		N/A	RO
14	Fault 13	4A4D	U16		N/A	RO
15	Fault 14	4A4E	U16		N/A	RO
16	Fault 15	4A4F	U16		N/A	RO
17	Fault 16	4A50	U16		N/A	RO
18	Reserve	4A51 -- 4A7F	N/A		N/A	N/A

5.6.3Real-time data address2

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	R-phase temperature	4A80	S16		0.1℃	RO
2	S-phase temperature	4A81	S16		0.1℃	RO
3	T-phase temperature	4A82	S16		0.1℃	RO
4	PV1_1 input temperature	4A83	S16		0.1℃	RO
5	PV1_1 output temperature	4A84	S16		0.1℃	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
6	PV1_2 input temperature	4A85	S16		0.1℃	RO
7	PV1_2 output temperature	4A86	S16		0.1℃	RO
8	PV1_3 input temperature	4A87	S16		0.1℃	RO
9	PV1_3 output temperature	4A88	S16		0.1℃	RO
10	PV1_4 input temperature	4A89	S16		0.1℃	RO
11	PV1_4 output temperature	4A8A	S16		0.1℃	RO
12	PV2_1 input temperature	4A8B	S16		0.1℃	RO
13	PV2_1 output temperature	4A8C	S16		0.1℃	RO
14	PV2_2 input temperature	4A8D	S16		0.1℃	RO
15	PV2_2 output temperature	4A8E	S16		0.1℃	RO
16	PV2_3 input temperature	4A8F	S16		0.1℃	RO
17	PV2_3 output temperature	4A90	S16		0.1℃	RO
18	PV2_4 input temperature	4A91	S16		0.1℃	RO
19	PV2_4 output temperature	4A92	S16		0.1℃	RO
20	PV3_1 input temperature	4A93	S16		0.1℃	RO
21	PV3_1 output temperature	4A94	S16		0.1℃	RO
22	PV3_2 input temperature	4A95	S16		0.1℃	RO
23	PV3_2 output temperature	4A96	S16		0.1℃	RO
24	PV3_3 input temperature	4A97	S16		0.1℃	RO
25	PV3_3 output temperature	4A98	S16		0.1℃	RO
26	PV3_4 input temperature	4A99	S16		0.1℃	RO
27	PV3_4 output temperature	4A9A	S16		0.1℃	RO
28	PV4_1 input temperature	4A9B	S16		0.1℃	RO
29	PV4_1 output temperature	4A9C	S16		0.1℃	RO
30	PV4_2 input temperature	4A9D	S16		0.1℃	RO
31	PV4_2 output temperature	4A9E	S16		0.1℃	RO
32	PV4_3 input temperature	4A9F	S16		0.1℃	RO
33	PV4_3 output temperature	4AA0	S16		0.1℃	RO
34	PV4_4 input temperature	4AA1	S16		0.1℃	RO
35	PV4_4 output temperature	4AA2	S16		0.1℃	RO
36	PV5_1 input temperature	4AA3	S16		0.1℃	RO
37	PV5_1 output temperature	4AA4	S16		0.1℃	RO
38	PV5_2 input temperature	4AA5	S16		0.1℃	RO
39	PV5_2 output temperature	4AA6	S16		0.1℃	RO
40	PV5_3 input temperature	4AA7	S16		0.1℃	RO
41	PV5_3 output temperature	4AA8	S16		0.1℃	RO
42	PV5_4 input temperature	4AA9	S16		0.1℃	RO
43	PV5_4 output temperature	4AAA	S16		0.1℃	RO
44	PV6_1 input temperature	4AAB	S16		0.1℃	RO
45	PV6_1 output temperature	4AAC	S16		0.1℃	RO
46	PV6_2 input temperature	4AAD	S16		0.1℃	RO
47	PV6_2 output temperature	4AAE	S16		0.1℃	RO
48	PV6_3 input temperature	4AAF	S16		0.1℃	RO
49	PV6_3 output temperature	4AB0	S16		0.1℃	RO
50	PV6_4 input temperature	4AB1	S16		0.1℃	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
51	PV6_4 output temperature	4AB2	S16		0.1℃	RO
52	PV7_1 input temperature	4AB3	S16		0.1℃	RO
53	PV7_1 output temperature	4AB4	S16		0.1℃	RO
54	PV7_2 input temperature	4AB5	S16		0.1℃	RO
55	PV7_2 output temperature	4AB6	S16		0.1℃	RO
56	PV7_3 input temperature	4AB7	S16		0.1℃	RO
57	PV7_3 output temperature	4AB8	S16		0.1℃	RO
58	PV7_4 input temperature	4AB9	S16		0.1℃	RO
59	PV7_4 output temperature	4ABA	S16		0.1℃	RO
60	PV8_1 input temperature	4ABB	S16		0.1℃	RO
61	PV8_1 output temperature	4ABC	S16		0.1℃	RO
62	PV8_2 input temperature	4ABD	S16		0.1℃	RO
63	PV8_2 output temperature	4ABE	S16		0.1℃	RO
64	PV8_3 input temperature	4ABF	S16		0.1℃	RO
65	PV8_3 output temperature	4AC0	S16		0.1℃	RO
66	PV8_4 input temperature	4AC1	S16		0.1℃	RO
67	PV8_4 output temperature	4AC2	S16		0.1℃	RO
68	Reserve	4AC3 -- 4ADF	N/A		N/A	N/A

5.6.4Parameter configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
10	Clear port temperature alarm fault	4AE0	U16	0: Dummy instruction 1: Clear alarm fault Other: invalid	N/A	RW
11	Set the sampling period for port temperature	4AE1	U16	Range: [50, 1000], default value: 100	1ms	RW
12	PV port temperature protection threshold	4AE2	U16		0.1℃	RW
13	AC port temperature protection threshold	4AE3	U16		0.1℃	RW
14	PV port temperature slope threshold	4AE4	U16		0.1℃	RW
15	AC port temperature slope threshold	4AE5	U16		0.1℃	RW
16	Port NTC fault detection time	4AE6	U16	Range：[1000, 65535]	1ms	RW
17	Port temperature fault detection time	4AE7	U16	Range：[1000, 65535]	1ms	RW
18	Port temperature secondary fault detection time	4AE8	U16	Range：[1000, 65535]	1ms	RW
19	Port temperature slope fault detection time	4AE9	U16	Range：[1000, 65535]	1ms	RW
20	Reserve	4AEA -- 4AEF	N/A		N/A	RW
21	Temperature fault enable control 1	4AF0	U16	Used in conjunction with "Fault 1-16",	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
22	Temperature fault enable control 2	4AF1	U16	sequentially indicating terminal temperature fault and terminal NTC fault enable. For example, if the read value from 0x4AF0 is 0x0007, it indicates that the fault alarm for bit 0, bit 1, and bit 2 of fault information 1 (register 0x4A41) is enabled	N/A	RW
23	Temperature fault enable control 3	4AF2	U16		N/A	RW
24	Temperature fault enable control 4	4AF3	U16		N/A	RW
25	Temperature fault enable control 5	4AF4	U16		N/A	RW
26	Temperature fault enable control 6	4AF5	U16		N/A	RW
27	Temperature fault enable control 7	4AF6	U16		N/A	RW
28	Temperature fault enable control 8	4AF7	U16		N/A	RW
29	Temperature fault enable control 9	4AF8	U16		N/A	RW
30	Temperature fault enable control 10	4AF9	U16		N/A	RW
31	Temperature fault enable control 11	4AFA	U16		N/A	RW
32	Temperature fault enable control 12	4AFB	U16		N/A	RW
33	Temperature fault enable control 13	4AFC	U16		N/A	RW
34	Temperature fault enable control 14	4AFD	U16		N/A	RW
35	Temperature fault enable control 15	4AFE	U16		N/A	RW
36	Temperature fault enable control 16	4AFF	U16		N/A	RW

5.7DCDC Data Area

5.7.1DCDC System information address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
4	ARM to inverter 485 protocol version	5004	U16		N/A	RO
5	Reserve	5005	N/A		N/A	N/A
6	ARM's internal protocol version for DSP	5006	U16		N/A	RO
7	DSP to ARM Internal Protocol Version	5007	U16		N/A	RO
8	ARM's CAN protocol version for BMS	5008	U16		N/A	RO
9	Reserve	5009	N/A		N/A	N/A
10	Protocol version between DCDC modules	500A	U16		N/A	RO
11	ARM app software version	500B	ASCII	As version number "V1.2.3-SF.RC1", register value: 0x5346 0x2E52 0x4331 0x0000, which is the string "SF.RC1"	N/A	RO
		500C	ASCII		N/A	RO
		500D	ASCII		N/A	RO
		500E	ASCII		N/A	RO
12	Control board hardware version	500F	ASCII	As version number "V101", register value: 0x5631 3031	N/A	RO
		5010	ASCII		N/A	RO
13	ARM bootloader software version	5011	ASCII	As version number "V1.2.3-SF.RC1", register value: 0x5346 0x2E52 0x4331 0x0000, which is the string "SF.RC1"	N/A	RO
		5012	ASCII		N/A	RO
		5013	ASCII		N/A	RO
		5014	ASCII		N/A	RO
14	ARM chip code	5015	ASCII	Example: 0x4730 represents "G0"	N/A	RO
15	DSP app software version	5016	ASCII	As version number "V1.2.3-SF.RC1", register value: 0x5346 0x2E52 0x4331 0x0000, which is the string "SF.RC1"	N/A	RO
		5017	ASCII		N/A	RO
		5018	ASCII		N/A	RO
		5019	ASCII		N/A	RO
16	Power board hardware version	501A	ASCII	As version number "V101", register value: 0x5631 3031	N/A	RO
		501B	ASCII		N/A	RO
17	DSP bootloader software version	501C	ASCII	As version number "V1.2.3-SF.RC1", register value: 0x5346 0x2E52 0x4331 0x0000, which is the string "SF.RC1"	N/A	RO
		501D	ASCII		N/A	RO
		501E	ASCII		N/A	RO
		501F	ASCII		N/A	RO
18	DSP chip code	5020	ASCII	Example: 0x4730 represents "G0"	N/A	RO
19	ARM APP version number supplement	5021	U16	For example, version number "V1.2.3-SF.RC1", register value: 0x0001 0x0203	N/A	RO
		5022	U16		N/A	RO
20	ARM BOOT version number supplement	5023	U16	For example, version number "V1.2.3-SF.RC1", register value: 0x0001 0x0203	N/A	RO
		5024	U16		N/A	RO
21	DSP APP version number supplement	5025	U16	For example, version number "V1.2.3-SF.RC1", register value: 0x0001 0x0203	N/A	RO
		5026	U16		N/A	RO
22	DSP BOOT version number supplement	5027	U16	For example, version number "V1.2.3-SF.RC1", register value: 0x0001 0x0203	N/A	RO
		5028	U16		N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
23	Device SN number	5029 - 5032	ASCII	Like SN number "SD1021080KC223140001"	N/A	RO
24	Reserve	5033 - 503F	N/A		N/A	N/A

5.7.2DCDC Real-time data address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
69	Current total number of parallel DCDC modules	5044	U16	Range:[1, 4]	N/A	RO
70	Startup countdown	5045	U16	Range:[0, 65535]	1S	RO
71	System running state	5046	U16	0: Standby 1: Discharging 2: Charging	N/A	RO
72	Power limiting state	5047	U16	0: Unrestricted 1: Command limiting power 2: SOC limiting power 3: Over temperature limiting power 4: Boost to load reduction ratio 5: High voltage side over-voltage load reduction 6: Fan malfunction load reduction	N/A	RO
73	SOC balanced power percentage	5048	S16	Range:[-1000, 1000]	0.1%	R
74	DSP work state	5049	U16	0: wait 1: Check 2: Normal 3: Recover Fault 4: Permanent Fault	N/A	RO
75	Maximum speed of external fan	504A	U16		1r/min	RO
76	Actual speed of external fan	504B	U16		1r/min	RO
77	Maximum speed of internal fan	504C	U16		1r/min	RO
78	Actual speed of internal fan	504D	U16		1r/min	RO
79	Low voltage side bus voltage	504E	S16		0.1V	RO
80	Front end voltage of low-voltage side relay	504F	S16		0.1V	RO
81	Low voltage side current	5050	S16		0.1A	RO
82	Low voltage side power	5051	S16		0.01kW	RO
83	Low voltage side total current	5052	S16		0.1A	RO
84	Number of parallel branches	5053	U16		N/A	RO
85	Branch current 1	5054	S16		0.1A	RO
86	Branch current 2	5055	S16		0.1A	RO
87	Branch current 3	5056	S16		0.1A	RO
88	Branch current 4	5057	S16		0.1A	RO
89	Branch flying capacitor voltage 1	5058	S16		0.1V	RO
90	Branch flying capacitor voltage 2	5059	S16		0.1V	RO
91	Branch flying capacitor voltage 3	505A	S16		0.1V	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
92	Branch flying capacitor voltage 4	505B	S16		0.1V	RO
93	High voltage bus voltage	505C	S16		0.1V	RO
94	Front end voltage of high-voltage side relay	505D	S16		0.1V	RO
95	Insulation impedance	505E	U16		1kΩ	RO
96	12V Auxiliary voltage	505F	U16		0.1V	RO
97	Internal environmental temperature of the chassis	5060	S16		1℃	RO
98	Radiator temperature 1	5061	S16		1℃	RO
99	Radiator temperature 2	5062	S16		1℃	RO
100	Reserve	5063 - 5064	N/A		N/A	N/A
101	Fault 1	5065	U16	More information refer “Description of DCDC fault information”	N/A	RO
102	Fault 2	5066	U16		N/A	RO
103	Fault 3	5067	U16		N/A	RO
104	Fault 4	5068	U16		N/A	RO
105	Fault 5	5069	U16		N/A	RO
106	Fault 6	506A	U16		N/A	RO
107	Fault 7	506B	U16		N/A	RO
108	Fault 8	506C	U16		N/A	RO
109	Fault 9	506D	U16		N/A	RO
110	Fault 10	506E	U16		N/A	RO
111	State 1	506F	U16	Bit0：ARM Reset(State1) Others：Reserve(State2 ~ 5) Clear the grid connected parameters or real-time control of the inverter after writing them.	N/A	RO
112	State 2	5070	U16		N/A	RO
113	State 3	5071	U16		N/A	RO
114	State 4	5072	U16		N/A	RO
115	State 5	5073	U16		N/A	RO
116	Charging capacity	5074 - 5075	U32		1Wh	RO
117	Discharging capacity	5076 - 5077	U32		1Wh	RO
118	Reserve	5078 - 507F	N/A		N/A	N/A

5.7.3DCDC Parameter configuration address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
37	The year of system	5084	U16	Range:[0, 99], base on 2000 year	N/A	RW
38	The month of system	5085	U16	Range:[1, 12]	N/A	RW
39	The date of system	5086	U16	Range:[1, 31]	N/A	RW
40	The hour of system	5087	U16	Range:[0, 23]	N/A	RW
41	The minute of system	5088	U16	Range:[0, 59]	N/A	RW
42	The second of system	5089	U16	Range:[0, 59]	N/A	RW
43	External RS485 address	508A	U16	Range:[1, 247]	N/A	RW
44	RS485 Baud rate	508B	U16	1：2400bps 2：4800bps 3：9600bps(default)	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				4：19200bps 5：38400bps 6：57600bps 7：115200bps		
45	RS485 Stop bit	508C	U16	0: 1 bit stop bit (default) 2: 2-bit stop bit	N/A	RW
46	RS485 Parity bit	508D	U16	0: No parity/None (default) 1: Even parity/Even 2: Odd parity/Odd	N/A	RW
47	Restore factory settings with safety signs	508E	U16	Writing 0xDCDC takes effect, other invalid	N/A	RW
48	Restore factory settings	508F	U16	Write value of 1 is valid Bit0: reserve Bit1: Clear event records Bit2: Restore all functional configuration parameters to default values When reading, return the status of the last write operation: 0x0000: Successful 0x0001-0x00FF: In operation	N/A	RW
49	Reserve	5090 - 50BF	N/A		N/A	N/A

5.7.4DCDC Parameter control address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Parallel DCDC quantity setting	5144	U16	Range:[1, 4]	N/A	RW
2	DCDC encoding	5145	U16	Range:[0, 3]	N/A	RW
3	Power on waiting time	5146	U16		1s	RW
4	Rated voltage of power grid	5147	U16		0.1V	RW
5	Rated frequency of power grid	5148	U16		0.01Hz	RW
6	Number of phases in the power grid	5149	U16	1: Single phase, 3: Three phase	N/A	RW
7	Leakage current protection value	514A	U16		1mA	RW
8	Time for 0 power to rise to rated power	514B	U16		1ms	RW
9	Time for rated power to decrease to 0 power	514C	U16		1ms	RW
10	Charge and discharge power conversion time	514D	U16		1ms	RW
11	Maximum allowable charging current	514E	S16		0.1A	RW
12	Maximum allowable discharge current	514F	S16		0.1A	RW
13	Reserve	5050 - 505F	N/A		N/A	N/A
14	Remote power on/off	5160	U16	0x55: Power on	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
15	Forced power on/off	5161	U16	0xAA: Power off	N/A	RW
16	Percentage of fan load limit power	5162	U16	Range:[0, 100]	1%	RW
17	Model power level	5163	U16	1：5KW 5：25KW Others: Reserve	N/A	RW
18	Charge and discharge mode	5164	U16	0x00: Standby 0x01: Autonomous charging and discharging 0x02: Force Charging 0x03: Force Discharging	N/A	RW
19	Real time power enable	5165	U16	0x55: Enable 0xAA: Prohibit	N/A	RW
20	Real time output power percentage	5166	U16	Cooperate with charging and discharging modes, Range:[0, 100]	1%	RW
21	Charge and discharge current command	5167	U16	Cooperate with charging and discharging modes to configure charging and discharging currents.	N/A	RW
22	Reserve	5168 - 517E	N/A		N/A	N/A
23	Clear battery statistics	517F	U16	DCDC power statistics from 0	N/A	RW

5.7.5DCDC Historical events data address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	History event ID 1st most recent	5180	U16		N/A	RO
2	The year and month of the most recent 1st historical event	5181	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
3	The day and hour of the most recent 1st historical event	5182	U16	High byte indicates day, range：[1 ~ 31] Low byte indicate hour,range：[0 ~ 23]	N/A	RO
4	The minute and second of the most recent 1st historical event	5183	U16	High byte indicates minute, range：[0 ~ 59] Low byte indicate second,range：[0 ~ 59]	N/A	RO
5	History event ID 2nd most recent	5184	U16		N/A	RO
6	The year and month of the most recent 2nd historical event	5185	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
7	The day and hour of the most recent 2nd historical event	5186	U16	High byte indicates day, range：[1 ~ 31] Low byte indicate hour,range：[0 ~ 23]	N/A	RO
8	The minute and second of the most recent 2nd historical event	5187	U16	High byte indicates minute, range：[0 ~ 59] Low byte indicate second,range：[0 ~ 59]	N/A	RO
9	...	...	...	...	...	...
10	History event ID 99th most recent	5308	U16		N/A	RO
11	The year and month of the most recent 99th historical event	5309	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
12	The day and hour of the most recent 99th historical event	530A	U16	High byte indicates day, range：[1 ~ 31] Low byte indicate hour,range：[0 ~ 23]	N/A	RO
13	The minute and second of the most recent 99th historical event	530B	U16	High byte indicates minute, range：[0 ~ 59] Low byte indicate second,range：[0 ~ 59]	N/A	RO
14	History event ID 100th most recent	530C	U16		N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
15	The year and month of the most recent 100th historical event	530D	U16	High byte indicates year, range：[0, 99] Low byte indicate month,range：[1, 12]	N/A	RO
16	The day and hour of the most recent 100th historical event	530E	U16	High byte indicates day, range：[1 ~ 31] Low byte indicate hour,range：[0 ~ 23]	N/A	RO
17	The minute and second of the most recent 100th historical event	530F	U16	High byte indicates minute, range：[0 ~ 59] Low byte indicate second,range：[0 ~ 59]	N/A	RO

Note: This block of data contains a total of 100 historical event data, arranged in address order. For detailed explanations of historical event IDs, please refer to the fault description corresponding to the fault ID in the "Description of DCDC fault information".

5.8PCU Data Area

5.8.1PCU Information data address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Total number of PCU online	6004	U16		N/A	RO
2	Reserve	6005 - 600F	N/A		N/A	N/A
3	PCU Code	6010	U16	Code of each PCU in a multi machine system	N/A	RO
4	PCU Serial number	6011 - 601A	ASCII	Example: 0x5043 represents "PC"	N/A	RO
5	PCU Software version number	601B - 601C	ASCII		N/A	RO
6	PCU Hardware version number	601D	U16		N/A	RO
7	PCU protocol CAN version number	601E	U16		N/A	RO
		601F	U16		N/A	RO
8	PCU Module state	6020	U16	0: Update 1: Check 2: Normal 3: Active 4: Fault 5: Permanent Fault	N/A	RO
9	Internal temperature of the chassis	6021	U16		1℃	RO
10	Radiator temperature 1	6022	S16		1℃	RO
11	Radiator temperature 2	6023	S16		1℃	RO
12	Alarm 1	6024	U16	More information refer "Description of PCU alarms information", example:Register 0x6024 reads back value 0x0002, indicating "Charging soft startup failed" alarm	N/A	RO
13	Alarm 2	6025	U16		N/A	RO
14	Alarm 3	6026	U16		N/A	RO
15	Alarm 4	6027	U16		N/A	RO
16	Fault 1	6028	U16	More information refer "Description of PCU fault information", example:Register 0x6028 read back value 0x0001 indicates a "Radiator 1 over temperature protection" fault	N/A	RO
17	Fault 2	6029	U16		N/A	RO
18	Fault 3	602A	U16		N/A	RO
19	Fault 4	602B	U16		N/A	RO
20	DC High-voltage side voltage	602C	U16	Example: 0x0FD3 represents 405.1V	0.1V	RO
21	DC Low-voltage side voltage	602D	U16	Example: 0x0214 represents 53.2V	0.1V	RO
22	DC Low-voltage side current	602E	S16	Charging is positive, discharging is negative Example: 0xFFCD represents discharge of 5.1A	0.1A	RO
23	DC Low-voltage side power	602F	S16	Charging is positive, discharging is negative Example: 0xF8C5 represents a discharge of 18.51kW	0.01kW	RO
24	Positive bus-bar voltage	6030	U16		0.1V	RO
25	Negative bus-bar voltage	6031	U16		0.1V	RO
26	Fly across capacitor voltage 1	6032	U16		0.1V	RO
27	Fly across capacitor voltage 2	6033	U16		0.1V	RO
28	Fly across capacitor voltage 3	6034	U16		0.1V	RO
29	Fly across capacitor voltage 4	6035	U16		0.1V	RO
30	Inductive current branch 1	6036	S16		0.1A	RO
31	Inductive current branch 2	6037	S16		0.1A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
32	Inductive current branch 3	6038	S16		0.1A	RO
33	Inductive current branch 4	6039	S16		0.1A	RO
34	Reserve	603A - 603F	N/A		N/A	N/A

5.8.2PCU Configuration parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	PCU query control word	6044	U16	Range:[1, 32] Note: Select the PCU module to be read or set by setting this register	N/A	RW
2	DC module remote power on/off	6045	U16	0: Power off 1: charging disabled 2: Discharge prohibited 3: Charging and discharging allowed	N/A	RW
3	Maximum charging current of the DC module	6046	U16	Range:[0, 65535]	0.1A	RW
4	Maximum discharging current of the DC module	6047	U16	Range:[0, 65535]	0.1A	RW
5	Calibration ratio for hvbus voltage on the high voltage side	6048	U16	Range:[950, 1050]	0.001p.u.	RW
6	High-voltage side pbus voltage calibration ratio	6049	U16	Range:[950, 1050]	0.001p.u.	RW
7	High-voltage side charge current calibration ratio	604A	U16	Range:[950, 1050]	0.001p.u.	RW
8	High-voltage side discharge current calibration ratio	604B	U16	Range:[950, 1050]	0.001p.u.	RW
9	Low voltage side voltage calibration ratio	604C	U16	Range:[950, 1050]	0.001p.u.	RW
10	Low voltage side charge current calibration ratio	604D	U16	Range:[950, 1050]	0.001p.u.	RW
11	Low voltage side discharge current calibration ratio	604E	U16	Range:[950, 1050]	0.001p.u.	RW
12	Reserve	604F - 607F	N/A		N/A	N/A

5.9BDU Data Area

5.9.1BDU Information data address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Total number of BDU online	6084	U16		N/A	RO
2	Reserve	6085 - 608F	N/A		N/A	N/A
3	BDU Code	6090	U16	Code starts from 1.	N/A	RO
4	BDU Serial number	6091 - 609A	ASCII		N/A	RO
5	BDU Version number	609B	ASCII	The lower 8 bits of the register hold the ASCII code. The default value for the official version is 'V'.	N/A	RO
6		609C	ASCII	Master Version Number	N/A	RO
7		609D	ASCII	Non-standard custom version number	N/A	RO
8		609E	ASCII	Software sub-version number	N/A	RO
9	BDU Fault information	609F	U16	Bit0: Reverse battery connection(Fault ID: 928) Bit1: Circuit failure(Fault ID: 929) Bit2 ~ Bit15：Reserve	N/A	RO
10	Number of cells in a single cluster	60A0	U16		N/A	RO
11	Total number of strings	60A1	U16		N/A	RO
12	Total number of battery circuits	60A2	U16		N/A	RO
13	Reserve	60A3 - 60BF				

5.9.2BDU Configuration parameter address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	BDU query control word	60C4	U16	Select the BDU module to be read or set by setting this register.	N/A	RW
2	Specify the route number	60C5	U16	Range:[1, 255]	N/A	RW
3	Rated power of string with specified number of channels	60C6	U16		1W	RO
4	Rated power of batteries for specified routes	60C7	U16		1W	RO
5	Reserve	60C8 - 60FF	N/A		N/A	N/A

5.10 BMS Data Area

5.10.1BMS Information data address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	BMS System clock	9004 - 9005	U32	Bit0~Bit5: Second, Range:[0, 59] Bit6~Bit11: Minute, Range:[0, 59] Bit12~Bit16: Hour, Range:[0~23] Bit17~Bit21: Day, Range:[1, 31] Bit22~Bit25: Month, Range:[1~12] Bit26~Bit31: Year, Range:[0~63], base on 2000 year	N/A	RO
2	CAN Protocol version number	9006	U16		N/A	RO
3	Manufacturer information	9007 - 900A	ASCII	Example: 0x414D represents "AM"	N/A	RO
4	BMS Version number	900B	U16		N/A	RO
5	Cell type	900C	U16		N/A	RO
6	Battery pack parameters	900D	U16	High 8 bits: number of battery pack cells in parallel Low 8 bits: number of strings of all battery packs	N/A	RO
7	Real-time remaining capacity	900E	U16	Range:[0, 100]	1%	RO
8	Total voltage	900F	U16		0.1V	R
9	Total current	9010	S16		0.1A	R
10	Average cell temperature	9011	S16		0.1℃	R
11	State of charge	9012	U16	Range:[0, 100]	1%	R
12	Health level	9013	U16	Range:[0, 100]	1%	R
13	BMS System protection information 0	9014	U16		N/A	RO
14	BMS System protection information 1	9015	U16		N/A	RO
15	BMS System alarm information 0	9016	U16		N/A	RO
16	BMS System alarm information 1	9017	U16		N/A	RO
17	BMS Version	9018	ASCII	The default value for the official version is' V '	N/A	RO
18	BMS Software major version	9019	U16	Software major version, Range:[0, 255]	N/A	RO
19	BMS Software non-standard version number	901A	U16	Software non-standard version number, Range:[0, 255]	N/A	RO
20	BMS Software minor version	901B	U16	Software minor version, Range:[0, 255]	N/A	RO
21	BMS System protection information 2	901C	U16		N/A	RO
22	BMS System protection information 3	901D	U16		N/A	RO
23	Reserve	901E - 901F	N/A		N/A	N/A
24	BMS Query control	9020	U16	Bit0~Bit7: Queried battery pack serial number, valid range 0~15, 0 means the 1st pack battery Bit8~Bit11: Queried battery pack serial number, valid range 0~15, 0 means the 1st pack	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				Bit12~Bit15： Query fault number, valid range 0~5, 0 means the latest fault		
25	BMS Function switch control	9021	U16	Battery function switch control word. Bit0~Bit1: Forced Sleep Control: Reserved for 0,1, 2 for hibernation Bit2~Bit3: Current running log saving control: 0-do not save, 1-do not act, 2-save Bit4~Bit7: Reserved； Bit8~Bit15: The address of the battery that needs to be controlled, FF is broadcast	N/A	RW
26	Battery online status table	9022	U16	Bit0 to bit15 represent the minimum to largest online addresses. Up to 16 batteries are online, 1 means online and 0 means offline. For example: 0x03 indicates that the BTS battery address 0x01 and 0x02 online	N/A	RO
27	Battery hibernation state table	9023	U16	The battery hibernation state table is only supported by BTS batteries for the time being, and all other batteries are 0. bit0 to bit15 indicate the minimum to maximum address sleep. 1 indicates hibernation. For example: 0x12 indicates that the 0x02 address and 0x11 address of the BTS battery are dormant	N/A	RO
28	BMS Serial number	9024 - 902D	U16		N/A	RO
29	System forced discharge command:	902E	U16	Bit0~Bit7 Available value: 0: Normal 1: Forced power replenishment 2: Forced standby 3: Forced filling Bit8: 0:Prohibit charge, 1:Allow charge Bit9: 0:Prohibit discharge, 1:Allow discharge	N/A	RO
30	BMS maximum discharge current limit value	902F	S16		1A	RO
31	BMS maximum charge current limit value	9030	S16		1A	RO
32	Reserve	9031 - 903F	N/A		N/A	N/A

5.10.2BMS Battery pack real-time data address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Real time data battery pack number	9044	U16	High 8 bits: battery pack serial number Low 8 bits: battery pack pack serial number	N/A	RO
2	Real-time data timestamp	9045 - 9046	U32	Bit0~Bit5: second, Range:[0, 59] Bit6~Bit11: minute, Range:[0, 59]	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				Bit12~Bit16: hour, Range:[0~23] Bit17~Bit21: day, Range:[1, 31] Bit22~Bit25: month, Range:[1~12] Bit26~Bit31: year, Range:[0~63], base on 2000year		
3	Battery pack serial number	9047 - 9050	ASCII		N/A	RO
4	Real time voltage of battery cell 1	9051	U16		0.001V	RO
5	Real time voltage of battery cell 2	9052	U16		0.001V	RO
6	...	...	...	...	...	...
7	Real time voltage of cell 23	9067	U16		0.001V	RO
8	Real time voltage of cell 24	9068	U16		0.001V	RO
9	Maximum voltage of battery cell	9069	U16		0.001V	RO
10	Minimum voltage of battery cell	906A	U16		0.001V	RO
11	Real time temperature of battery pack 1	906B	S16		0.1℃	RO
12	Real time temperature of battery pack 2	906C	S16		0.1℃	RO
13	Real time temperature of battery pack 3	906D	S16		0.1℃	RO
14	Real time temperature of battery pack 4	906E	S16		0.1℃	RO
15	MOS tube temperature	906F	S16		0.1℃	RO
16	Environmental temperature inside the battery pack	9070	S16		0.1℃	RO
17	Battery pack current	9071	S16		0.1A	RO
18	Remaining capacity	9072	U16		0.1Ah	RO
19	Full charge capacity	9073	U16		0.1Ah	RO
20	Number of cycles	9074	U16		1cycle	RO
21	Equilibrium state	9075	U16	Bit0~Bit15: Corresponding to the equilibrium state of the 1-16 battery cells Note: Bit 0 indicates off, 1 indicates on	N/A	RO
22	Alarm status	9076	U16		N/A	RO
23	Protection status	9077	U16		N/A	RO
24	Fault status	9078	U16		N/A	RO
25	Total voltage	9079	U16		0.1V	RO
26	SOC	907A	U16	Range:[0, 100]	1%	RO
27	The total number of battery packs in this group of batteries	907B	U16		N/A	RO
28	The number of battery cell strings in a single battery pack of this group	907C	U16		N/A	RO
29	Reserve	907D - 907F	N/A		N/A	N/A

Note: This block of data contains a total of 24 real-time cell voltage data, arranged in address order.

5.10.3BMS Battery pack fault data address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Real-time faulty battery pack number	9084	U16	Bit0-Bit7: Queried battery Pack serial number, valid range 0~15, 0 means the 1st battery pack Bit8-Bit11: Queried battery pack serial number, valid range 0~15, 0 means the 1st battery pack Bit12-Bit15: Queried fault serial number, valid range 0~5, 0 means the latest fault	N/A	RO
2	Fault data time stamp	9085 - 9086	U32	Bit0~Bit5: second, Range:[0, 59] Bit6~Bit11: minute, Range:[0, 59] Bit12~Bit16: hour, Range:[0~23] Bit17~Bit21: day, Range:[1, 31] Bit22~Bit25: month, Range:[1~12] Bit26~Bit31: year, Range:[0~63], base on 2000year	N/A	RO
3	Battery pack serial number	9087 - 9090	ASCII		N/A	RO
4	Real time voltage of battery cell 1	9091	U16		0.001V	RO
5	Real time voltage of battery cell 2	9092	U16		0.001V	RO
6	...	...	...	...	...	...
7	Real time voltage of cell 23	90A7	U16		0.001V	RO
8	Real time voltage of cell 24	90A8	U16		0.001V	RO
9	Maximum voltage of battery cell	90A9	U16		0.001V	RO
10	Minimum voltage of battery cell	90AA	U16		0.001V	RO
11	Battery pack Fault temperature 1	90AB	S16		0.1℃	RO
12	Battery pack Fault temperature 2	90AC	S16		0.1℃	RO
13	Battery pack Fault temperature 3	90AD	S16		0.1℃	RO
14	Battery pack Fault temperature 4	90AE	S16		0.1℃	RO
15	MOS tube temperature	90AF	S16		0.1℃	RO
16	Environmental temperature inside the battery pack	90B0	S16		0.1℃	RO
17	Battery pack current	90B1	S16		0.1A	RO
18	Remaining capacity	90B2	U16		0.1Ah	RO
19	Full charge capacity	90B3	U16		0.1Ah	RO
20	Total voltage	90B4	U16		0.01v	RO
21	Alarm status	90B5	U16		N/A	RO
22	Protection status	90B6	U16		N/A	RO
23	Fault status	90B7	U16		N/A	RO
24	The total number of battery packs in this group of batteries	90B8	U16		N/A	RO
25	The number of battery cell strings in a single battery pack of this group	90B9	U16		0.1V	RO
26	Cell information	90BA	U16	Bit0~Bit7: Cell serial number, 0 represents the first cell Bit8~Bit15: enumerated type, indicating the cell data type	N/A	RW

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				(0 indicates cell temperature, unit 0.1° C, I16) (1 indicates cell voltage, unit 0.001V, U16) (Other - Reserved)		
27	Cell data	90BB	U16	Data content, used in combination with 0x90BA	N/A	RO
28	Real time temperature of battery pack 5	90BC	S16		0.1℃	RO
29	Real time temperature of battery pack 6	90BD	S16		0.1℃	RO
30	Real time temperature of battery pack 7	90BE	S16		0.1℃	RO
31	Real time temperature of battery pack 8	90BF	S16		0.1℃	RO
32	Battery pack malfunction temperature 5	90C0	S16		0.1℃	RO
33	Battery pack malfunction temperature 6	90C1	S16		0.1℃	RO
34	Battery pack malfunction temperature 7	90C2	S16		0.1℃	RO
35	Battery pack malfunction temperature 8	90C3	S16		0.1℃	RO
36	Reserve	90C4 - 90FF	N/A		N/A	N/A

Note: This block of data contains a total of 24 battery cell fault voltage data, arranged in address order.

5.10.4BMS Battery pack information address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	Balanced bus voltage	9104	U16		0.1V	RO
2	Balanced current	9105	S16		0.1A	RO
3	Battery cell manufacturer	9106 - 9109	ASCII		N/A	RO
4	SOH	910A	U16	Range:[0, 1000]	0.1%	RO
5	Pack rated capacity	910B	U16		0.1Ah	RO
6	Battery cell temperature 1	910C	S16		0.1℃	RO
7	Battery cell temperature 2	910D	S16		0.1℃	RO
8	...	...	...	...	...	...
9	Battery cell temperature 15	911A	S16		0.1℃	RO
10	Battery cell temperature 16	911B	S16		0.1℃	RO
11	Total discharge Ah during the life-cycle	911C - 911D	U32		0.01Ah	RO
12	Total charging Ah during the life-cycle	911E - 911F	U32		0.01Ah	RO
13	Total discharge Wh during the life-cycle	9120 - 9121	U32		1Wh	RO
14	Total charging Wh during the life-cycle	9122 - 9123	U32		1Wh	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
15	Alarm status 2	9124	U16		N/A	RO
16	Protection status 2	9125	U16		N/A	RO
17	Fault status 2	9126	U16		N/A	RO

Note: This block of data contains a total of 16 battery cell temperature data items, arranged in address order.

5.10.5Battery cluster 1 address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	BDU Serial number	9400 - 9409	ASCII		N/A	RO
2	BDU Software version number	940A - 940D	ASCII	Example: "0V100011"	N/A	RO
3	Battery number	940E	U16	Number of batteries connected to the port	N/A	RO
4	Reserve	940F - 941F	N/A		N/A	N/A
5	PCU1 Software version number	9420 - 9423	ASCII	Example: "0V010010"	N/A	RO
6	BMS1 Serial number	9424 - 942D	ASCII		N/A	RO
7	BMS1 Software version number	942E - 9431	ASCII	Example: "0V010111"	N/A	RO
8	BMS1 Voltage	9432	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
9	BMS1 Current	9433	S16	High voltage side, displaying current to the outside world	0.01A	RO
10	BMS1 Power	9434	S16	External display of power	0.01kW	RO
11	BMS1 State of charge	9435	U16		1%	RO
12	BMS1 Environmental temperature of battery	9436	S16		0.1℃	RO
13	BMS1 Current state	9437	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
14	Reserve	9438 - 943F	N/A		N/A	N/A
15	PCU2 Software version number	9440 - 9443	ASCII	Example: "0V010010"	N/A	RO
16	BMS2 Serial number	9444 - 944D	ASCII		N/A	RO
17	BMS2 Software version number	944E - 9451	ASCII	Example: "0V010111"	N/A	RO
18	BMS2 Voltage	9452	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
19	BMS2 Current	9453	S16	High voltage side, displaying current to the outside world	0.01A	RO
20	BMS2 Power	9454	S16	External display of power	0.01kW	RO
21	BMS2 State of charge	9455	U16		1%	RO
22	BMS2 Environmental temperature of battery	9456	S16		0.1℃	RO
23	BMS2 Current state	9457	U16	1：Charging state 2：Discharging state 3：Sleeping state	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				4：Fault state 5：Loss reduction state Other：Unknown state		
24	Reserve	9458 - 945F	N/A		N/A	N/A
25	PCU3 Software version number	9460 - 9463	ASCII	Example: "0V010010"	N/A	RO
26	BMS3 Serial number	9464 - 946D	ASCII		N/A	RO
27	BMS3 Software version number	946E - 9471	ASCII	Example: "0V010111"	N/A	RO
28	BMS3 Voltage	9472	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
29	BMS3 Current	9473	S16	High voltage side, displaying current to the outside world	0.01A	RO
30	BMS3 Power	9474	S16	External display of power	0.01kW	RO
31	BMS3 State of charge	9475	U16		1%	RO
32	BMS3 Environmental temperature of battery	9476	S16		0.1℃	RO
33	BMS3 Current state	9477	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
34	Reserve	9478 - 947F	N/A		N/A	N/A
35	PCU4 Software version number	9480 - 9483	ASCII	Example: "0V010010"	N/A	RO
36	BMS4 Serial number	9484 - 948D	ASCII			
37	BMS4 Software version number	948E - 9491	ASCII	Example: "0V010111"	N/A	RO
38	BMS4 Voltage	9492	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
39	BMS4 Current	9493	S16	High voltage side, displaying current to the outside world	0.01A	RO
40	BMS4 Power	9494	S16	External display of power	0.01kW	RO
41	BMS4 State of charge	9495	U16		1%	RO
42	BMS4 Environmental temperature of battery	9496	S16		0.1℃	RO
43	BMS4 Current state	9497	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
44	Reserve	9498 - 949F	N/A		N/A	N/A
45	PCU5 Software version number	94A0 - 94A3	ASCII	Example: "0V010010"	N/A	RO
46	BMS5 Serial number	94A4 - 94AD	ASCII			
47	BMS5 Software version number	94AE - 94B1	ASCII	Example: "0V010111"	N/A	RO
48	BMS5 Voltage	94B2	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
49	BMS5 Current	94B3	S16	High voltage side, displaying current to the	0.01A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				outside world		
50	BMS5 Power	94B4	S16	External display of power	0.01kW	RO
51	BMS5 State of charge	94B5	U16		1%	RO
52	BMS5 Environmental temperature of battery	94B6	S16		0.1℃	RO
53	BMS5 Current state	94B7	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
54	Reserve	94B8 - 94BF	N/A		N/A	N/A
55	PCU6 Software version number	94C0 - 94C3	ASCII	Example: "0V010010"	N/A	RO
56	BMS6 Serial number	94C4 - 94CD	ASCII			
57	BMS6 Software version number	94CE - 94D1	ASCII	Example: "0V010111"	N/A	RO
58	BMS6 Voltage	94D2	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
59	BMS6 Current	94D3	S16	High voltage side, displaying current to the outside world	0.01A	RO
60	BMS6 Power	94D4	S16	External display of power	0.01kW	RO
61	BMS6 State of charge	94D5	U16		1%	RO
62	BMS6 Environmental temperature of battery	94D6	S16		0.1℃	RO
63	BMS6 Current state	94D7	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
64	Reserve	94D8 - 94DF	N/A		N/A	N/A
65	PCU7 Software version number	94E0 - 94E3	ASCII	Example: "0V010010"	N/A	RO
66	BMS7 Serial number	94E4 - 94ED	ASCII			
67	BMS7 Software version number	94EE - 94F1	ASCII	Example: "0V010111"	N/A	RO
68	BMS7 Voltage	94F2	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
69	BMS7 Current	94F3	S16	High voltage side, displaying current to the outside world	0.01A	RO
70	BMS7 Power	94F4	S16	External display of power	0.01kW	RO
71	BMS7 State of charge	94F5	U16		1%	RO
72	BMS7 Environmental temperature of battery	94F6	S16		0.1℃	RO
73	BMS7 Current state	94F7	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
				Other：Unknown state		
74	Reserve	94F8 - 94FF	N/A		N/A	N/A
75	PCU8 Software version number	9500 - 9503	ASCII	Example: "0V010010"	N/A	RO
76	BMS8 Serial number	9504 - 950D	ASCII			
77	BMS8 Software version number	950E - 9511	ASCII	Example: "0V010111"	N/A	RO
78	BMS8 Voltage	9512	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
79	BMS8 Current	9513	S16	High voltage side, displaying current to the outside world	0.01A	RO
80	BMS8 Power	9514	S16	External display of power	0.01kW	RO
81	BMS8 State of charge	9515	U16		1%	RO
82	BMS8 Environmental temperature of battery	9516	S16		0.1℃	RO
83	BMS8 Current state	9517	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
84	Reserve	9518 - 95FF	N/A		N/A	N/A

5.10.6Battery cluster 2 address

Index	Name	Address（HEX）	Type	Description	Unit	Properties
1	BDU Serial number	9600 - 9609	ASCII		N/A	RO
2	BDU Software version number	960A - 960D	ASCII	Example: "0V100011"	N/A	RO
3	Battery number	960E	U16	Number of batteries connected to the port	N/A	RO
4	Reserve	960F - 961F	N/A		N/A	N/A
5	PCU1 Software version number	9620 - 9623	ASCII	Example: "0V010010"	N/A	RO
6	BMS1 Serial number	9624 - 962D	ASCII		N/A	RO
7	BMS1 Software version number	962E - 9631	ASCII	Example: "0V010111"	N/A	RO
8	BMS1 Voltage	9632	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
9	BMS1 Current	9633	S16	High voltage side, displaying current to the outside world	0.01A	RO
10	BMS1 Power	9634	S16	External display of power	0.01kW	RO
11	BMS1 State of charge	9635	U16		1%	RO
12	BMS1 Environmental temperature of battery	9636	S16		0.1℃	RO
13	BMS1 Current state	9637	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
14	Reserve	9638 - 963F	N/A		N/A	N/A
15	PCU2 Software version number	9640 - 9643	ASCII	Example: "0V010010"	N/A	RO
16	BMS2 Serial number	9644 - 964D	ASCII		N/A	RO
17	BMS2 Software version number	964E - 9651	ASCII	Example: "0V010111"	N/A	RO
18	BMS2 Voltage	9652	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
19	BMS2 Current	9653	S16	High voltage side, displaying current to the outside world	0.01A	RO
20	BMS2 Power	9654	S16	External display of power	0.01kW	RO
21	BMS2 State of charge	9655	U16		1%	RO
22	BMS2 Environmental temperature of battery	9656	S16		0.1℃	RO
23	BMS2 Current state	9657	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
24	Reserve	9658 - 965F	N/A		N/A	N/A
25	PCU3 Software version number	9660 - 9663	ASCII	Example: "0V010010"	N/A	RO
26	BMS3 Serial number	9664 - 966D	ASCII		N/A	RO
27	BMS3 Software version number	966E - 9671	ASCII	Example: "0V010111"	N/A	RO
28	BMS3 Voltage	9672	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
29	BMS3 Current	9673	S16	High voltage side, displaying current to the outside world	0.01A	RO
30	BMS3 Power	9674	S16	External display of power	0.01kW	RO
31	BMS3 State of charge	9675	U16		1%	RO
32	BMS3 Environmental temperature of battery	9676	S16		0.1℃	RO
33	BMS3 Current state	9677	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
34	Reserve	9678 - 967F	N/A		N/A	N/A
35	PCU4 Software version number	9680 - 9683	ASCII	Example: "0V010010"	N/A	RO
36	BMS4 Serial number	9684 - 968D	ASCII			
37	BMS4 Software version number	968E - 9691	ASCII	Example: "0V010111"	N/A	RO
38	BMS4 Voltage	9692	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
39	BMS4 Current	9693	S16	High voltage side, displaying current to the outside world	0.01A	RO
40	BMS4 Power	9696	S16	External display of power	0.01kW	RO
41	BMS4 State of charge	9697	U16		1%	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
42	BMS4 Environmental temperature of battery	9696	S16		0.1℃	RO
43	BMS4 Current state	9697	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
44	Reserve	9698 - 969F	N/A		N/A	N/A
45	PCU5 Software version number	96A0 - 96A3	ASCII	Example: "0V010010"	N/A	RO
46	BMS5 Serial number	96A4 - 96AD	ASCII			
47	BMS5 Software version number	96AE - 96B1	ASCII	Example: "0V010111"	N/A	RO
48	BMS5 Voltage	96B2	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
49	BMS5 Current	96B3	S16	High voltage side, displaying current to the outside world	0.01A	RO
50	BMS5 Power	96B4	S16	External display of power	0.01kW	RO
51	BMS5 State of charge	96B5	U16		1%	RO
52	BMS5 Environmental temperature of battery	96B6	S16		0.1℃	RO
53	BMS5 Current state	96B7	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
54	Reserve	96B8 - 96BF	N/A		N/A	N/A
55	PCU6 Software version number	96C0 - 96C3	ASCII	Example: "0V010010"	N/A	RO
56	BMS6 Serial number	96C4 - 96CD	ASCII			
57	BMS6 Software version number	96CE - 96D1	ASCII	Example: "0V010111"	N/A	RO
58	BMS6 Voltage	96D2	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
59	BMS6 Current	96D3	S16	High voltage side, displaying current to the outside world	0.01A	RO
60	BMS6 Power	96D4	S16	External display of power	0.01kW	RO
61	BMS6 State of charge	96D5	U16		1%	RO
62	BMS6 Environmental temperature of battery	96D6	S16		0.1℃	RO
63	BMS6 Current state	96D7	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
64	Reserve	96D8 - 96DF	N/A		N/A	N/A
65	PCU7 Software version number	96E0 - 96E3	ASCII	Example: "0V010010"	N/A	RO

Index	Name	Address（HEX）	Type	Description	Unit	Properties
66	BMS7 Serial number	96E4 - 96ED	ASCII			
67	BMS7 Software version number	96EE - 96F1	ASCII	Example: "0V010111"	N/A	RO
68	BMS7 Voltage	96F2	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
69	BMS7 Current	96F3	S16	High voltage side, displaying current to the outside world	0.01A	RO
70	BMS7 Power	96F4	S16	External display of power	0.01kW	RO
71	BMS7 State of charge	96F5	U16		1%	RO
72	BMS7 Environmental temperature of battery	96F6	S16		0.1℃	RO
73	BMS7 Current state	96F7	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
74	Reserve	96F8 - 96FF	N/A		N/A	N/A
75	PCU8 Software version number	9700 - 9703	ASCII	Example: "0V010010"	N/A	RO
76	BMS8 Serial number	9704 - 970D	ASCII			
77	BMS8 Software version number	970E - 9711	ASCII	Example: "0V010111"	N/A	RO
78	BMS8 Voltage	9712	U16	High voltage side, displaying voltage to the outside world	0.1V	RO
79	BMS8 Current	9713	S16	High voltage side, displaying current to the outside world	0.01A	RO
80	BMS8 Power	9714	S16	External display of power	0.01kW	RO
81	BMS8 State of charge	9715	U16		1%	RO
82	BMS8 Environmental temperature of battery	9716	S16		0.1℃	RO
83	BMS8 Current state	9717	U16	1：Charging state 2：Discharging state 3：Sleeping state 4：Fault state 5：Loss reduction state Other：Unknown state	N/A	RO
84	Reserve	9718 - 97FF	N/A		N/A	N/A

6.Appendix to the Agreement

6.1Description of system fault information

The Domain	Byte	Bit	Fault ID	Description
Fault 1 （0x0405）	byte1	bit0	1	Grid over-voltage
		bit1	2	Grid under-voltage
		bit2	3	Grid over-frequency
		bit3	4	Grid under-frequency
		bit4	5	Leakage current faults
		bit5	6	High penetration error
		bit6	7	Low penetration error
		bit7	8	island error
	byte2	bit0	9	Grid instantaneous value over-voltage 1
		bit1	10	Grid instantaneous value over-voltage2
		bit2	11	Grid line voltage error
		bit3	12	Inverter voltage error
		bit4	13	Anti-backflow overload
		bit5	14	Grid voltage unbalance
		bit6	15	Inverter transient over-voltage
		bit7	16	Sudden grid phase change
Fault 2 （0x0406）	byte1	bit0	17	Grid current sampling error
		bit1	18	DCI sampling error(AC)
		bit2	19	Network voltage sampling error(DC)
		bit3	20	Network voltage sampling error(AC)
		bit4	21	GFCI sampling error(DC)
		bit5	22	GFCI sampling error(AC)
		bit6	23	DCV sampling error
		bit7	24	Input current sampling error
	byte2	bit0	25	DCI sampling error(DC)
		bit1	26	Branch current sampling error
		bit2	27	PV low impedance to ground
		bit3	28	PID abnormal output
		bit4	29	Leakage current consistency error
		bit5	30	Network voltage consistency error
		bit6	31	DCI consistency error
		bit7	32	Neutral ground fault
Fault 3 （0x0407）	byte1	bit0	33	SPI communication error(DC)
		bit1	34	SPI communication error(AC)
		bit2	35	Chip error(DC)
		bit3	36	Chip error(AC)
		bit4	37	Auxiliary power error
		bit5	38	Inverter soft start failure
		bit6	39	Arc shutdown protection

The Domain	Byte	Bit	Fault ID	Description
		bit7	40	Weak light detection failure
	byte2	bit0	41	Relay detection failure
		bit1	42	Low insulation impedance
		bit2	43	Grounding error
		bit3	44	Wrong input mode setting
		bit4	45	CT error
		bit5	46	Input reversal error
		bit6	47	Parallel error
		bit7	48	Serial number error
Fault 4（0x0408）	byte1	bit0	49	Battery temperature protection
		bit1	50	Radiator 1 temperature protection
		bit2	51	Radiator 2 temperature protection
		bit3	52	Radiator 3 temperature protection
		bit4	53	Radiator 4 temperature protection
		bit5	54	Radiator 5 temperature protection
		bit6	55	Radiator 6 temperature protection
		bit7	56	NTC error
	byte2	bit0	57	Ambient temperature 1 protection
		bit1	58	Ambient temperature 2 protection
		bit2	59	Module 1 temperature protection
		bit3	60	Module 2 temperature protection
		bit4	61	Module 3 temperature protection
		bit5	62	Module temperature difference too large
		bit6	63	N/A
		bit7	64	N/A
Fault 5（0x0409）	byte1	bit0	65	Bus-bar valid value unbalanced
		bit1	66	Bus-bar instantaneous value unbalance
		bit2	67	Bus-bar under-voltage during grid connection
		bit3	68	Bus-bar low voltage
		bit4	69	PV over-voltage
		bit5	70	Battery over-voltage
		bit6	71	LLC-Bus over-voltage protection
		bit7	72	Bus-bar Rms software over-voltage
	byte2	bit0	73	Bus-bar transient software over-voltage
		bit1	74	Fly-span capacitor over-voltage protection
		bit2	75	Fly-span capacitor under-voltage protection
		bit3	76	PV under-voltage
		bit4	77	N/A
		bit5	78	N/A
		bit6	79	N/A
		bit7	80	N/A
Fault 6（0x040A）	byte1	bit0	81	Battery over-current software protection
		bit1	82	DCI over-current protection
		bit2	83	Output transient current protection
		bit3	84	BckBst software over-current

The Domain	Byte	Bit	Fault ID	Description
		bit4	85	Inverter Rms current protection
		bit5	86	PV transient software over-current
		bit6	87	PV parallel uneven current
		bit7	88	Output current unbalance
	byte2	bit0	89	PV software over-current protection
		bit1	90	Balanced circuit over-current protection
		bit2	91	Resonance protection
		bit3	92	Software wave-by-wave current limit protection
		bit4	93	PV branch software over-current 1(enabled by default)
		bit5	94	Primary DC charge and discharge over current protection
		bit6	95	Software BUS over-current protection
		bit7	96	EPS short circuit protection
Fault 7（0x040B）	byte1	bit0	97	LLC bus-bar hardware over-voltage
		bit1	98	Inverter bus hardware over-voltage
		bit2	99	BckBst hardware over-current
		bit3	100	Battery hardware over-current
		bit4	101	Primary DC hardware GaN tube error
		bit5	102	PV hardware over-current
		bit6	103	AC output hardware over-current
		bit7	104	Hardware differential over-current
	byte2	bit0	105	Meter communication error
		bit1	106	Serial number model error
		bit2	107	Hardware version mismatch
		bit3	108	Generator startup failure
		bit4	109	Generator overload protection
		bit5	110	Overload protection 1
		bit6	111	Overload protection 2
		bit7	112	Overload protection 3
Fault 8（0x040C）	byte1	bit0	113	Over-temperature load-down
		bit1	114	Frequency load-down
		bit2	115	Frequency loading
		bit3	116	Voltage load-down
		bit4	117	Voltage loading
		bit5	118	Low temperature load-down
		bit6	119	N/A
		bit7	120	N/A
	byte2	bit0	121	Lightning Protection Fault(DC)
		bit1	122	Lightning Protection Fault(AC)
		bit2	123	Output short circuit error
		bit3	124	Low battery power
		bit4	125	Battery prohibit discharge alarm
		bit5	126	N/A
		bit6	127	N/A
		bit7	128	Battery input reverse protection
Fault 9	byte1	bit0	129	AC hardware over-current fault

The Domain	Byte	Bit	Fault ID	Description
（0x040D）		bit1	130	Bus over-voltage fault
		bit2	131	Bus hardware over-voltage fault
		bit3	132	PV uneven flow fault
		bit4	133	EPS battery over-current fault
		bit5	134	Output transient over-current fault
		bit6	135	AC current unbalance fault
		bit7	136	Inverter soft start failure fault
	byte2	bit0	137	Input mode setting fault
		bit1	138	Input over-current fault
		bit2	139	Input hardware over-current fault
		bit3	140	Relay permanent failure
		bit4	141	Bus unbalance fault
		bit5	142	Lightning Protection Fault(DC)
		bit6	143	Lightning Protection Fault(AC)
		bit7	144	Grid relay fault
Fault 10 （0x040E）	byte1	bit0	145	USB fault
		bit1	146	WIFI fault
		bit2	147	Bluetooth fault
		bit3	148	RTC clock fault
		bit4	149	Communication board EEPROM error
		bit5	150	Communication board FLASH error
		bit6	151	The battery is partially disconnected
		bit7	152	Safety version error
	byte2	bit0	153	SCI error(DC)
		bit1	154	SCI error(AC)
		bit2	155	SCI error(Fuse)
		bit3	156	Software version inconsistency
		bit4	157	Lithium battery 1 communication failure
		bit5	158	Lithium battery 2 communication failure
		bit6	159	Lithium battery 3 communication failure
		bit7	160	Lithium battery 4 communication failure
Fault 11 （0x040F）	byte1	bit0	161	Forced shutdown
		bit1	162	Remote shutdown
		bit2	163	Drms0 shutdown
		bit3	164	Power station communication failure shutdown
		bit4	165	N/A
		bit5	166	N/A
		bit6	167	N/A
		bit7	168	AFCI force shutdown
	byte2	bit0	169	Fan 1 fault
		bit1	170	Fan 2 fault
		bit2	171	Fan 3 fault
		bit3	172	Fan 4 fault
		bit4	173	Fan 5 fault
		bit5	174	Fan 6 fault

The Domain	Byte	Bit	Fault ID	Description
		bit6	175	Fan 7 fault
		bit7	176	Heating film fault
Fault 12 (0x0410)	byte1	bit0	177	BMS over-voltage protection
		bit1	178	BMS under-voltage protection
		bit2	179	BMS high temperature protection
		bit3	180	BMS low temperature protection
		bit4	181	BMS over-current protection
		bit5	182	BMS short-circuit protection
		bit6	183	BMS version inconsistency
		bit7	184	BMS CAN version inconsistency
	byte2	bit0	185	BMS CAN version too low
		bit1	186	Battery discharge over-temperature protection
		bit2	187	Battery discharge low temperature protection
		bit3	188	Battery charging over-temperature protection
		bit4	189	Arc device communication failure
		bit5	190	Battery charging low temperature protection
		bit6	191	PID repair failure
		bit7	192	PLC module heartbeat lost
Fault 13 (0x0411)	byte1	bit0	193	Group string insurance open circuit 1-1
		bit1	194	Group string insurance open circuit 1-2
		bit2	195	Group string insurance open circuit 2-1
		bit3	196	Group string insurance open circuit 2-2
		bit4	197	Group string insurance open circuit 3-1
		bit5	198	Group string insurance open circuit 3-2
		bit6	199	Group string insurance open circuit 4-1
		bit7	200	Group string insurance open circuit 4-2
	byte2	bit0	201	Group string insurance open circuit 5-1
		bit1	202	Group string insurance open circuit 5-2
		bit2	203	Group string insurance open circuit 6-1
		bit3	204	Group string insurance open circuit 6-2
		bit4	205	Group string insurance open circuit 7-1
		bit5	206	Group string insurance open circuit 7-2
		bit6	207	Group string insurance open circuit 8-1
		bit7	208	Group string insurance open circuit 8-2
Fault 14 (0x0412)	byte1	bit0	209	Group string insurance open circuit 9-1
		bit1	210	Group string insurance open circuit 9-2
		bit2	211	Group string insurance open circuit 10-1
		bit3	212	Group string insurance open circuit 10-2
		bit4	213	Group string insurance open circuit 11-1
		bit5	214	Group string insurance open circuit 11-2
		bit6	215	Group string insurance open circuit 12-1
		bit7	216	Group string insurance open circuit 12-2
	byte2	bit0	217	Group string insurance open circuit 13-1
		bit1	218	Group string insurance open circuit 13-2
		bit2	219	Group string insurance open circuit 14-1

The Domain	Byte	Bit	Fault ID	Description
		bit3	220	Group string insurance open circuit 14-2
		bit4	221	Group string insurance open circuit 15-1
		bit5	222	Group string insurance open circuit 15-2
		bit6	223	Group string insurance open circuit 16-1
		bit7	224	Group string insurance open circuit 16-2
Fault 15（0x0413）	byte1	bit0	225	Input insurance open circuit 0
		bit1	226	Input insurance open circuit 1
		bit2	227	Input insurance open circuit 2
		bit3	228	Input insurance open circuit 3
		bit4	229	Input insurance open circuit 4
		bit5	230	Input insurance open circuit 5
		bit6	231	Input insurance open circuit 6
		bit7	232	Input insurance open circuit 7
	byte2	bit0	233	Input insurance open circuit 8
		bit1	234	Input insurance open circuit 9
		bit2	235	Input insurance open circuit 10
		bit3	236	Input insurance open circuit 11
		bit4	237	Input insurance open circuit 12
		bit5	238	Input insurance open circuit 13
		bit6	239	Input insurance open circuit 14
		bit7	240	Input insurance open circuit 15
Fault 16（0x0414）	byte1	bit0	241	N/A
		bit1	242	N/A
		bit2	243	N/A
		bit3	244	N/A
		bit4	245	N/A
		bit5	246	N/A
		bit6	247	N/A
		bit7	248	N/A
	byte2	bit0	249	N/A
		bit1	250	N/A
		bit2	251	N/A
		bit3	252	N/A
		bit4	253	N/A
		bit5	254	N/A
		bit6	255	N/A
		bit7	256	N/A
Fault 17（0x0415）	byte1	bit0	257	N/A
		bit1	258	N/A
		bit2	259	N/A
		bit3	260	N/A
		bit4	261	N/A
		bit5	262	N/A
		bit6	263	N/A
		bit7	264	N/A

The Domain	Byte	Bit	Fault ID	Description
	byte2	bit0	265	N/A
		bit1	266	N/A
		bit2	267	N/A
		bit3	268	N/A
		bit4	269	N/A
		bit5	270	N/A
		bit6	271	N/A
		bit7	272	N/A
Fault 18 (0x0416)	byte1	bit0	273	N/A
		bit1	274	N/A
		bit2	275	N/A
		bit3	276	N/A
		bit4	277	N/A
		bit5	278	N/A
		bit6	279	N/A
		bit7	280	N/A
	byte2	bit0	281	N/A
		bit1	282	N/A
		bit2	283	N/A
		bit3	284	N/A
		bit4	285	N/A
		bit5	286	N/A
		bit6	287	N/A
		bit7	288	N/A
Fault 19 (0x0432)	byte1	bit0	289	N/A
		bit1	290	N/A
		bit2	291	N/A
		bit3	292	N/A
		bit4	293	N/A
		bit5	294	N/A
		bit6	295	N/A
		bit7	296	N/A
	byte2	bit0	297	N/A
		bit1	298	N/A
		bit2	299	N/A
		bit3	300	N/A
		bit4	301	N/A
		bit5	302	N/A
		bit6	303	N/A
		bit7	304	N/A
Fault 20 (0x0433)	byte1	bit0	305	Control unit over temperature warning
		bit1	306	Control unit over temperature protection
		bit2	307	Control unit low temperature warning
		bit3	308	Temperature sensor fault
		bit4	309	Abnormal DC port temperature (including temperature and NTC abnormalities)

The Domain	Byte	Bit	Fault ID	Description
		bit5	310	Abnormal AC port temperature (including temperature and NTC abnormalities)
		bit6	311	N/A
		bit7	312	N/A
	byte2	bit0	313	Module 4 temperature protection
		bit1	314	Module 5 temperature protection
		bit2	315	Module 6 temperature protection
		bit3	316	N/A
		bit4	317	N/A
		bit5	318	N/A
		bit6	319	N/A
		bit7	320	N/A
Fault 21（0x0434）	byte1	bit0	321	Grid phase sequence error
		bit1	322	Phase line to ground failure
		bit2	323	Anti-short-circuit protection
		bit3	324	Anti-short-circuit device fault
		bit4	325	DC relay fault
		bit5	326	Anti-short-circuit temperature error
		bit6	327	Module temperature low protection
		bit7	328	AC start timeout
	byte2	bit0	329	Over modulation
		bit1	330	Device ID mismatch
		bit2	331	Internal communicate fault
		bit3	332	Carrier synchronization signal fault
		bit4	333	Bus short circuit protection
		bit5	334	Boost short circuit fault
		bit6	335	PV+ ground short circuit fault
		bit7	336	N/A
Fault 22（0x0435）	byte1	bit0	337	N/A
		bit1	338	N/A
		bit2	339	N/A
		bit3	340	N/A
		bit4	341	N/A
		bit5	342	N/A
		bit6	343	N/A
		bit7	344	N/A
	byte2	bit0	345	N/A
		bit1	346	N/A
		bit2	347	N/A
		bit3	348	N/A
		bit4	349	N/A
		bit5	350	N/A
		bit6	351	N/A
		bit7	352	N/A
Fault 23（0x0436）	byte1	bit0	353	N/A
		bit1	354	N/A

The Domain	Byte	Bit	Fault ID	Description
		bit2	355	N/A
		bit3	356	N/A
		bit4	357	N/A
		bit5	358	N/A
		bit6	359	N/A
		bit7	360	N/A
	byte2	bit0	361	N/A
		bit1	362	N/A
		bit2	363	N/A
		bit3	364	N/A
		bit4	365	N/A
		bit5	366	N/A
		bit6	367	N/A
		bit7	368	N/A
Fault 24（0x0437）	byte1	bit0	369	N/A
		bit1	370	N/A
		bit2	371	N/A
		bit3	372	N/A
		bit4	373	N/A
		bit5	374	N/A
		bit6	375	N/A
		bit7	376	N/A
	byte2	bit0	377	Inverter hardware MOS tube fault
		bit1	378	EPS hardware output over-current protection
		bit2	379	AFCI self-checking error
		bit3	380	Load short circuit protection
		bit4	381	DC switch 1 trip
		bit5	382	DC switch 2 trip
		bit6	383	DC switch 3 trip
		bit7	384	DC switch 4 trip
Fault 25（0x0438）	byte1	bit0	385	Hardware AD sampling bias error
		bit1	386	Fan 8 fault
		bit2	387	N/A
		bit3	388	N/A
		bit4	389	N/A
		bit5	390	N/A
		bit6	391	N/A
		bit7	392	N/A
	byte2	bit0	393	N/A
		bit1	394	N/A
		bit2	395	N/A
		bit3	396	N/A
		bit4	397	N/A
		bit5	398	N/A
		bit6	399	N/A

The Domain	Byte	Bit	Fault ID	Description
		bit7	400	N/A
Fault 26 （0x0439）	byte1	bit0	401	Arc failure 0
		bit1	402	Arc failure 1
		bit2	403	Arc failure 2
		bit3	404	Arc failure 3
		bit4	405	Arc failure 4
		bit5	406	Arc failure 5
		bit6	407	Arc failure 6
		bit7	408	Arc failure 7
	byte2	bit0	409	Arc failure 8
		bit1	410	Arc failure 9
		bit2	411	Arc failure 10
		bit3	412	Arc failure 11
		bit4	413	Arc failure 12
		bit5	414	Arc failure 13
		bit6	415	Arc failure 14
		bit7	416	Arc failure 15
Fault 27 （0x043A）	byte1	bit0	417	Arc failure 16
		bit1	418	Arc failure 17
		bit2	419	Arc failure 18
		bit3	420	Arc failure 19
		bit4	421	Arc failure 20
		bit5	422	Arc failure 21
		bit6	423	Arc failure 22
		bit7	424	Arc failure 23
	byte2	bit0	425	Arc failure 24
		bit1	426	Arc failure 25
		bit2	427	Arc failure 26
		bit3	428	Arc failure 27
		bit4	429	Arc failure 28
		bit5	430	Arc failure 29
		bit6	431	Arc failure 30
		bit7	432	Arc failure 31
Fault 28 （0x043B）	byte1	bit0	433	AFCI 1 communication fault
		bit1	434	AFCI 2 communication fault
		bit2	435	N/A
		bit3	436	N/A
		bit4	437	N/A
		bit5	438	N/A
		bit6	439	N/A
		bit7	440	N/A
	byte2	bit0	441	Modify over voltage protection
		bit1	442	ARM_DSP protocol version inconsistent
		bit2	443	ARM_AFCI protocol version inconsistent
		bit3	444	ARM_DCDC protocol version inconsistent

The Domain	Byte	Bit	Fault ID	Description
		bit4	445	N/A
		bit5	446	N/A
		bit6	447	N/A
		bit7	448	N/A
Fault 29（0x043C）	byte1	bit0	449	The ground zero voltage is too high
		bit1	450	PV branch software over-current 2(disabled by default)
		bit2	451	N-line open circuit fault
		bit3	452	Grid frequency change protection
		bit4	453	Phase-loss failure
		bit5	454	PV branch software current regurgitation
		bit6	455	Inverter voltage sampling error
		bit7	456	Inverter EPS sampling failure fault
	byte2	bit0	457	Bus voltage consistency error
		bit1	458	Bus current sampling error
		bit2	459	Parallel off grid start synchronization fault
		bit3	460	Primary hardware over-current protection
		bit4	461	Off-grid prohibited discharge failure
		bit5	462	Live wire and ground wire short circuit fault
		bit6	463	Permanent fault of live wire and ground wire short circuit
		bit7	464	N/A
Fault 30（0x043D）	byte1	bit0	465	DCDC Fault
		bit1	466	BCU Fault
		bit2	467	BMU Fault
		bit3	468	PID Fault
		bit4	469	N/A
		bit5	470	N/A
		bit6	471	N/A
		bit7	472	N/A
	byte2	bit0	473	Discharge Mos fault
		bit1	474	Charge Mos fault
		bit2	475	Battery NTC fault
		bit3	476	Cell fault
		bit4	477	Cell voltage sampling fault
		bit5	478	Storage fault
		bit6	479	N/A
		bit7	480	N/A

6.2Description of system status information

The Domain	Byte	Bit	Alarm ID	Description
Status 1 (0x0477)	byte1	bit0	1	Grid low voltage drop load
		bit1	2	Over temperature drop load
		bit2	3	Bus over voltage drop load
		bit3	4	Reactive power limit active power
		bit4	5	Command active load reduction
		bit5	6	DRMs load reduction
		bit6	7	Anti-backflow load reduction
		bit7	8	Remote active power control
	byte2	bit0	9	Power grid under-voltage safety regulation load
		bit1	10	Over-frequency load reduction
		bit2	11	Under-frequency loading
		bit3	12	Internal (other) load reduction
		bit4	13	PV branch over-current limited load
		bit5	14	Grid high voltage alarm
		bit6	15	N/A
		bit7	16	N/A
Status 2 (0x0478)	byte1	bit0	17	Remote PF control
		bit1	18	Remote reactive power (reactive power percentage) control
		bit2	19	Independent reactive power control
		bit3	20	Running state at night
		bit4	21	N/A
		bit5	22	N/A
		bit6	23	N/A
		bit7	24	N/A
	byte2	bit0	25	Forced shutdown
		bit1	26	Remote shutdown
		bit2	27	DRMs0 shutdown
		bit3	28	Anti-backflow communication protection shutdown
		bit4	29	Emergency shut down
		bit5	30	N/A
		bit6	31	N/A
		bit7	32	N/A
Status 3 (0x0479)	byte1	bit0	33	N/A
		bit1	34	Grid disconnection prohibit status
		bit2	35	Grid connection prohibit charge and discharge
		bit3	36	EMS shutdown
		bit4	37	AC-bus precharge failure
		bit5	38	BMS fault
		bit6	39	N/A
		bit7	40	Heating film heating in progress
	byte2	bit0	41	Heating film enable
		bit1	42	BMS requests heating
		bit2	43	DSP allows heating

The Domain	Byte	Bit	Alarm ID	Description
		bit3	44	PV meter communication disconnected
		bit4	45	N/A
		bit5	46	N/A
		bit6	47	N/A
		bit7	48	N/A
Status 4（0x047A）	byte1	bit0	49	Boost tube 1 breakdown
		bit1	50	Boost tube 2 breakdown
		bit2	51	Boost tube 3 breakdown
		bit3	52	Boost tube 4 breakdown
		bit4	53	N/A
		bit5	54	N/A
		bit6	55	N/A
		bit7	56	N/A
	byte2	bit0	57	N/A
		bit1	58	N/A
		bit2	59	N/A
		bit3	60	N/A
		bit4	61	N/A
		bit5	62	N/A
		bit6	63	N/A
		bit7	64	N/A
Status 5（0x047B）	byte1	bit0	65	N/A
		bit1	66	N/A
		bit2	67	N/A
		bit3	68	N/A
		bit4	69	N/A
		bit5	70	N/A
		bit6	71	N/A
		bit7	72	N/A
	byte2	bit0	73	N/A
		bit1	74	N/A
		bit2	75	N/A
		bit3	76	N/A
		bit4	77	N/A
		bit5	78	N/A
		bit6	79	N/A
		bit7	80	N/A
Status 6（0x047C）	byte1	bit0	81	Remote load reduction enable
		bit1	82	Logic interface load reduction enable
		bit2	83	Anti-backflow load reduction enable
		bit3	84	N/A
		bit4	85	N/A
		bit5	86	N/A
		bit6	87	N/A
		bit7	88	N/A

The Domain	Byte	Bit	Alarm ID	Description
	byte2	bit0	89	N/A
		bit1	90	N/A
		bit2	91	N/A
		bit3	92	N/A
		bit4	93	N/A
		bit5	94	N/A
		bit6	95	N/A
		bit7	96	N/A
Status 7（0x047D）	byte1	bit0	97	ISO warning
		bit1	98	N/A
		bit2	99	N/A
		bit3	100	N/A
		bit4	101	N/A
		bit5	102	N/A
		bit6	103	N/A
		bit7	104	N/A
	byte2	bit0	105	N/A
		bit1	106	N/A
		bit2	107	Fan 1 alarm
		bit3	108	N/A
		bit4	109	N/A
		bit5	110	N/A
		bit6	111	N/A
		bit7	112	N/A
Status 8（0x047E）	byte1	bit0	113	N/A
		bit1	114	N/A
		bit2	115	N/A
		bit3	116	N/A
		bit4	117	N/A
		bit5	118	N/A
		bit6	119	N/A
		bit7	120	N/A
	byte2	bit0	121	N/A
		bit1	122	N/A
		bit2	123	N/A
		bit3	124	N/A
		bit4	125	N/A
		bit5	126	N/A
		bit6	127	N/A
		bit7	128	N/A
Status 9（0x047F）	byte1	bit0	129	N/A
		bit1	130	N/A
		bit2	131	N/A
		bit3	132	N/A
		bit4	133	N/A

The Domain	Byte	Bit	Alarm ID	Description
		bit5	134	N/A
		bit6	135	N/A
		bit7	136	N/A
	byte2	bit0	137	N/A
		bit1	138	N/A
		bit2	139	N/A
		bit3	140	N/A
		bit4	141	N/A
		bit5	142	N/A
		bit6	143	N/A
		bit7	144	N/A

6.3Description of DCDC fault information

The Domain	Byte	Bit	Fault ID	Description
Fault 1 (0x5065)	byte1	bit0	1	Low-voltage side under-voltage
		bit1	2	Low-voltage side over-voltage software
		bit2	3	High-voltage side under-voltage
		bit3	4	How-voltage side over-voltage software
		bit4	5	Fly-span capacitance imbalance
		bit5	6	Low-voltage side over-current software
		bit6	7	ARM chip failure
		bit7	8	High voltage reverse connection fault
	byte2	bit0	9	High voltage side over-voltage hardware
		bit1	10	Low voltage side over-voltage hardware
		bit2	11	Low voltage side over-current hardware
		bit3	12	IGBT driver failure
		bit4	13	Emergency version number error
		bit5	14	Insulation fault
		bit6	15	Ground fault
		bit7	16	Relay fault
Fault 2 (0x5066)	byte1	bit0	17	ARM communication failure
		bit1	18	PCS communication failure
		bit2	19	PCU1 communication failure
		bit3	20	PCU2 communication failure
		bit4	21	PCU3 communication failure
		bit5	22	BMS1 communication failure
		bit6	23	BMS2 communication failure
		bit7	24	BMS3 communication failure
	byte2	bit0	25	DC1 communication failure
		bit1	26	DC2 communication failure
		bit2	27	DC3 communication failure
		bit3	28	DC4 communication failure
		bit4	29	DC5 communication failure
		bit5	30	DC6 communication failure
		bit6	31	Module over temperature
		bit7	32	Zero drift
Fault 3 (0x5067)	byte1	bit0	33	High-voltage fuse fault
		bit1	34	Low-voltage side wake-up fault
		bit2	35	Low-voltage side wake-up timeout
		bit3	36	Environmental over-temperature protection
		bit4	37	Fly-span capacitor over-voltage hardware protection
		bit5	38	Low-voltage side reverse fault
		bit6	39	High-voltage side relay short circuit fault
		bit7	40	High-voltage side relay open circuit fault
	byte2	bit0	41	Low-voltage side relay short circuit fault
		bit1	42	Low-voltage side relay open circuit fault
		bit2	43	N/A

The Domain	Byte	Bit	Fault ID	Description
		bit3	44	High and low voltage mismatch fault
		bit4	45	Low-voltage side precharge failure
		bit5	46	High-voltage side precharge failure
		bit6	47	N/A
		bit7	48	N/A
Fault 4 （0x5068）	byte1	bit0	49	N/A
		bit1	50	N/A
		bit2	51	N/A
		bit3	52	N/A
		bit4	53	N/A
		bit5	54	N/A
		bit6	55	N/A
		bit7	56	N/A
	byte2	bit0	57	N/A
		bit1	58	N/A
		bit2	59	N/A
		bit3	60	N/A
		bit4	61	N/A
		bit5	62	N/A
		bit6	63	N/A
		bit7	64	N/A
Fault 5 （0x5069）	byte1	bit0	65	N/A
		bit1	66	N/A
		bit2	67	N/A
		bit3	68	N/A
		bit4	69	N/A
		bit5	70	N/A
		bit6	71	N/A
		bit7	72	N/A
	byte2	bit0	73	N/A
		bit1	74	N/A
		bit2	75	N/A
		bit3	76	N/A
		bit4	77	N/A
		bit5	78	N/A
		bit6	79	N/A
		bit7	80	N/A
Fault 6 （0x506A）	byte1	bit0	81	N/A
		bit1	82	N/A
		bit2	83	N/A
		bit3	84	N/A
		bit4	85	N/A
		bit5	86	N/A
		bit6	87	N/A
		bit7	88	N/A

The Domain	Byte	Bit	Fault ID	Description
	byte2	bit0	89	N/A
		bit1	90	N/A
		bit2	91	N/A
		bit3	92	N/A
		bit4	93	N/A
		bit5	94	N/A
		bit6	95	N/A
		bit7	96	N/A
Fault 7 （0x506B）	byte1	bit0	97	N/A
		bit1	98	N/A
		bit2	99	N/A
		bit3	100	N/A
		bit4	101	N/A
		bit5	102	N/A
		bit6	103	N/A
		bit7	104	N/A
	byte2	bit0	105	N/A
		bit1	106	N/A
		bit2	107	N/A
		bit3	108	N/A
		bit4	109	N/A
		bit5	110	N/A
		bit6	111	N/A
		bit7	112	N/A
Fault 8 （0x506C）	byte1	bit0	113	N/A
		bit1	114	N/A
		bit2	115	N/A
		bit3	116	N/A
		bit4	117	N/A
		bit5	118	N/A
		bit6	119	N/A
		bit7	120	N/A
	byte2	bit0	121	N/A
		bit1	122	N/A
		bit2	123	N/A
		bit3	124	N/A
		bit4	125	N/A
		bit5	126	N/A
		bit6	127	N/A
		bit7	128	N/A
Fault 9 （0x506D）	byte1	bit0	129	Software version inconsistency
		bit1	130	SCI communication failure
		bit2	131	CAN1 communication failure(DCDC Intranet)
		bit3	132	CAN2 communication failure(DCDC to battery)
		bit4	133	N/A

The Domain	Byte	Bit	Fault ID	Description
		bit5	134	N/A
		bit6	135	N/A
		bit7	136	N/A
	byte2	bit0	137	N/A
		bit1	138	N/A
		bit2	139	N/A
		bit3	140	N/A
		bit4	141	N/A
		bit5	142	N/A
		bit6	143	N/A
		bit7	144	N/A
Fault 10（0x506E）	byte1	bit0	145	DCDC count error
		bit1	146	PCU count error
		bit2	147	BMS count error
		bit3	148	Fan 1 alarm (inside)
		bit4	149	Fan 2 alarm (outside)
		bit5	150	Fan 3 alarm (outside)
		bit6	151	MAC address conflict
		bit7	152	N/A
	byte2	bit0	153	N/A
		bit1	154	N/A
		bit2	155	N/A
		bit3	156	N/A
		bit4	157	N/A
		bit5	158	N/A
		bit6	159	N/A
		bit7	160	N/A

6.4Description of PCU alarms information

The Domain	Byte	Bit	Fault ID	Description
Alarm 1 (0x6024)	byte1	bit0	800	N/A
		bit1	801	Charging soft startup failed
		bit2	802	Discharging soft startup failed
		bit3	803	Fan 1 fault
		bit4	804	Fan 2 fault
		bit5	805	Fan 3 fault
		bit6	806	Fan 4 fault
		bit7	807	PCU version inconsistent
	byte2	bit0	808	Radiator 1 over-temperature alarm
		bit1	809	Environment over temperature alarm
		bit2	810	Radiator 2 over-temperature alarm
		bit3	811	N/A
		bit4	812	N/A
		bit5	813	Prohibit charging alarm
		bit6	814	Prohibit discharging alarm
		bit7	815	Battery imbalance alarm
Alarm 2 (0x6025)	byte1	bit0	816	Bus-bar over-voltage alarm
		bit1	817	Bus-bar under-voltage alarm
		bit2	818	Battery over-voltage alarm
		bit3	819	Battery under-voltage alarm
		bit4	820	N/A
		bit5	821	N/A
		bit6	822	N/A
		bit7	823	N/A
	byte2	bit0	824	N/A
		bit1	825	N/A
		bit2	826	N/A
		bit3	827	N/A
		bit4	828	N/A
		bit5	829	N/A
		bit6	830	N/A
		bit7	831	N/A
Alarm 3 (0x6026)	byte1	bit0	832	N/A
		bit1	833	N/A
		bit2	834	N/A
		bit3	835	N/A
		bit4	836	N/A
		bit5	837	N/A
		bit6	838	N/A
		bit7	839	N/A
	byte2	bit0	840	N/A
		bit1	841	N/A
		bit2	842	N/A

The Domain	Byte	Bit	Fault ID	Description
		bit3	843	N/A
		bit4	844	N/A
		bit5	845	N/A
		bit6	846	N/A
		bit7	847	N/A
Alarm 4 （0x6027）	byte1	bit0	848	N/A
		bit1	849	N/A
		bit2	850	N/A
		bit3	851	N/A
		bit4	852	N/A
		bit5	853	N/A
		bit6	854	N/A
		bit7	855	N/A
	byte2	bit0	856	N/A
		bit1	857	N/A
		bit2	858	N/A
		bit3	859	N/A
		bit4	860	N/A
		bit5	861	N/A
		bit6	862	N/A
		bit7	863	N/A

6.5Description of PCU fault information

The Domain	Byte	Bit	Fault ID	Description
Fault 1（0x6028）	byte1	bit0	864	Radiator 1 over temperature protection
		bit1	865	Environment over temperature protection
		bit2	866	SCI communication fault
		bit3	867	CAN1 communication fault
		bit4	868	Relay 1 fault
		bit5	869	Radiator 2 over temperature protection
		bit6	870	CAN2 communication fault
		bit7	871	Relay 3 fault
	byte2	bit0	872	Bus-bar software over-voltage
		bit1	873	Bus-bar software under-voltage
		bit2	874	Battery software over-voltage
		bit3	875	Battery software under-voltage
		bit4	876	Battery software over-current
		bit5	877	Fly-span capacitor 1 over-voltage
		bit6	878	Fly-span capacitor 2 over-voltage
		bit7	879	Hardware over-current
Fault 2（0x6029）	byte1	bit0	880	Permanent bus-bar over-voltage
		bit1	881	Permanent battery under-voltage
		bit2	882	Permanent over-current
		bit3	883	Permanent hardware over-current
		bit4	884	Fly-span capacitor 3 over-voltage
		bit5	885	Fly-span capacitor 4 over-voltage
		bit6	886	Fly-span capacitor 1 under-voltage
		bit7	887	Fly-span capacitor 2 under-voltage
	byte2	bit0	888	Relay 1 permanent fault
		bit1	889	Relay 2 permanent fault
		bit2	890	Fly-span capacitor 3 under-voltage
		bit3	891	Fly-span capacitor 4 under-voltage
		bit4	892	N/A
		bit5	893	Permanent short-circuit protection
		bit6	894	Permanent battery activation failed
		bit7	895	Permanent BUS reverse connection
Fault 3（0x602A）	byte1	bit0	896	Battery status error
		bit1	897	PWM mode error
		bit2	898	BMS version error
		bit3	899	BMS over-voltage and over-current fault
		bit4	900	Battery average over-current protection
		bit5	901	Average overload protection
		bit6	902	Bus-bar software over-current
		bit7	903	Software CBC over-current protection
	byte2	bit0	904	Pack ID error
		bit1	905	Startup bus-bar short-circuit protection
		bit2	906	Bus-bar average under-voltage

The Domain	Byte	Bit	Fault ID	Description
		bit3	907	Chip clock fault
		bit4	908	Inverter CAN communication fault
		bit5	909	Radiator low temperature fault
		bit6	910	Environment low temperature fault
		bit7	911	Sampling bias calibration fault
Fault 4（0x602B）	byte1	bit0	912	N/A
		bit1	913	N/A
		bit2	914	N/A
		bit3	915	N/A
		bit4	916	N/A
		bit5	917	N/A
		bit6	918	N/A
		bit7	919	N/A
	byte2	bit0	920	N/A
		bit1	921	N/A
		bit2	922	N/A
		bit3	923	N/A
		bit4	924	N/A
		bit5	925	N/A
		bit6	926	N/A
		bit7	927	N/A

6.6Description of port temperature detection fault

The Domain	Byte	Bit	Fault ID	Description
Fault1（0x4A41）	byte1	bit0	1	R-phase temperature fault
		bit1	2	S-phase temperature fault
		bit2	3	T-phase temperature fault
		bit3	4	N/A
		bit4	5	N/A
		bit5	6	N/A
		bit6	7	N/A
		bit7	8	N/A
	byte2	bit0	9	R-phase NTC fault
		bit1	10	S-phase NTC fault
		bit2	11	T-phase NTC fault
		bit3	12	N/A
		bit4	13	N/A
		bit5	14	N/A
		bit6	15	N/A
		bit7	16	N/A
Fault2（0x4A42）	byte1	bit0	17	PV1_1 input port temperature fault
		bit1	18	PV1_1 output port temperature fault
		bit2	19	PV1_2 input port temperature fault

The Domain	Byte	Bit	Fault ID	Description
		bit3	20	PV1_2 output port temperature fault
		bit4	21	PV1_3 input port temperature fault
		bit5	22	PV1_3 output port temperature fault
		bit6	23	PV1_4 input port temperature fault
		bit7	24	PV1_4 output port temperature fault
	byte2	bit0	25	PV2_1 input port temperature fault
		bit1	26	PV2_1 output port temperature fault
		bit2	27	PV2_2 input port temperature fault
		bit3	28	PV2_2 output port temperature fault
		bit4	29	PV2_3 input port temperature fault
		bit5	30	PV2_3 output port temperature fault
		bit6	31	PV2_4 input port temperature fault
		bit7	32	PV2_4 output port temperature fault
Fault 3 （0x4A43）	byte1	bit0	33	PV3_1 input port temperature fault
		bit1	34	PV3_1 output port temperature fault
		bit2	35	PV3_2 input port temperature fault
		bit3	36	PV3_2 output port temperature fault
		bit4	37	PV3_3 input port temperature fault
		bit5	38	PV3_3 output port temperature fault
		bit6	39	PV3_4input port temperature fault
		bit7	40	PV3_4 output port temperature fault
	byte2	bit0	41	PV4_1 input port temperature fault
		bit1	42	PV4_1 output port temperature fault
		bit2	43	PV4_2 input port temperature fault
		bit3	44	PV4_2 output port temperature fault
		bit4	45	PV4_3 input port temperature fault
		bit5	46	PV4_3 output port temperature fault
		bit6	47	PV4_4 input port temperature fault
		bit7	48	PV4_4 output port temperature fault
Fault 4 （0x4A44）	byte1	bit0	49	PV5_1input port temperature fault
		bit1	50	PV5_1 output port temperature fault
		bit2	51	PV5_2 input port temperature fault
		bit3	52	PV5_2 output port temperature fault
		bit4	53	PV5_3input port temperature fault
		bit5	54	PV5_3 output port temperature fault
		bit6	55	PV5_4 input port temperature fault
		bit7	56	PV5_4 output port temperature fault
	byte2	bit0	57	PV6_1 input port temperature fault
		bit1	58	PV6_1 output port temperature fault
		bit2	59	PV6_2 input port temperature fault
		bit3	60	PV6_2 output port temperature fault
		bit4	61	PV6_3 input port temperature fault
		bit5	62	PV6_3 output port temperature fault
		bit6	63	PV6_4 input port temperature fault
		bit7	64	PV6_4 output port temperature fault

The Domain	Byte	Bit	Fault ID	Description
Fault 5 （0x4A45）	byte1	bit0	65	PV7_1 input port temperature fault
		bit1	66	PV7_1 output port temperature fault
		bit2	67	PV7_2 input port temperature fault
		bit3	68	PV7_2 output port temperature fault
		bit4	69	PV7_3input port temperature fault
		bit5	70	PV7_3 output port temperature fault
		bit6	71	PV7_4 input port temperature fault
		bit7	72	PV7_4 output port temperature fault
	byte2	bit0	73	PV8_1 input port temperature fault
		bit1	74	PV8_1 output port temperature fault
		bit2	75	PV8_2 input port temperature fault
		bit3	76	PV8_2 output port temperature fault
		bit4	77	PV8_3 input port temperature fault
		bit5	78	PV8_3 output port temperature fault
		bit6	79	PV8_4 input port temperature fault
		bit7	80	PV8_4 output port temperature fault
Fault 6 （0x4A46）	byte1	bit0	81	N/A
		bit1	82	N/A
		bit2	83	N/A
		bit3	84	N/A
		bit4	85	N/A
		bit5	86	N/A
		bit6	87	N/A
		bit7	88	N/A
	byte2	bit0	89	N/A
		bit1	90	N/A
		bit2	91	N/A
		bit3	92	N/A
		bit4	93	N/A
		bit5	94	N/A
		bit6	95	N/A
		bit7	96	N/A
Fault 7 （0x4A47）	byte1	bit0	97	N/A
		bit1	98	N/A
		bit2	99	N/A
		bit3	100	N/A
		bit4	101	N/A
		bit5	102	N/A
		bit6	103	N/A
		bit7	104	N/A
	byte2	bit0	105	N/A
		bit1	106	N/A
		bit2	107	N/A
		bit3	108	N/A
		bit4	109	N/A

The Domain	Byte	Bit	Fault ID	Description
		bit5	110	N/A
		bit6	111	N/A
		bit7	112	N/A
Fault 8 （0x4A48）	byte1	bit0	113	N/A
		bit1	114	N/A
		bit2	115	N/A
		bit3	116	N/A
		bit4	117	N/A
		bit5	118	N/A
		bit6	119	N/A
		bit7	120	N/A
	byte2	bit0	121	N/A
		bit1	122	N/A
		bit2	123	N/A
		bit3	124	N/A
		bit4	125	N/A
		bit5	126	N/A
		bit6	127	N/A
		bit7	128	N/A
Fault 9 （0x4A49）	byte1	bit0	129	N/A
		bit1	130	N/A
		bit2	131	N/A
		bit3	132	N/A
		bit4	133	N/A
		bit5	134	N/A
		bit6	135	N/A
		bit7	136	N/A
	byte2	bit0	137	N/A
		bit1	138	N/A
		bit2	139	N/A
		bit3	140	N/A
		bit4	141	N/A
		bit5	142	N/A
		bit6	143	N/A
		bit7	144	N/A
Fault 10 （0x4A4A）	byte1	bit0	145	PV1_1 input port NTC fault
		bit1	146	PV1_1 output port NTC fault
		bit2	147	PV1_2 input port NTC fault
		bit3	148	PV1_2 output port NTC fault
		bit4	149	PV1_3 input port NTC fault
		bit5	150	PV1_3 output port NTC fault
		bit6	151	PV1_4 input port NTC fault
		bit7	152	PV1_4 output port NTC fault
	byte2	bit0	153	PV2_1 input port NTC fault
		bit1	154	PV2_1 output port NTC fault

The Domain	Byte	Bit	Fault ID	Description
		bit2	155	PV2_2 input port NTC fault
		bit3	156	PV2_2 output port NTC fault
		bit4	157	PV2_3 input port NTC fault
		bit5	158	PV2_3 output port NTC fault
		bit6	159	PV2_4 input port NTC fault
		bit7	160	PV2_4 output port NTC fault
Fault 11 （0x4A4B）	byte1	bit0	161	PV3_1 input port NTC fault
		bit1	162	PV3_1 output port NTC fault
		bit2	163	PV3_2 input port NTC fault
		bit3	164	PV3_2 output port NTC fault
		bit4	165	PV3_3 input port NTC fault
		bit5	166	PV3_3 output port NTC fault
		bit6	167	PV3_4 input port NTC fault
		bit7	168	PV3_4 output port NTC fault
	byte2	bit0	169	PV4_1 input port NTC fault
		bit1	170	PV4_1 output port NTC fault
		bit2	171	PV4_2 input port NTC fault
		bit3	172	PV4_2 output port NTC fault
		bit4	173	PV4_3 input port NTC fault
		bit5	174	PV4_3 output port NTC fault
		bit6	175	PV4_4 input port NTC fault
		bit7	176	PV4_4 output port NTC fault
Fault 12 （0x4A4C）	byte1	bit0	177	PV5_1 input port NTC fault
		bit1	178	PV5_1 output port NTC fault
		bit2	179	PV5_2 input port NTC fault
		bit3	180	PV5_2 output port NTC fault
		bit4	181	PV5_3 input port NTC fault
		bit5	182	PV5_3 output port NTC fault
		bit6	183	PV5_4 input port NTC fault
		bit7	184	PV5_4 output port NTC fault
	byte2	bit0	185	PV6_1 input port NTC fault
		bit1	186	PV6_1 output port NTC fault
		bit2	187	PV6_2 input port NTC fault
		bit3	188	PV6_2 output port NTC fault
		bit4	189	PV6_3 input port NTC fault
		bit5	190	PV6_3 output port NTC fault
		bit6	191	PV6_4 input port NTC fault
		bit7	192	PV6_4 output port NTC fault
Fault 13 （0x4A4D）	byte1	bit0	193	PV7_1 input port NTC fault
		bit1	194	PV7_1 output port NTC fault
		bit2	195	PV7_2 input port NTC fault
		bit3	196	PV7_2 output port NTC fault
		bit4	197	PV7_3 input port NTC fault
		bit5	198	PV7_3 output port NTC fault
		bit6	199	PV7_4 input port NTC fault

The Domain	Byte	Bit	Fault ID	Description
		bit7	200	PV7_4 output port NTC fault
	byte2	bit0	201	PV8_1 input port NTC fault
		bit1	202	PV8_1 output port NTC fault
		bit2	203	PV8_2 input port NTC fault
		bit3	204	PV8_2 output port NTC fault
		bit4	205	PV8_3 input port NTC fault
		bit5	206	PV8_3 output port NTC fault
		bit6	207	PV8_4 input port NTC fault
		bit7	208	PV8_4 output port NTC fault
Fault 14（0x4A4E）	byte1	bit0	209	N/A
		bit1	210	N/A
		bit2	211	N/A
		bit3	212	N/A
		bit4	213	N/A
		bit5	214	N/A
		bit6	215	N/A
		bit7	216	N/A
	byte2	bit0	217	N/A
		bit1	218	N/A
		bit2	219	N/A
		bit3	220	N/A
		bit4	221	N/A
		bit5	222	N/A
		bit6	223	N/A
		bit7	224	N/A
Fault 15（0x4A4F）	byte1	bit0	225	N/A
		bit1	226	N/A
		bit2	227	N/A
		bit3	228	N/A
		bit4	229	N/A
		bit5	230	N/A
		bit6	231	N/A
		bit7	232	N/A
	byte2	bit0	233	N/A
		bit1	234	N/A
		bit2	235	N/A
		bit3	236	N/A
		bit4	237	N/A
		bit5	238	N/A
		bit6	239	N/A
		bit7	240	N/A
Fault 16（0x4A50）	byte1	bit0	241	N/A
		bit1	242	N/A
		bit2	243	N/A
		bit3	244	N/A

The Domain	Byte	Bit	Fault ID	Description
		bit4	245	N/A
		bit5	246	N/A
		bit6	247	N/A
		bit7	248	N/A
	byte2	bit0	249	N/A
		bit1	250	N/A
		bit2	251	N/A
		bit3	252	N/A
		bit4	253	N/A
		bit5	254	N/A
		bit6	255	N/A
		bit7	256	N/A